UNIT 5

Impact of Computing

100-Question Priority-Starred Master Set

- Questions 1–90: Multiple Choice (Full Priority Ratings & Distractor Breakdowns)
- Questions 91–100: Create Performance Task Style Written Responses (9 Points Each)
- Complete Answer Key and Detailed Worked Solutions at the End

Examination Structure

This paper contains 100 continuously numbered questions designed for complete mastery of AP Computer Science Principles Unit 5: Impact of Computing (Big Idea 5: IOC, weighted 21%–26% of the AP Exam). All answers, step-by-step worked solutions, and 9-point scoring rubrics appear after Question 100.

★ PRIORITY RATING SYSTEM

★★★★★ **Very Important**	High-frequency exam concepts: Beneficial vs harmful unintended effects, Digital Divide socio-economic root causes, Computing bias in historical training data and proxy variables, Crowdsourcing citizen science & consensus validation, Creative Commons licensing matrix (BY, NC, SA, ND), PII & re-identification linkage attacks, Phishing mechanics vs Spear-Phishing, Multi-Factor Authentication (MFA distinct factor categories), and Cryptographic password salting/hashing.
★★★★ **High Priority**	Core impact standards: Automatic copyright upon creation vs Public Domain (CC0), Open source vs proprietary licensing, Algorithmic bias audits & disaggregated evaluations, Passive digital footprints & browser fingerprinting, Mass surveillance vs 4th Amendment privacy, Keyloggers, Ransomware extortion, Password entropy, and Virtual Private Networks (VPNs).
★★★ **Mandatory**	Essential social & technical mechanics: Gig economy labor impacts, Web Content Accessibility Guidelines (WCAG), Astroturfing in crowdsourced ratings, DMCA safe harbor rules, Cryptojacking background resource theft, and Software patch lifecycles.

Topic Coverage (Big Idea 5: IOC-1 through IOC-2)

5.1 Beneficial & Harmful Computing Innovations, Unintended Consequences, Gig Economy & Automation

5.2 Digital Divide Geographic, Socioeconomic, Homework Gap, Mesh Networks & Subsidies

5.3 Computing Bias Training Data Biases, Facial Recognition Disparities, Proxies & Auditing

5.4 Crowdsourcing Citizen Science (eBird/Galaxy Zoo), Consensus Filters, Quality vs Astroturfing

5.5 Legal & Ethics Copyright, Creative Commons (BY/NC/SA/ND), Open Source, Fair Use & DMCA

5.6 Safe Computing PII, Mosaic Re-Identification, Cookies, Fingerprinting, Phishing, Malware & MFA

AP Exam Pro Tip: Remember: Computing innovations always produce *both beneficial and harmful effects*, often unanticipated by creators! **Computing bias** is not fixed merely by deleting demographic labels—algorithms easily reconstruct bias through **proxy variables** (e.g., zip codes, browsing habits). In **Intellectual Property**, copyright attaches automatically upon fixation; **Creative Commons** grants flexible advance permissions without surrender of ownership. And in **Cybersecurity**: *Multi-Factor Authentication (MFA) requires at least two distinct categories (know, have, are)*—entering a password plus a security question is NOT true MFA!

- **Questions 1–90:** Multiple choice, four options (A–D), exactly one correct answer.
- **Questions 91–100:** Create Performance Task Written Response scenarios, 9 points each, with full rubrics.
- **Difficulty Mix:** Authentic Medium AP Exam Level ❖ Rigorously 3-Pass Verified.
- **Answer Section:** Begins after Question 100 — master answer key, detailed MCQ solutions with distractor failure analysis, and model FRQ solutions with official point allocations.

Question 01

★★★★★ VERY IMPORTANT

BENEFICIAL VS. HARMFUL EFFECTS OF COMPUTING INNOVATIONS

A computing innovation is defined as an innovation that includes a computer or program code as an integral part of its functionality.

Which of the following statements best characterizes the societal impacts of computing innovations according to the AP Computer Science Principles curriculum?

(A) Computing innovations almost always produce both beneficial and harmful effects, some of which are intentional while others are completely unintended by the original developers.

(B) Computing innovations only impact individuals who personally own a smartphone or desktop computer.

(C) Computing innovations only produce harmful effects if the developers intentionally write malicious software code.

(D) Computing innovations provide exclusively beneficial effects because automated technology is inherently superior to human judgment.

Question 02

★★★★★ VERY IMPORTANT

UNINTENDED CONSEQUENCES OF SOCIAL MEDIA ALGORITHMS

Engineers at a social media company develop an automated recommendation algorithm designed solely to maximize **user engagement** (measured by time spent on the app, clicks, and video shares).

Which of the following describes an **unintended harmful consequence** of this optimization goal?

(A) The algorithm prevents the phone battery from charging overnight.

(B) The algorithm charges users double the subscription fee whenever they post photographs.

(C) The algorithm amplifies sensationalized, controversial, and false news stories because outrage and conflict generate higher user engagement and longer viewing times.

(D) The algorithm automatically deletes all user accounts after 30 days of inactivity.

Question 03

★★★★ HIGH PRIORITY

SOCIOECONOMIC IMPACTS OF RIDE-SHARING & GIG ECONOMY PLATFORMS

Mobile ride-sharing applications connect vehicle drivers directly with passengers via GPS-enabled smartphones.

Which of the following correctly pairs a **beneficial effect** and a **harmful effect** of this computing innovation?

(A) Beneficial: Completely eliminates the need for roads and highways | Harmful: Prevents drivers from ever buying personal vehicles.

(B) Beneficial: Makes all gasoline free of charge worldwide | Harmful: Restricts vehicle speeds to five miles per hour.

(C) Beneficial: Automatically repairs flat tires via wireless Bluetooth signals | Harmful: Forces passengers to walk to their final destination.

(D) Beneficial: Provides flexible on-demand income for drivers and affordable transit for passengers | Harmful: Contributes to urban traffic congestion and offers unstable earnings without traditional employee healthcare benefits.

Question 04

★★★★ HIGH PRIORITY

TRADE-OFFS IN SMART HOME INTERNET OF THINGS (IOT) TECHNOLOGY

A family installs Internet-connected smart locks, security cameras, and voice-activated virtual assistant speakers throughout their home.

What is the primary **security and privacy trade-off** introduced by adopting these computing innovations?

(A) Smart locks guarantee that household members will never forget their homework.

(B) Smart home devices completely eliminate the need for home electrical wiring.

(C) Smart devices make the house physically invisible to neighbors.

(D) Greater convenience and remote household automation come at the expense of creating network vulnerabilities that attackers could exploit to spy on private conversations or unlock physical doors.

Question 05

★★★★★ VERY IMPORTANT

SOCIETAL IMPACTS OF AUTONOMOUS VEHICLES

Automotive companies are developing self-driving autonomous vehicles using computer vision, LIDAR sensors, and deep learning algorithms.

Which of the following describes a major **beneficial effect** alongside a major **economic disruption** resulting from this innovation?

(A) Beneficial: Drastically reduces automobile crashes caused by human driver fatigue and distraction | Economic Disruption: Threatens the livelihood of millions of commercial truck, taxi, and delivery drivers through job displacement.

(B) Beneficial: Doubles the speed of sound inside the vehicle | Economic Disruption: Destroys all computers in the country.

(C) Beneficial: Eliminates gravity so vehicles can fly across oceans | Economic Disruption: Forces all roads to be repaved with solid gold.

(D) Beneficial: Prevents rain from falling on highways | Economic Disruption: Replaces gasoline engines with steam boilers.

Question 06

★★★★ HIGH PRIORITY

TARGET POPULATION DESIGN AND DEMOGRAPHIC EXCLUSION

A software startup designs a mobile banking app exclusively optimized for the latest flagship smartphones with high-speed 5G connectivity and facial recognition biometrics.

Which segment of the population is most likely to be **unintentionally excluded** from accessing financial services through this innovation?

(A) Software engineers who work at technology companies.

(B) Executives who purchase the latest electronic devices every year.

(C) Low-income and elderly individuals who use older, inexpensive smartphones without biometric sensors or who live in areas lacking 5G cellular infrastructure.

(D) College students studying computer science.

Question 07

★★★★★ VERY IMPORTANT

ACCESSIBILITY AND ASSISTIVE TECHNOLOGIES IN SOFTWARE DESIGN

Under the Web Content Accessibility Guidelines (WCAG), software developers are encouraged to include text descriptions (`alt` text) for all images, ensure high color contrast ratios, and provide keyboard navigation shortcuts.

What is the primary objective of implementing these **Accessibility Features**?

(A) To prevent search engines from indexing website content.

(B) To increase the download size of web pages to test server bandwidth limits.

(C) To ensure that computing innovations can be effectively used by individuals with diverse physical, sensory, and cognitive abilities, including those who rely on screen readers.

(D) To restrict website access exclusively to government employees.

Question 08

★★★★ HIGH PRIORITY

IMPACT OF E-COMMERCE ON TRADITIONAL BRICK-AND-MORTAR RETAIL

Global e-commerce platforms allow consumers to order millions of manufactured goods online with automated one-day home delivery.

Which of the following represents an **adverse cultural and economic impact** of this computing innovation on local communities?

(A) A permanent global ban on consumer credit cards.

(B) The decline and closure of local brick-and-mortar storefronts, leading to reduced municipal sales tax revenue and the loss of vibrant neighborhood commercial centers.

(C) An increase in the price of physical paper dictionaries.

(D) A complete worldwide shortage of cardboard shipping boxes.

Question 09

★★★★ HIGH PRIORITY

HEALTHCARE IMPACTS: TELEMEDICINE AND REMOTE PATIENT MONITORING

During public health crises, healthcare systems rapidly deploy **Telemedicine platforms** allowing physicians to consult with patients via secure video calls and remote vitals monitoring.

Which of the following describes both a major **benefit** and a major **limitation** of telemedicine?

(A) Benefit: Replaces all nurses with robots | Limitation: Only works during rainy weather.

(B) Benefit: Cures all genetic diseases instantly | Limitation: Prevents doctors from writing prescription medicines.

(C) Benefit: Eliminates the need for medical schools | Limitation: Causes computer screens to emit medical radiation.

(D) Benefit: Expands access to specialized doctors for rural patients who live far from major hospitals | Limitation: Cannot perform physical examinations or emergency surgical procedures, and excludes patients without reliable broadband internet.

Question 10

★★★★★ VERY IMPORTANT

COMPREHENSIVE INNOVATION IMPACT SYNTHESIS TABLE

An AP Computer Science Principles class analyzes the intended purposes, benefits, and unintended harmful impacts of four major computing innovations:

Innovation	Primary Intended Purpose	Unintended Harmful Consequence
Innovation 1: GPS Navigation	Provide turn-by-turn driving directions	Reroutes thousands of commuter cars through quiet residential neighborhoods, causing congestion and safety hazards
Innovation 2: Facial Recognition	Enhance security and airport boarding speed	Exhibits higher error and misidentification rates for women and individuals with darker skin tones
Innovation 3: Digital Streaming	Offer on-demand entertainment to consumers	Severely diminishes artist compensation from traditional physical album sales
Innovation 4: Smart Wearable Monitors	Track personal fitness and heart metrics	Health insurers may access biometric data to deny medical coverage or raise premiums

Which overarching takeaway from Big Idea 5 is best illustrated by this table?

(A) Engineers can easily eliminate all negative consequences simply by writing programs in the Python language.

(B) Only Innovation 2 is a true computing innovation because it uses artificial intelligence.

(C) Computing innovations should be completely prohibited by law to prevent any possibility of negative consequences.

(D) Computing innovations designed with beneficial intentions can generate complex, widespread negative societal side-effects that require ethical evaluation and regulatory oversight.

Question 11

★★★★★ VERY IMPORTANT

DEFINITION OF THE DIGITAL DIVIDE

The **Digital Divide** is one of the most significant socioeconomic challenges examined in computer science.

Which of the following best defines the Digital Divide according to the AP CSP curriculum?

(A) The physical division between the CPU chip and the RAM memory stick inside a desktop computer case.

(B) The mathematical difference between 32-bit IPv4 addresses and 128-bit IPv6 addresses.

(C) The gap between individuals, households, businesses, and geographic areas at different socioeconomic levels with regard both to their opportunities to access information and communication technologies and to their use of the Internet for a wide variety of activities.

(D) The software wall separating the frontend user interface from the backend database server.

Question 12

★★★★★ VERY IMPORTANT

GEOGRAPHIC INEQUITIES IN BROADBAND DEPLOYMENT

A family living in a remote rural farming valley has access only to slow, expensive satellite internet with severe monthly data caps, while families in the nearby metropolitan city enjoy cheap, unlimited gigabit fiber-optic service.

Which economic and technical factor best explains why telecommunications companies are often reluctant to install high-speed fiber-optic lines in rural communities?

(A) Fiber-optic cables cannot physically transmit light through rural air or farmland.

(B) Satellite radio signals automatically dissolve underground glass cables.

(C) The high upfront financial cost of laying miles of physical fiber-optic cables across sparsely populated terrain yields too few paying customers to make the private infrastructure investment profitable.

(D) Federal law prohibits rural citizens from learning computer science or using the Internet.

Question 13

★★★★★ VERY IMPORTANT

THE 'HOMEWORK GAP' AND EDUCATIONAL INEQUITIES

During a school year, a high school teacher requires students to conduct online research, write essays in cloud-based collaborative word processors, and submit homework via a web portal.

Which educational disparity, commonly termed the **'Homework Gap'**, is most likely to occur in this classroom?

(A) Students from low-income households who lack reliable home internet or dedicated personal laptops will fall behind academically compared to wealthier peers who have seamless home access.

(B) Students with fast home internet will receive lower grades because computer screens reduce reading comprehension.

(C) All students will automatically achieve perfect test scores regardless of study habits.

(D) Students will be unable to turn in physical handwritten paper assignments during morning homeroom.

Question 14

★★★★ HIGH PRIORITY

AGE, DEMOGRAPHICS, AND DIGITAL LITERACY DISPARITIES

A state government migrates its voter registration, unemployment benefits, and driver's license renewal services to an online-only web portal to reduce municipal administrative costs.

Which group of citizens is most disproportionately disadvantaged by this transition?

(A) Elderly senior citizens who may lack familiarity with web browsers, digital navigation skills, or personal computing devices.

(B) Corporate technology executives.

(C) Teenagers who play multiplayer video games.

(D) College students majoring in software engineering.

Question 15

★★★★ HIGH PRIORITY

THE GLOBAL DIGITAL DIVIDE BETWEEN NATIONS

While over 90% of households in developed nations possess high-speed broadband connections, large portions of the population in developing nations lack basic Internet access.

Which combination of obstacles contributes most heavily to this **Global Digital Divide**?

(A) The refusal of modern computers to process non-English text characters.

(B) Global treaties that forbid developing countries from importing computer silicon.

(C) A complete absence of radio waves in equatorial regions of the Earth.

(D) Unreliable electrical power grids, high costs of cellular data relative to local average income, and a scarcity of educational content published in local indigenous languages.

Question 16

★★★★★ VERY IMPORTANT

MOBILE-ONLY ACCESS VS. DEDICATED BROADBAND COMPUTING

A census survey reveals that 100% of families in a neighborhood own at least one smartphone with cellular data, but only 35% have a desktop/laptop computer connected to fixed home broadband.

Why does relying **exclusively on a smartphone** still represent a significant digital divide disadvantage for students and job seekers?

(A) Smartphones cannot connect to the global Internet or send emails.

(B) Cellular networks are legally prohibited from displaying website text.

(C) Smartphones automatically erase user data every 24 hours.

(D) Smartphones have small screens, touch keyboards unsuited for long essays, restrictive mobile operating systems that hinder software development, and strict monthly cellular data caps that throttle video lectures.

Question 17

★★★★ HIGH PRIORITY

COMMUNITY WIRELESS MESH NETWORKS AS AN EQUITY SOLUTION

In an underserved urban neighborhood where major commercial ISPs refuse to upgrade old copper phone lines, local residents install directional Wi-Fi antennas on their rooftops. The antennas connect to one another in a decentralized peer-to-peer web, sharing a single high-speed fiber connection across dozens of homes.

What is this community-driven solution called?

(A) A centralized mainframe computer array.

(B) A physical dial-up telegraph line.

(C) A **Community Wireless Mesh Network**.

(D) A commercial satellite television broadcast.

Question 18

★★★★ HIGH PRIORITY

THE ROLE OF PUBLIC LIBRARIES AND COMMUNITY CENTERS

Many municipalities fund public libraries to provide free high-speed Wi-Fi, loaner laptop computers, portable mobile hotspots, and free digital skills workshops for residents.

How do these public institutions directly combat the **Digital Divide**?

(A) They serve as essential public access points and safety nets, ensuring that individuals without financial means can access government services, apply for jobs, and develop essential digital literacy skills.

(B) They replace the need for schools, colleges, and teachers in the community.

(C) They prevent private tech companies from manufacturing electronic devices.

(D) They physically confiscate smartphones from citizens to force them to read physical books.

Question 19

★★★★ HIGH PRIORITY

GOVERNMENT POLICIES, SUBSIDIES, AND UNIVERSAL SERVICE INITIATIVES

Programs such as the federal **E-Rate program** and affordable broadband vouchers provide government funding to connect schools and libraries to high-speed internet and subsidize monthly broadband bills for low-income households.

What is the underlying philosophy behind treating high-speed Internet as a **public necessity (utility)** rather than a luxury commodity?

(A) Because access to high-speed Internet is now essential for participating in modern economic life, obtaining an education, accessing healthcare, and exercising democratic rights, lack of access perpetuates systemic poverty.

(B) Because computers run out of digital bits unless subsidized by government taxes.

(C) Because the government wants to monitor every keystroke made by every citizen in real time.

(D) Because high-speed internet eliminates the need for roads, bridges, and clean drinking water.

Question 20

★★★★★ VERY IMPORTANT

ANALYZING COMMUNITY DIGITAL DIVIDE PROFILES AND INTERVENTIONS

A state technology council analyzes broadband access profiles for four diverse school districts:

District	Geography	Median Household Income	Home Broadband Access	Primary Barrier
District W	Remote Mountainous Rural	$42,000	28%	No physical telecommunication lines exist within 15 miles
District X	Dense Urban Center	$26,000	45%	Fiber cables exist, but monthly $80 subscription fees are unaffordable
District Y	Suburban Tech Corridor	$110,000	97%	Negligible barrier
District Z	Agricultural Farmland	$38,000	32%	Sparse population makes commercial cable installation unprofitable

Which of the following policy actions would most effectively address the specific barrier in **District X**?

(A) Replacing all computers in District X with physical encyclopedias.

(B) Providing income-eligible families with direct monthly broadband financial subsidies or vouchers, because the physical fiber infrastructure is already present and only the financial cost is prohibitive.

(C) Banning all telecommunications companies from operating in District X.

(D) Laying 50 miles of new undersea fiber cables across the city streets.

Question 21

★★★★★ VERY IMPORTANT

DEFINITION OF COMPUTING BIAS

In computer science, **Computing Bias** occurs when an automated system or algorithm produces systematically unfair or prejudiced outcomes.

Which of the following statements correctly describes how computing bias originates in software systems?

(A) Computing bias only affects software written in older programming languages like Fortran.

(B) Computing bias is physically impossible because mathematical algorithms and computer code are inherently objective and free of all prejudice.

(C) Computing bias can be embedded directly by human programmer choices, but more frequently emerges when machine learning models are trained on historical datasets that reflect human prejudices or demographic underrepresentation.

(D) Computing bias only occurs when computer hardware microprocessors overheat due to excessive electrical current.

Question 22

★★★★★ VERY IMPORTANT

BIASED TRAINING DATA IN AUTOMATED RESUME SCREENING

A major technology corporation trains an Artificial Intelligence model to review thousands of job resumes and automatically select top applicants for interviews. The model is trained on resumes submitted to the company over the preceding 10 years, during which the vast majority of hired software engineers were men.

What unintended behavior is the hiring algorithm most likely to exhibit?

(A) The algorithm automatically hires all human beings who live on Earth.

(B) The algorithm prints every resume in reverse alphabetical order.

(C) The algorithm deletes the company's website from the Internet.

(D) The algorithm penalizes resumes containing words associated with women (such as 'women's chess club captain' or graduates of women's colleges) because it learned to associate past male candidates with hiring success.

Question 23

★★★★★ VERY IMPORTANT

DEMOGRAPHIC DISPARITIES IN FACIAL RECOGNITION ALGORITHMS

Academic researchers benchmark several commercial facial recognition systems used by law enforcement and airport security. They discover that while the systems achieve 99% accuracy on lighter-skinned males, error rates jump to nearly 35% when classifying darker-skinned females.

What is the primary technical cause of this severe performance disparity?

(A) The image datasets used to train the computer vision neural networks consisted predominantly of lighter-skinned male faces, leaving the algorithms with insufficient training samples to learn facial features of darker-skinned women.

(B) Darker-skinned individuals emit magnetic frequencies that block camera sensors.

(C) Facial recognition algorithms are programmed to only work during the morning hours.

(D) Computer monitors cannot display darker skin tones.

Question 24

★★★★★ VERY IMPORTANT

ALGORITHMIC RISK ASSESSMENT AND PROXY VARIABLES IN CRIMINAL JUSTICE

Courts in several jurisdictions utilize automated algorithmic risk assessment tools (such as COMPAS) to predict the likelihood that an arrested defendant will commit a future crime before determining bail.

Even though the software code explicitly excludes 'race' as an input variable, why can the algorithm still produce racially biased risk scores?

(A) The algorithm uses other input variables (such as neighborhood zip code, socio-economic factors, and prior arrest history) that act as statistical **proxy variables** closely correlated with systemic racial disparities.

(B) The algorithm physically reads the defendant's thoughts through Wi-Fi signals.

(C) Computers always output random numbers when calculating legal bail amounts.

(D) The algorithm is required by law to flip a digital coin for every defendant.

Question 25

★★★★ HIGH PRIORITY

HEALTHCARE INEQUITIES IN DIAGNOSTIC MACHINE LEARNING

A biomedical research team trains an AI diagnostic app to detect malignant skin melanoma from smartphone photos. The training dataset consists of 50,000 photos collected almost entirely from clinics in Scandinavian countries.

When deployed globally, why does this diagnostic tool fail to reliably protect patients in Mediterranean, African, and Latin American populations?

(A) The app only works if the patient speaks a Scandinavian language.

(B) Melanomas present with different visual contrast and pigmentation patterns on darker skin types, but the model never learned these patterns due to the homogeneous Scandinavian training data.

(C) Skin cancer cannot be detected using digital cameras outside of Scandinavia.

(D) Sunlight in equatorial regions erases the software code from smartphones.

Question 26

★★★★ HIGH PRIORITY

SOFTWARE ENGINEERING TEAM DIVERSITY AND DEVELOPER BLINDSPOTS

When a major smartphone company first released its comprehensive health and fitness tracking software, the app tracked blood pressure, dietary calories, and steps, but completely omitted female menstrual cycle tracking—a basic health metric for half the human population.

Which software engineering design reality does this omission illustrate?

(A) Tracking steps consumes all available smartphone battery power.

(B) The operating system was programmed to prevent female users from downloading apps.

(C) A lack of diversity on software development teams can lead to **developer blindspots**, where features essential to underrepresented groups are unintentionally overlooked during the design phase.

(D) Smartphones do not have enough internal memory to track calendar dates for women.

Question 27

★★★★ HIGH PRIORITY

ALGORITHMIC DISCRIMINATION IN ONLINE ADVERTISING DELIVERY

Researchers studying online targeted advertising algorithms find that ads for high-paying STEM and executive jobs ($200,000+ salaries) are delivered to male profiles significantly more frequently than to female profiles with identical educational credentials.

Which factor often causes this advertising delivery disparity?

(A) Federal regulations require tech companies to hide executive job openings from women.

(B) Women are forbidden from using web browsers to search for employment.

(C) The ad platform's automated bidding and optimization algorithm determines that female consumers are more commercially valuable to retail advertisers, thereby outbidding job ads and steering women toward consumer goods rather than career opportunities.

(D) The Internet protocols (TCP/IP) only route job listings to computers registered to men.

Question 28

★★★★★ VERY IMPORTANT

TECHNIQUES FOR AUDITING AND MITIGATING ALGORITHMIC BIAS

Before deploying an automated credit-scoring model that determines whether loan applicants are approved for home mortgages, a bank hires an independent team to conduct an **Algorithmic Audit**.

Which methodology should the auditors employ to verify that the model is fair across demographic groups?

(A) Ensure that the credit-scoring program is printed on recycled paper.

(B) Test the model only on the developers who wrote the computer code.

(C) Verify that the software compiles without syntax errors in less than one second.

(D) Disaggregate performance metrics (such as approval rates, false positive rates, and false negative rates) across distinct demographic subgroups (race, gender, age) to verify equitable accuracy and outcomes.

Question 29

★★★★ HIGH PRIORITY

DECONSTRUCTING THE MYTH OF ALGORITHMIC NEUTRALITY

A city official claims: '*Our new predictive policing software cannot be biased because it is run entirely by a computer algorithm, and numbers do not have feelings or personal prejudices.*'

Why is the official's reasoning flawed according to computational ethics?

(A) While computers do not have feelings, algorithms are designed by humans, use datasets selected by humans, and learn from historical human records that contain deeply ingrained societal biases.

(B) Numbers in computer science are always subjective and change their mathematical value daily.

(C) Computers develop personal emotional feelings whenever they run predictive models.

(D) City officials are legally barred from reading computer software output.

Question 30

★★★★★ VERY IMPORTANT

COMPARATIVE ALGORITHMIC BIAS SCENARIOS AND MITIGATIONS

An ethics committee reviews four automated decision systems displaying biased outcomes:

System	Context	Observed Bias	Primary Root Cause
System 1	College Admissions Screening	Downgrades applications from non-traditional high schools	Historical training data lacked diverse school models
System 2	Voice Recognition Assistant	Struggles to comprehend Scottish and Southern regional accents	Training audio datasets recorded only standard American accents
System 3	Automated Translation Tool	Automatically assigns male pronouns to 'doctor' and female to 'nurse'	Reflects gender stereotypes embedded in historical web text corpora
System 4	Credit Card Credit Line Assessor	Awards lower credit limits to married women than their husbands	Uses joint spending history without disaggregating individual income

Which universal corrective action is most essential across all four systems to eliminate bias?

(A) Forcing all global citizens to speak with the exact same voice accent.

(B) Retraining and validating the models using comprehensively diverse, representative, and balanced datasets that accurately reflect all demographic and linguistic subgroups.

(C) Permanently shutting down all electrical power to colleges, hospitals, and banks.

(D) Replacing all digital computer code with analog paper filing cabinets.

Question 31

★★★★★ VERY IMPORTANT

DEFINITION AND MECHANICS OF CROWDSOURCING

With the rise of interconnected global networks, organizations increasingly solve complex challenges through **Crowdsourcing**.

Which of the following best defines crowdsourcing according to the AP CSP curriculum?

(A) The process of packing as many people as possible into a physical server computer room.

(B) A database indexing technique that groups identical numbers into a single memory block.

(C) The practice of obtaining needed services, data, ideas, or content by soliciting contributions from a large, distributed group of online people rather than relying solely on traditional employees or contractors.

(D) A network cyberattack where thousands of compromised bots flood a target web server with traffic.

Question 32

★★★★★ VERY IMPORTANT

CITIZEN SCIENCE AND COLLABORATIVE RESEARCH

In projects like **Galaxy Zoo** and **eBird**, hundreds of thousands of amateur volunteers classify telescope images of distant galaxies or record daily observations of migrating birds in an online portal.

What is the primary scientific advantage of this **Citizen Science** approach?

(A) It completely eliminates the need for professional astronomers or biologists.

(B) It allows volunteers to travel physically to distant galaxies via Internet cables.

(C) It allows researchers to analyze massive, petabyte-scale datasets that are far too large for a small team of scientists to manually examine, accelerating scientific discovery at minimal financial cost.

(D) It guarantees that every single volunteer observation is 100% free of human error.

Question 33

★★★★★ VERY IMPORTANT

REAL-TIME CROWDSOURCED TRAFFIC AND SENSOR TELEMETRY

Navigation applications like **Waze** and **OpenStreetMap** rely on crowdsourcing to provide turn-by-turn routing.

How do these applications combine **passive** and **active** crowdsourcing to improve driver commutes?

(A) Passively collecting real-time GPS speed telemetry from all active user smartphones to detect traffic slow-downs, while actively allowing drivers to report road hazards, construction, and police presence.

(B) Passively erasing maps while actively disconnecting the car battery.

(C) Passively downloading music files while actively changing traffic light colors via Bluetooth.

(D) Passively charging drivers' credit cards while actively disabling vehicle headlights.

Question 34

★★★★ HIGH PRIORITY

DECENTRALIZED PEER PRODUCTION VS. TRADITIONAL PUBLISHING

Unlike traditional print encyclopedias authored by a closed editorial board of invited scholars, **Wikipedia** is created through open collaborative peer production where anyone with Internet access can create and edit articles.

What is a major strength and a major challenge of this crowdsourced model?

(A) Strength: Operates without requiring any computer servers | Challenge: Can only be read on physical printed paper.

(B) Strength: Eliminates all spelling mistakes automatically | Challenge: Charges users $50 per viewed article.

(C) Strength: Guarantees that no article will ever change once written | Challenge: Only accessible by university professors.

(D) Strength: Rapidly incorporates new global events and covers niche topics in hundreds of languages | Challenge: Vulnerable to intentional vandalism, misinformation, and edit wars that require active moderation.

Question 35

★★★★ HIGH PRIORITY

ECONOMIC DEMOCRATIZATION THROUGH CROWDFUNDING

An independent inventor creates an affordable solar-powered water purifier. Instead of pitching to Wall Street venture capitalists or taking out high-interest bank loans, the inventor launches a campaign on a **Crowdfunding** platform (such as Kickstarter or GoFundMe).

How does crowdfunding democratize access to financial capital?

(A) It guarantees that every funded product will be manufactured without any production delays.

(B) It allows creators to raise funds through small financial contributions from thousands of everyday backers worldwide, validating market demand without giving up company ownership to traditional gatekeepers.

(C) It forces the federal government to print new paper money specifically for the inventor.

(D) It makes the product completely free of charge to all consumers permanently.

Question 36

★★★★★ VERY IMPORTANT

DATA VERIFICATION AND QUALITY CONTROL IN CROWDSOURCED SYSTEMS

Because crowdsourced platforms accept contributions from unvetted public users, they inevitably receive corrupted, malicious, or erroneous submissions.

Which of the following architectural strategies is most commonly used to maintain **data integrity** in large crowdsourced datasets?

(A) Deleting all submissions received on weekends.

(B) Allowing only users whose first names start with the letter 'A' to submit data.

(C) Implementing multi-contributor consensus algorithms, community reputation scores, and automated statistical anomaly detection to cross-verify submissions before publishing.

(D) Discarding 99% of all submitted data at random without inspection.

Question 37

★★★★ HIGH PRIORITY

RISKS OF CROWDSOURCE MANIPULATION: ASTROTURFING AND FAKE REVIEWS

An e-commerce business hires a 'click farm' of hundreds of low-wage online workers to post thousands of glowing 5-star reviews on its new product while posting 1-star negative reviews on competing products.

Which manipulation tactic is demonstrated in this scenario?

(A) **Astroturfing (Sockpuppeting)**, which creates a deceptive, manufactured illusion of grassroots public support or consensus through coordinated fake accounts.

(B) A distributed volunteer protein folding simulation.

(C) An automated optical character recognition scan.

(D) Open-source software license attribution.

Question 38

★★★★ HIGH PRIORITY

INCENTIVE STRUCTURES IN CROWDSOURCING PLATFORMS

To motivate diverse public participation, some crowdsourcing initiatives use **Gamification** (badges, leaderboards, and achievement levels), while others use **Micro-payments** (such as Amazon Mechanical Turk paying small monetary amounts per completed micro-task).

What is the primary trade-off between these two incentive models?

(A) Gamification relies on intrinsic hobbyist enthusiasm and produces high-quality voluntary engagement for specialized causes, whereas micro-payments enable rapid completion of massive repetitive tasks but risk attracting workers who rush through tasks to maximize hourly earnings.

(B) Gamification causes computers to crash due to excess video memory usage.

(C) Micro-payments make all software illegal to distribute.

(D) Gamification requires physical gold coins, while micro-payments can only be paid in physical paper stock certificates.

Question 39

★★★★ HIGH PRIORITY

DATA PRIVACY CONCERNS IN CROWDSOURCED TELEMETRY

A popular fitness tracking app publishes a global 'Heatmap' crowdsourcing the GPS jogging routes of millions of users worldwide to help runners find popular trails. Security analysts quickly discover that the heatmap clearly reveals the secret geographic perimeters and patrol routes of military bases in remote foreign deserts.

What critical computing lesson was demonstrated by this incident?

(A) Jogging outdoors is forbidden by international telecommunications standards.

(B) Aggregating and publishing crowdsourced location telemetry, even when anonymized, can inadvertently expose sensitive personal habits, military installations, and confidential security perimeters.

(C) GPS satellites only broadcast coordinates to runners during full moon nights.

(D) Fitness tracking apps should only be installed on desktop desktop computers that remain in one room.

Question 40

★★★★★ VERY IMPORTANT

COMPREHENSIVE SYNTHESIS OF CROWDSOURCING MODELS

An AP Computer Science Principles class summarizes four leading crowdsourcing frameworks:

Model	Example Platform	Primary Community Benefit	Inherent Operational Risk
Model 1: Citizen Science	Foldit / Zooniverse	Enables massive-scale scientific data processing	Inconsistent amateur skill levels require cross-verification
Model 2: Collaborative Knowledge	Wikipedia	Provides free, global, multilingual encyclopedic information	Susceptible to bias, edit disputes, and intentional vandalism
Model 3: Dynamic Mapping	Waze / OpenStreetMap	Delivers live, optimized vehicular navigation	May reveal sensitive private or secure geographic routes
Model 4: Crowdfunding	Kickstarter / GoFundMe	Democratizes access to capital for creative projects	Backers risk project delivery delays or product failure

Which overarching conclusion accurately characterizes these four crowdsourcing models?

(A) Crowdsourcing models are only legal in North America and are banned in all other continents.

(B) Crowdsourcing requires all participants to possess a Ph.D. in computer science.

(C) None of these models can function without using facial recognition artificial intelligence.

(D) All four models demonstrate that distributed human collaboration over the Internet can achieve tasks that exceed the capacity of single institutions, provided appropriate verification, moderation, and privacy safeguards are enforced.

Question 41

★★★★★ VERY IMPORTANT

FORMS OF INTELLECTUAL PROPERTY IN COMPUTER SCIENCE

A software developer invents a novel sorting algorithm, writes 10,000 lines of original program code, names the application 'QuickStream', and designs a distinctive logo icon.

Which legal protection automatically protects the **specific source code text** written by the developer against unauthorized copying?

(A) **Trademark**, which legally requires all users to pay cash directly to the United Nations.

(B) **Patent**, which only protects the physical copper wires inside the computer monitor.

(C) **Copyright**, which grants the creator exclusive legal rights to reproduce, distribute, display, and create derivative works of an original creative expression.

(D) **Public Domain**, which makes it illegal for anyone to ever run the software.

Question 42

★★★★★ VERY IMPORTANT

AUTOMATIC INCEPTION OF DIGITAL COPYRIGHT

A student writes a Python program for an AP project and posts the code snippet on a personal public blog without registering with any government office or adding a copyright symbol (©).

Under modern copyright law, what is the copyright status of this code?

(A) The code is **automatically copyrighted** by the student the moment it is created and saved; other programmers cannot legally copy or republish the code without the student's permission.

(B) The code instantly becomes Public Domain because it was posted on the Internet without a formal legal seal.

(C) The code is owned exclusively by the student's Internet Service Provider.

(D) The code is legally invalid because only corporate companies can hold copyright.

Question 43

★★★★★ VERY IMPORTANT

CREATIVE COMMONS LICENSING: CC BY (ATTRIBUTION)

A graphic designer creates a library of vector icons and publishes them under a **Creative Commons Attribution (CC BY)** license.

What does this license permit a student building a mobile application to do with the icons?

(A) The student may freely copy, distribute, modify, and even use the icons commercially, provided that the original designer is credited and given proper attribution.

(B) The student must pay a monthly royalty fee of $500 directly to the designer.

(C) The student can only view the icons on Mondays and Tuesdays.

(D) The student is forbidden from altering the icons in any way.

Question 44

★★★★★ VERY IMPORTANT

CREATIVE COMMONS MODIFIERS: SHAREALIKE (SA) AND NONCOMMERCIAL (NC)

An educational non-profit releases a computer science curriculum textbook under a **CC BY-NC-SA** license.

A for-profit tutoring company downloads the textbook, translates it into Spanish, and sells physical printed copies for $40 each.

Did the tutoring company violate the license, and why?

(A) No; translating text into another language automatically erases all copyright licenses.

(B) Yes; the company violated the **NC (NonCommercial)** clause by selling the book for commercial profit, and violated the **SA (ShareAlike)** clause if their derivative translation was not licensed under identical terms.

(C) Yes; but only because the book was printed on paper rather than viewed on a monitor.

(D) No; Creative Commons licenses legally require all products to be sold for commercial profit.

Question 45

★★★★★ VERY IMPORTANT

CREATIVE COMMONS: NODERIVATIVES (ND) VS. PUBLIC DOMAIN (CC0)

Consider two digital photographs published online:
- **Photo 1:** Published under **CC BY-ND (NoDerivatives)**.
- **Photo 2:** Published under **CC0 (Public Domain Dedication)**.

Which of the following actions is legally permitted for Photo 2 but strictly **prohibited** for Photo 1?

(A) Sending the photo link to a friend in an email.

(B) Cropping the photo, applying color filters, and incorporating it into a remixed digital collage without permission.

(C) Downloading the photo to a school laptop.

(D) Viewing the photo on a computer monitor.

Question 46

★★★★★ VERY IMPORTANT

OPEN SOURCE SOFTWARE VS. COMMERCIAL PROPRIETARY SOFTWARE

Software can be distributed under an **Open Source** model (e.g., Linux, Python, Blender) or a **Proprietary / Closed-Source** model (e.g., Microsoft Windows, Adobe Photoshop).

Which of the following describes a key societal and technical advantage of Open Source Software?

(A) Open source software can only be run on government-approved supercomputers.

(B) Open source software is physically guaranteed to never contain any programming errors or bugs.

(C) Open source software prevents users from connecting to the Internet.

(D) The underlying source code is publicly accessible, allowing programmers worldwide to inspect for security vulnerabilities, fix bugs collaboratively, and customize features freely.

Question 47

★★★★ HIGH PRIORITY

PLAGIARISM VS. ETHICAL ATTRIBUTION IN PROGRAMMING

A high school student enters a coding competition. While building their web app, the student imports an open-source JavaScript chart library licensed under the MIT license, includes the original copyright notice in their project files, and documents in their project report that the charting feature relies on this library.

Did the student commit **Plagiarism**?

(A) No; the student engaged in ethical code reuse because they adhered to the open-source license terms and provided explicit, honest **attribution** identifying the original author.

(B) Yes; programmers are legally required to invent every single line of code, operating system, and programming language from scratch.

(C) Yes; using open-source libraries is a federal felony punishable by imprisonment.

(D) No; but only because the competition judges did not notice the chart library.

Question 48

★★★★ HIGH PRIORITY

THE DIGITAL MILLENNIUM COPYRIGHT ACT (DMCA) AND SAFE HARBOR

A video-sharing website allows millions of users to upload home videos. A user uploads a full copyrighted Hollywood movie without the film studio's permission. Under the **Digital Millennium Copyright Act (DMCA) Safe Harbor** provisions, why is the website platform not immediately held financially liable for copyright infringement?

(A) Movies uploaded to the Internet automatically lose all copyright protection after three minutes.

(B) The platform operates as a passive online service provider that does not prescreen every upload, and maintains immunity as long as it promptly removes or disables access to the infringing material upon receiving a formal takedown notice from the copyright owner.

(C) The DMCA only applies to video games, not movies.

(D) Federal law grants video-sharing websites total immunity to sell pirated movies.

Question 49

★★★★ HIGH PRIORITY

THE FAIR USE DOCTRINE IN DIGITAL MEDIA

A student creates an educational YouTube video analyzing and critiquing cinematography techniques in modern cinema. The video includes three short, 10-second clips from major studio films with the student's voiceover analysis commentary overlaid on the clips.

Which legal principle protects the student's use of these copyrighted film clips against infringement claims?

(A) The International Patent Treaty.

(B) The **Fair Use Doctrine**, which permits limited use of copyrighted material without permission for purposes such as criticism, commentary, news reporting, teaching, and scholarship.

(C) The Right to Piracy statute.

(D) The Commercial Broadcast Mandate.

Question 50

★★★★★ VERY IMPORTANT

COMPREHENSIVE SOFTWARE AND MEDIA LICENSING SYNTHESIS TABLE

An AP Computer Science Principles class evaluates four digital assets and their licensing constraints:

Asset	License / Legal Status	Permitted Action	Prohibited Action
Asset 1: Linux Kernel	Open Source (GPL)	Modify and distribute source code	Distribute closed-source binary derivatives without source
Asset 2: Wikipedia Article	CC BY-SA	Reuse and translate the text	Sell copies under a proprietary, closed copyright license
Asset 3: NASA Space Photo	Public Domain (U.S. Govt)	Modify, print, sell, or remix with zero attribution required	None (Free for universal reuse)
Asset 4: Commercial OS	Proprietary Commercial	Run the software on licensed hardware	Disassemble, reverse-engineer, or distribute copies

Which of the following general conclusions is best supported by this table?

(A) All digital computer files on Earth are legally required to be licensed under Asset 4.

(B) Open source software licenses forbid anyone from learning how to program computers.

(C) Modern digital copyright spans a broad continuum from total closed restriction (Proprietary) to conditional open collaboration (Creative Commons / Open Source) and complete open freedom (Public Domain).

(D) Asset 3 is illegal to use in high school science projects.

Question 51

★★★★★ VERY IMPORTANT

DEFINITION AND SCOPE OF PERSONALLY IDENTIFIABLE INFORMATION (PII)

In cybersecurity and digital privacy, safeguarding **Personally Identifiable Information (PII)** is a paramount legal and technical priority.

Which of the following best defines Personally Identifiable Information?

(A) The color of the physical plastic casing surrounding a computer keyboard.

(B) The number of public parks located within a metropolitan area.

(C) Information about an individual that can be used directly or indirectly to distinguish, trace, or identify a specific person's identity, such as Social Security Numbers, biometric records, full legal names, and home addresses.

(D) The random temporary clock frequency of a computer processor, measured in gigahertz.

Question 52

★★★★★ VERY IMPORTANT

DATA AGGREGATION AND RE-IDENTIFICATION RISKS

A hospital releases an 'anonymized' health research dataset with patient names and social security numbers removed. However, the dataset still retains three fields: **Date of Birth**, **Gender**, and **5-digit Zip Code**.

Why is this dataset still extremely vulnerable to privacy violations according to data science research?

(A) Removing patient names causes the data to erase itself from memory.

(B) The hospital's computer hard drives will catch fire if zip codes are stored without names.

(C) Zip codes automatically convert all medical illnesses into computer viruses.

(D) Through **Data Aggregation**, combining just Date of Birth, Gender, and 5-digit Zip Code is mathematically sufficient to uniquely re-identify approximately 87% of the entire United States population when matched against public voter registration databases.

Question 53

★★★★ HIGH PRIORITY

ACTIVE VS. PASSIVE DIGITAL FOOTPRINTS

Every Internet user leaves a trail of digital data traces known as a **Digital Footprint**.

Which of the following correctly distinguishes an **Active Digital Footprint** from a **Passive Digital Footprint**?

(A) An active footprint is created while running on a treadmill; a passive footprint is created while sleeping.

(B) An active footprint is encrypted; a passive footprint can never be encrypted.

(C) An active footprint only exists on smartphones; a passive footprint only exists on desktop computers.

(D) An active footprint is created when a user deliberately shares data (such as posting on social media or filling out an online form); a passive footprint is collected automatically without the user's explicit action (such as server IP logging, browsing history, and device telemetry).

Question 54

★★★★★ VERY IMPORTANT

CONTINUOUS GEOLOCATION TRACKING AND INTIMATE PRIVACY EXPOSURE

A free flashlight utility mobile app requests permission to access the smartphone's **Precise GPS Location**. The app secretly records the user's location every three minutes and uploads the coordinates to commercial advertising servers.

Why does continuous geolocation tracking present an acute privacy threat beyond simple physical coordinates?

(A) GPS tracking turns the smartphone flashlight into a dangerous laser beam.

(B) Continuous location data reveals intimate personal details of a user's life—such as home address, workplace, places of religious worship, medical clinics visited, and political protests attended—allowing third parties to infer sensitive personal beliefs and health conditions.

(C) Continuous GPS tracking causes the smartphone to be magnetically attracted to police cars.

(D) Continuous location logging prevents the user from receiving phone calls.

Question 55

★★★★ HIGH PRIORITY

THIRD-PARTY TRACKING COOKIES AND CROSS-SITE WEB BEACONS

After searching for hiking boots on an outdoor gear website, a user notices that advertisements for those exact hiking boots follow them across completely unrelated news sites, cooking blogs, and social media feeds.

Which web technology enables advertising networks to track users across different independent websites?

(A) Computer memory chips are magnetically tuned to only think about hiking boots.

(B) The user's computer monitor stores photographs of boots in its glass screen.

(C) **Third-Party Tracking Cookies and Web Beacons (Pixel Tags)** embedded in third-party advertising scripts across multiple websites, allowing the ad network to build a persistent browsing profile tied to a unique browser identifier.

(D) The Internet Service Provider sends a physical postcard to every website the user visits.

Question 56

★★★★★ VERY IMPORTANT

DEVICE AND BROWSER FINGERPRINTING WITHOUT COOKIES

Even when a user clears all browser cookies, enables private browsing mode, and blocks third-party trackers, commercial tracking scripts can still reliably re-identify the user's browser across sessions.

Which advanced tracking methodology enables this identification without using cookies?

(A) Satellite photography zooming in through the user's window.

(B) Physical biometric fingerprinting scanned through the laptop keyboard keys.

(C) **Browser Fingerprinting**, which queries browser APIs for unique configurations (such as screen resolution, installed system fonts, audio hardware context, canvas graphics rendering quirks, and browser plugins) to generate a statistically unique device fingerprint.

(D) Submarines monitoring the transatlantic fiber-optic cables with periscopes.

Question 57

★★★★ HIGH PRIORITY

THE ROLE OF COMMERCIAL DATA BROKERS AND CONSUMER PROFILING

Companies known as **Data Brokers** operate multi-billion-dollar business models without having any direct consumer relationship with the individuals in their databases.

How do data brokers obtain consumer information, and what do they do with it?

(A) They harvest, purchase, and aggregate data from public records, smartphone app trackers, loyalty store cards, and credit reporting bureaus to construct detailed consumer dossiers that they sell to advertisers, insurers, and political campaigns.

(B) They provide free Wi-Fi routers to rural farms.

(C) They manually interview every global citizen in person every five years.

(D) They manufacture physical computers and donate them to public elementary schools.

Question 58

★★★★★ VERY IMPORTANT

CONSEQUENCES OF ENTERPRISE PII DATA BREACHES

A national credit reporting agency suffers a major cybersecurity breach in which hackers steal the names, Social Security Numbers, dates of birth, and home addresses of 147 million citizens.

What is the most severe, long-term consequence of this breach for the impacted individuals?

(A) Victims' smartphone batteries will permanently lose the ability to charge past 50%.

(B) All physical paper dollars owned by the victims will instantly turn into digital coins.

(C) Victims will be legally prohibited from ever moving to a new home address.

(D) Victims face long-term risks of **Identity Theft**, where criminals use the permanent stolen PII to fraudulently open bank accounts, obtain credit cards, take out loans, and file fraudulent tax returns in the victims' names.

Question 59

★★★★ HIGH PRIORITY

MODERN PRIVACY REGULATIONS: GDPR AND CCPA CONSUMER RIGHTS

In response to pervasive corporate data harvesting, governments enacted legal frameworks such as the European Union's **General Data Protection Regulation (GDPR)** and the **California Consumer Privacy Act (CCPA)**.

Which of the following fundamental rights do these regulations grant to digital citizens?

(A) The right to send computer viruses to commercial companies without legal penalty.

(B) The right to access the Internet without paying for electrical power.

(C) The right to obtain free brand-new smartphones every six months at government expense.

(D) The right to know what personal data companies collect about them, the right to demand that companies delete their personal data (the 'Right to be Forgotten'), and the right to opt out of having their data sold.

Question 60

★★★★★ VERY IMPORTANT

EVALUATING LEGITIMATE NEED VS. PRIVACY OVERREACH IN APP PERMISSIONS

A cybersecurity analyst reviews the system permission requests for four mobile applications:

Application	Advertised Purpose	Requested Permissions	Legitimate Functional Need?
App 1: Audio Recorder	Record voice memos	Microphone, Internal Storage	**Yes** (Essential for core functionality)
App 2: Turn-by-Turn GPS	Provide driving navigation	Precise Location, Speaker	**Yes** (Essential for real-time routing)
App 3: Calculator Utility	Perform basic math arithmetic	Precise Location, Contacts, Camera	**No** (Privacy Overreach & Data Harvesting)
App 4: Photo Editor	Apply visual filters to pictures	Camera, Photo Gallery	**Yes** (Essential for photo processing)

Why is **App 3** classified as an unacceptable privacy overreach?

(A) The calculator app is using too many mathematical numbers at the same time.

(B) Calculators are legally forbidden from performing division operations on smartphones.

(C) Basic arithmetic calculations require only local CPU mathematical operations; demanding access to the user's GPS location, private address book, and camera indicates unauthorized data harvesting unrelated to the app's functionality.

(D) Calculators can only run on solar-powered handheld devices, not smartphones.

Question 61

★★★★★ VERY IMPORTANT

THE FUNDAMENTAL TENSION: DIGITAL PRIVACY VS. PUBLIC SAFETY

One of the most consequential societal debates in modern computing is the tension between **Individual Digital Privacy** and **National Security / Law Enforcement**.

Which of the following best articulates the core conflict in this debate?

(A) National security agencies are legally prohibited from using computers or electronic databases.

(B) Digital privacy requires all computers to be unplugged, while national security requires all computers to be painted yellow.

(C) Robust end-to-end encryption shields innocent citizens and dissidents from authoritarian surveillance and identity theft, but simultaneously prevents law enforcement from intercepting communications between terrorists and violent criminals.

(D) Digital privacy is only needed by criminals, while law-abiding citizens have no legitimate interest in data privacy.

Question 62

★★★★★ VERY IMPORTANT

THE CRYPTOGRAPHIC IMPOSSIBILITY OF 'SAFE' ENCRYPTION BACKDOORS

Government officials propose requiring all smartphone manufacturers to build an 'exceptional access' **backdoor** into their encryption software so police can decrypt phones with a judge's warrant. Cryptographers and cybersecurity experts overwhelmingly oppose this proposal.

What is the primary technical reason cybersecurity experts argue backdoors are dangerous?

(A) Encryption backdoors cause smartphone screens to shatter spontaneously.

(B) Mathematically, there is no way to create an encryption backdoor that only 'good people' can use; any intentional architectural vulnerability or master key will inevitably be discovered, leaked, or exploited by hostile cybercriminals and foreign adversaries.

(C) Smartphone companies would be legally forced to stop manufacturing all electronic devices.

(D) Backdoors require computers to consume ten times more electrical voltage.

Question 63

★★★★★ VERY IMPORTANT

MASS BULK DATA COLLECTION AND TELEPHONY METADATA

Intelligence disclosures revealed that security agencies engaged in the bulk collection of millions of domestic telephone records daily. Officials defended the program by stating: '*We only collect the **metadata** (who called whom, when, and for how long), not the actual audio content of the conversations.*'

Why do civil liberties experts argue that bulk **metadata** collection is just as invasive as listening to content?

(A) Metadata converts phone calls into physical printed books.

(B) Metadata patterns over time reveal deeply intimate personal facts—such as calls to an addiction helpline at 2:00 AM, followed by a divorce attorney at 9:00 AM, and an oncologist at 11:00 AM—painting an unmistakable picture of an individual's private life without hearing a single word spoken.

(C) Metadata can only be analyzed by human beings writing with ink pens.

(D) Metadata physically forces the phone speaker to play static noise.

Question 64

★★★★ HIGH PRIORITY

CONSTITUTIONAL LAW AND DIGITAL SEARCH WARRANTS: RILEY V. CALIFORNIA

In the landmark Supreme Court case *Riley v. California* (2014), the Court ruled unanimously that police officers may **not** search the digital contents of an arrested person's smartphone without obtaining a search warrant from a judge.

What computational rationale did the Court provide for treating a smartphone differently from physical items in a suspect's pockets (such as a wallet or paper notepad)?

(A) Searching a phone requires police officers to learn assembly language.

(B) Smartphones automatically explode if opened by a police officer.

(C) Smartphones are made of glass, which is legally considered sacred under maritime law.

(D) A modern smartphone is not a mere physical container, but a digital repository holding millions of pages of personal text messages, financial records, medical histories, photos, and location tracking spanning years, holding vastly more intimate information than an entire physical house.

Question 65

★★★★★ VERY IMPORTANT

CHILLING EFFECT OF PUBLIC FACIAL RECOGNITION SURVEILLANCE

A city installs high-resolution automated facial recognition cameras on every street corner that continuously scan and log the identities of pedestrians entering public squares and parks.

What civil liberty risk, known as a **Chilling Effect**, is created by this pervasive public surveillance?

(A) Citizens will be prohibited from wearing shoes in public parks.

(B) Pedestrians' faces will physically turn blurry due to optical camera flash reflection.

(C) Camera lenses will freeze during winter months and shatter.

(D) Citizens may become fearful of participating in peaceful political protests, attending controversial religious services, or visiting legal medical clinics because their presence is permanently logged in a government tracking database.

Question 66

★★★★ HIGH PRIORITY

AUTOMATED LICENSE PLATE READERS (ALPR) AND PERVASIVE VEHICULAR TRACKING

Police departments and private neighborhood networks deploy **Automated License Plate Readers (ALPR)** that photograph passing cars, capture license plate numbers via optical character recognition, and record the exact GPS coordinates and timestamps of every vehicle.

What is the primary civil liberties concern when millions of these non-suspect vehicle logs are pooled into searchable national databases?

(A) It makes license plates unreadable to human eyes.

(B) It causes car engines to run out of motor oil faster.

(C) It allows government agencies and private investigators to reconstruct the complete historical travel routes, habits, and associations of innocent citizens who are suspected of no crime.

(D) It automatically charges drivers a $10 fee every time they turn left.

Question 67

★★★★ HIGH PRIORITY

WORKPLACE DIGITAL SURVEILLANCE AND 'BOSSWARE'

During the shift to remote work, a corporate employer installs monitoring software on employee laptops that logs every keystroke, records periodic webcam screenshots, tracks mouse movements, and flags employees as 'idle' if they stop typing for more than two minutes.

Which psychological and workplace impact is most directly caused by this **workplace surveillance**?

(A) A complete elimination of the need for employees to sleep at night.

(B) Increased employee stress, burnout, and anxiety, as workers feel constantly distrusted and perform performative busywork (like jiggling the mouse) rather than focusing on creative, high-value problem solving.

(C) Employees' fingers become permanently bonded to the computer keys.

(D) An automatic 500% increase in the speed of the company's network connection.

Question 68

★★★★ HIGH PRIORITY

PREDICTIVE POLICING SOFTWARE AND SELF-FULFILLING FEEDBACK LOOPS

A city utilizes predictive policing software that sends police patrols to geographic blocks forecasted to have the highest future crime rates. The model is trained on historic drug arrest records.

How does this automated deployment create a **self-fulfilling feedback loop**?

(A) The computer deletes all police radios from existence.

(B) Because police are dispatched more heavily to those specific blocks, they naturally observe and make more arrests there, which feeds new arrest records back into the model, causing the algorithm to predict even more crime in those same neighborhoods.

(C) The algorithm physically relocates houses and buildings to different parts of the city.

(D) The algorithm automatically teaches criminals how to become police officers.

Question 69

★★★★★ VERY IMPORTANT

PSYCHOGRAPHIC PROFILING, MICRO-TARGETING, AND DEMOCRATIC INTEGRITY

In political campaigns (such as the Cambridge Analytica scandal), political consulting firms harvested private data from tens of millions of social media profiles to construct **psychographic profiles** (assessing personality traits like neuroticism, openness, and fear).

How was this data exploited to influence democratic elections?

(A) Voters were segmented and served hyper-targeted, fear-based political advertisements and disinformation tailored precisely to their specific psychological vulnerabilities and anxieties to depress voter turnout or manipulate vote choices.

(B) The firm automatically elected a computer server to become the President of the United States.

(C) The firm erased the concept of voting from human memory.

(D) The consulting firm used radio waves to rewrite the paper ballots inside physical ballot boxes.

Question 70

★★★★★ VERY IMPORTANT

COMPREHENSIVE SYNTHESIS: SURVEILLANCE TECHNOLOGIES AND CIVIL LIBERTIES

An AP Computer Science Principles seminar reviews four modern surveillance technologies:

Technology	Operational Justification	Primary Civil Liberties / Ethical Hazard
Tech 1: End-to-End Encryption	Shields personal communication from cyber criminals and spies	Limits law enforcement intelligence gathering during active investigations
Tech 2: Facial Recognition Cams	Rapidly identifies wanted fugitive suspects in airports	Chilling effect on public assembly and high error rates for minority groups
Tech 3: Bulk Metadata Logging	Detects covert communication webs among foreign terror cells	Enables warrantless, pervasive profiling of innocent domestic citizens
Tech 4: Predictive Policing	Optimizes scarce municipal police patrol resources	Creates self-reinforcing racial feedback loops based on biased historic arrest data

Which overarching engineering conclusion is most accurate regarding these technologies?

(A) The Bill of Rights was repealed by the Internet Engineering Task Force in 1995.

(B) None of these technologies use computer software or digital algorithms.

(C) Computer scientists are legally required to ignore all societal and ethical consequences of their work.

(D) Every surveillance technology entails profound trade-offs between public security and fundamental civil liberties; responsible computer scientists must analyze both technical capabilities and societal consequences when designing and deploying systems.

Question 71

★★★★★ VERY IMPORTANT

SOCIAL ENGINEERING AND THE HUMAN FACTOR IN CYBERSECURITY

Cybersecurity professionals frequently note that the easiest way to breach a secure corporate network is not by cracking complex mathematical encryption algorithms, but by exploiting **Social Engineering**.

Which of the following best defines social engineering in computer security?

(A) The process of programming robots to assemble social media servers in a factory.

(B) The construction of physical bridges and highways to connect computer science departments.

(C) A software protocol that automatically translates binary code into spoken human language.

(D) The psychological manipulation of human individuals into divulging confidential information, clicking malicious links, or performing actions that compromise security systems.

Question 72

★★★★★ VERY IMPORTANT

MECHANICS AND INDICATORS OF PHISHING ATTACKS

A student receives an urgent email appearing to come from their bank, stating: '*Security Alert: Your account will be permanently closed in 24 hours unless you verify your password immediately by clicking here.*' The link leads to a login webpage that looks identical to the real bank, but has the URL `http://www.secure-bank-login-update.biz`.

Which cyberattack is described, and what is its objective?

(A) A distributed volunteer citizen science protein simulation.

(B) A **Phishing** attack, designed to trick the student into entering their private banking username and password on an attacker-controlled clone website.

(C) An automated operating system security patch installation.

(D) A legitimate routine customer survey administered by the bank.

Question 73

★★★★★ VERY IMPORTANT

SPEAR-PHISHING: TARGETED SOCIAL ENGINEERING

Unlike generic mass phishing emails sent to millions of random addresses, a cybercriminal conducts research on the chief financial officer (CFO) of a solar power company. The attacker crafts a single, highly customized email referencing the CFO's recent attendance at an industry conference in Denver and attaches a malicious PDF disguised as an invoice from the conference organizer.

What specialized form of cyberattack is demonstrated here?

(A) An open-source software license audit.

(B) **Spear-Phishing**, which is a highly targeted, personalized phishing attack directed at a specific individual or organization using researched personal information.

(C) A volumetric DDoS bandwidth flood.

(D) A physical cable severance.

Question 74

★★★★★ VERY IMPORTANT

ROGUE WI-FI ACCESS POINTS AND EVIL TWIN EXPLOITS

While waiting at a busy metropolitan train station, a passenger opens their laptop's Wi-Fi menu and connects to an open, unencrypted network named `TrainStation_Complimentary_WiFi`.

Unknown to the passenger, this wireless network is an **Evil Twin (Rogue Access Point)** deployed by an attacker sitting nearby with a battery-powered Wi-Fi router. What immediate danger does the passenger face?

(A) The attacker can eavesdrop on and capture all unencrypted data packets transmitted by the laptop (packet sniffing), intercept cleartext passwords, and redirect web queries to malicious phishing sites.

(B) The laptop screen will permanently turn purple.

(C) The laptop's hard drive will instantly vaporize into thin air.

(D) The attacker physically controls the speed of the passenger's train.

Question 75

★★★★★ VERY IMPORTANT

KEYLOGGERS: KEYSTROKE LOGGING MALWARE AND HARDWARE DEVICES

A student secretly inserts a tiny physical USB dongle between the computer tower and the keyboard cable in a public school computer lab. The dongle quietly records every key pressed by every user throughout the day.

What type of surveillance tool is this, and why is it dangerous to cybersecurity?

(A) An anti-virus cleaner that disinfects physical keyboard plastic.

(B) A network router that connects the school to satellite radio.

(C) A **Hardware Keylogger**; it captures passwords, usernames, and private messages at the physical input level before the operating system or software encryption can protect the data.

(D) A wireless mouse booster that speeds up cursor movement.

Question 76

★★★★★ VERY IMPORTANT

RANSOMWARE ATTACKS AND CRITICAL INFRASTRUCTURE EXTORTION

An employee at a regional hospital opens an email attachment disguised as a medical supply invoice. Within minutes, malicious software silently traverses the hospital's internal network, using military-grade cryptography to encrypt all patient medical records, surgical schedules, and database files. A text file appears on all screens demanding $2 million in Bitcoin for the decryption key.

Which type of malware has infected the hospital?

(A) An open-source spreadsheet editor.

(B) A public utility broadband subsidy.

(C) A volunteer galaxy classification algorithm.

(D) **Ransomware**, which locks and encrypts a victim's critical data, holding it hostage until an extortion ransom is paid.

Question 77

★★★★ HIGH PRIORITY

DISTINGUISHING MALWARE TYPOLOGIES: TROJAN, VIRUS, AND WORM

A student downloads a file claiming to be a free full-version download of a popular commercial video game from an untrusted file-sharing forum. When the student runs the installer, the computer displays a game title screen, but secretly installs a background remote-access backdoor giving hackers control over the webcam.

Which category of malware best describes this deceptive program?

(A) A harmless digital certificate issued by an accredited authority.

(B) A physical electrical fuse inside the power supply.

(C) An open Creative Commons public attribution stamp.

(D) A **Trojan Horse**, which disguises itself as desirable, legitimate software while concealing malicious hidden functionality.

Question 78

★★★★ HIGH PRIORITY

CRYPTOJACKING: UNAUTHORIZED COMPUTATIONAL RESOURCE HIJACKING

A student notices that their laptop fan is constantly spinning at maximum speed, the battery drains in 30 minutes, and web pages take forever to load. An IT technician discovers a hidden script running in a background browser tab that solves cryptographic proof-of-work puzzles to generate digital coins for an anonymous cybercriminal.

What form of cyber exploitation has occurred?

(A) **Cryptojacking**, the unauthorized hijacking of a victim's computing hardware (CPU/GPU) and electricity to mine cryptocurrency for an attacker.

(B) A hardware upgrade making the processor run ten times faster.

(C) An automated file compression algorithm reducing hard drive space.

(D) A software feature designed to clean dust from the internal cooling fans.

Question 79

★★★★ HIGH PRIORITY

WATERING HOLE CYBERATTACKS

Instead of directly attacking an aerospace defense contractor whose internal firewalls are nearly impenetrable, a sophisticated hacking group identifies a small, niche lunch catering website frequently visited by the aerospace engineers. The hackers compromise the catering website and inject zero-day malware that infects the engineers' computers when they view the daily lunch menu.

What cyberattack strategy is demonstrated here?

(A) A **Watering Hole Attack**, which targets a specific group of users by compromising a trusted third-party website that the targets are known to visit regularly.

(B) An open-access scientific journal publication.

(C) A physical bank burglary.

(D) An automated spelling proofreader.

Question 80

★★★★★ VERY IMPORTANT

COMPREHENSIVE SYNTHESIS OF CYBERSECURITY THREAT VECTORS

A high school cybersecurity club summarizes four common threat vectors and their primary mechanisms:

Threat Vector	Attack Mechanism	Primary Attack Target	Most Effective Defense
Threat 1: Phishing	Deceptive emails mimicking trusted brands	User login credentials	User skepticism, email filters, and Multi-Factor Authentication
Threat 2: Keylogger	Secretly records physical keyboard keystrokes	Cleartext passwords as typed	Physical device inspection and virtual scrambled keyboards
Threat 3: Ransomware	Cryptographically encrypts local and network files	Institutional data availability	Immutable offline air-gapped backups and patch management
Threat 4: Evil Twin Wi-Fi	Rogue wireless access point spoofing real SSID	In-transit unencrypted packets	Using encrypted Virtual Private Networks (VPNs)

Which fundamental security principle is best reinforced by this comprehensive synthesis?

(A) Users should write their passwords on sticky notes attached to their monitors to prevent digital theft.

(B) Only Threat 1 is real; the other three threats were invented for science fiction movies.

(C) Cybersecurity requires **Defense-in-Depth**—a multi-layered defense combining technical cryptography, network security architecture, automated monitoring, and continuous human user education.

(D) Installing an antivirus program makes a computer completely immune to all future cyber threats forever.

Question 81

★★★★★ VERY IMPORTANT

MULTI-FACTOR AUTHENTICATION (MFA) AND AUTHENTICATION FACTORS

A secure corporate web portal requires employees to log in using both an alphanumeric password and a time-based 6-digit verification code sent to an authenticator app installed on their registered mobile smartphone.

Why does this system qualify as **Multi-Factor Authentication (MFA)** according to cybersecurity standards?

(A) It completely eliminates the need for mathematical encryption during data transmission across fiber-optic cables.

(B) It combines two fundamentally different categories of authentication factors: 'something you know' (the password) and 'something you have' (the physical smartphone running the authenticator application).

(C) It converts the user's password into an unencrypted plain text string accessible to any public internet service.

(D) It requires the user to type two separate passwords that both belong strictly to the 'something you know' category.

Question 82

★★★★★ VERY IMPORTANT

PASSWORD SECURITY: COMBINATORIAL ENTROPY AND BRUTE-FORCE RESISTANCE

A cybersecurity analyst advises a school district to revise its password policy. The analyst recommends using a 16-character passphrase composed of four randomly selected common dictionary words (e.g., `banana-rocket-purple-pencil`) rather than an 8-character password with mixed cases and symbols (e.g., `Tr@7#k9!`).

What mathematical and security principle justifies this recommendation?

(A) Automated cracking algorithms are programmed by law to skip any password containing common English words.

(B) Password search space (combinatorial entropy) increases exponentially with string length, making longer passphrases exponentially more resistant to automated brute-force cracking tools while remaining easier for humans to remember.

(C) Four-word passphrases disable all network communication between the user's browser and web servers.

(D) Shorter passwords with special characters are mathematically impossible for computers to test using automated dictionary programs.

Question 83

★★★★★ VERY IMPORTANT

CREDENTIAL STUFFING AND THE DANGER OF PASSWORD REUSE ACROSS SERVICES

An online retailer experiences a data breach where user account credentials (email addresses and passwords) are leaked onto the dark web. Within 48 hours, several victims find that their unrelated personal email and cloud storage accounts on completely different platforms have also been accessed by unauthorized parties.

Which cyberattack occurred, and what user practice enabled it?

(A) A computer virus that physically altered the fiber-optic cables connecting the user's home router.

(B) A distributed citizen science project where volunteers voluntarily donated their cloud passwords.

(C) A hardware electromagnetic pulse attack that wiped all remote internet databases simultaneously.

(D) A **Credential Stuffing** attack, enabled by users reusing the exact same username and password combination across multiple unrelated websites and services.

Question 84

★★★★ HIGH PRIORITY

BIOMETRIC AUTHENTICATION: BENEFITS, LIMITATIONS, AND IRREVOCABILITY

A smartphone manufacturer replaces PIN-based screen unlock with 3D facial recognition biometrics. While users appreciate unlocking their devices instantly without typing, privacy and security experts highlight a fundamental limitation of biometric credentials compared to alphanumeric passwords.

Which of the following describes a major security limitation inherent to biometric authentication?

(A) Unlike passwords, biometric identifiers (such as fingerprints or facial geometry) cannot be reset or revoked once the underlying digital template data is compromised or breached.

(B) Biometric scans can only be processed when the smartphone is completely powered off.

(C) Biometric sensors require continuous wired fiber-optic connections to national government headquarters to function.

(D) Biometric data automatically transfers all legal copyrights of user photos to the smartphone vendor.

Question 85

★★★★★ VERY IMPORTANT

CRYPTOGRAPHIC PASSWORD STORAGE: ONE-WAY HASHING AND SALTING

A web application stores user passwords in its central database. To ensure security in case the database is ever stolen, the developers run each password through a cryptographic one-way hash function combined with a unique, randomly generated cryptographic **salt** before saving the record.

Why is adding a unique salt to each password critical for defending against database breaches?

(A) It allows the server administrators to reverse the mathematical hash function and read the plain text password whenever needed.

(B) It automatically transmits the user's password directly to public internet search engines for public archiving.

(C) It ensures that two users with identical passwords will produce completely different stored hash values, preventing attackers from using precomputed Rainbow Tables to instantly crack large batches of passwords.

(D) It speeds up user login times by disabling all mathematical verification checks on the server.

Question 86

★★★★★ VERY IMPORTANT

PASSWORD MANAGERS AND MODERN CREDENTIAL HYGIENE

Cybersecurity agencies strongly encourage internet users to adopt a reputable **Password Manager** application.

Which of the following is a primary security benefit of using a password manager?

(A) It sends all stored user passwords in unencrypted plain text via public email every week for backup.

(B) It generates, encrypts, and stores unique, high-entropy passwords for every distinct website, requiring the user to memorize only a single strong master passphrase.

(C) It guarantees that websites visited by the user will never experience server-side hardware failures.

(D) It replaces all websites' security firewalls with a single local text file on the user's desktop.

Question 87

★★★★★ VERY IMPORTANT

NETWORK SECURITY ON UNTRUSTED NETWORKS: HTTPS AND VIRTUAL PRIVATE NETWORKS (VPNS)

A remote employee works from an open public Wi-Fi network at a coffee shop. To protect their network traffic from eavesdropping (packet sniffing) by other users connected to the same unencrypted local wireless access point, the employee activates a **Virtual Private Network (VPN)**.

How does a VPN protect the employee's data on this untrusted local network?

(A) It converts all web pages into audio broadcasts transmitted over public FM radio frequencies.

(B) It creates an encrypted tunnel that encapsulates all data packets transmitted between the employee's computer and the VPN server, preventing local eavesdroppers from viewing packet payloads and browsing activity.

(C) It physically disconnects the coffee shop's wireless router from the public power grid.

(D) It permanently blocks all outgoing electrical signals from the computer's battery.

Question 88

★★★★ HIGH PRIORITY

SOFTWARE UPDATES, PATCH MANAGEMENT, AND VULNERABILITY LIFECYCLE

Operating system developers regularly issue automated software updates labeled as 'Security Patches'. Many users postpone installing these updates because they require restarting the computer.

Why is delaying the installation of software security patches dangerous from a cybersecurity perspective?

(A) When a software vendor releases a security patch, malicious actors analyze the patch code to discover the underlying vulnerability and immediately launch automated attacks against unpatched systems.

(B) Unpatched systems are legally confiscated by international telecommunications regulatory agencies.

(C) Security patches delete all user files to make room for newer versions of the operating system.

(D) Delaying updates causes computer monitors to physically lose their liquid crystal display properties.

Question 89

★★★★ HIGH PRIORITY

PRINCIPLE OF LEAST PRIVILEGE IN SYSTEM ADMINISTRATION

A network administrator configures workstations for office staff. Instead of granting all employees full 'Administrator' (root) access privileges, the administrator gives them standard 'User' accounts that allow running approved productivity applications but restrict installing software or modifying system kernel configuration files.

Which cybersecurity principle is being implemented, and what is its protective benefit?

(A) The Principle of Hardware Redundancy, which doubles the number of hard drives inside each monitor.

(B) The Principle of Total Surveillance, which livestreams all employee keystrokes to social media platforms.

(C) The Principle of Maximum Bandwidth, which increases the physical clock speed of employee microprocessors.

(D) The **Principle of Least Privilege**, which minimizes the potential damage ('blast radius') if an employee accidentally executes malware or falls victim to phishing.

Question 90

★★★★★ VERY IMPORTANT

COMPREHENSIVE SYNTHESIS: MULTI-LAYER CYBERSECURITY DEFENSES

A school district's IT department designs a comprehensive cybersecurity defense policy to mitigate multiple attack vectors. The table below outlines proposed defense layers:

Defense Layer	Implemented Control	Target Threat Mitigated
Layer 1: Identity & Access	Multi-Factor Authentication (MFA)	Stolen / Phished Passwords
Layer 2: Network Transmission	Encrypted VPN & TLS 1.3	Unencrypted Wi-Fi Eavesdropping
Layer 3: Endpoint Software	Automated Vulnerability Patching	Exploits of Known Software Flaws
Layer 4: Human Factor	Mandatory Phishing Awareness Training	Social Engineering Deception

Based on defense-in-depth principles in AP Computer Science Principles, why is implementing all four layers together substantially more effective than relying solely on Layer 1 (MFA)?

(A) Implementing multiple layers allows the school to turn off all firewalls and operate without passwords.

(B) Layer 1 (MFA) is strictly illegal under international computing standards unless supported by Layer 4.

(C) No single defensive control can mitigate all threat categories; an attacker can bypass MFA by exploiting an unpatched software vulnerability (Layer 3) or eavesdropping on unencrypted local network traffic (Layer 2) without needing user credentials.

(D) Automated patching (Layer 3) renders all mathematical encryption in Layer 2 completely unnecessary.

Question 91 · Written Response Scenario

★★★★★ 9 POINTS

ALGORITHMIC BIAS AUDIT IN AI HIRING SOFTWARE

An enterprise software firm builds an automated resume screening algorithm trained on 10 years of historical company hiring decisions. An independent audit reveals that female and minority engineering applicants are systematically ranked lower despite having equivalent credentials.

Respond to the following four parts based on computing bias, data ethics, and algorithmic design:

- Part A: Identify the root cause of the algorithmic bias and explain how historical training data embeds systemic discrimination.
- Part B: Propose a specific technical procedure or data preprocessing step to mitigate the bias before training.
- Part C: Describe a program algorithm and data abstraction to monitor candidate scoring distributions across demographic groups.
- Part D: Discuss the ethical responsibilities of software engineers and corporate accountability mechanisms for automated hiring.

Question 92 · Written Response Scenario

★★★★★ 9 POINTS

PRIVACY-PRESERVING CONTACT TRACING AND GEOLOCATION TRACKING

During a public health crisis, a municipal health department commissions a smartphone app to notify citizens if they have been exposed to an infectious pathogen. Developers must choose between a centralized GPS tracking system and a decentralized Bluetooth proximity beacon protocol.

Analyze the security, privacy, and architectural trade-offs of this mobile application:

- Part A: Contrast the privacy risks of centralized GPS tracking with decentralized Bluetooth proximity beacons.
- Part B: Specify an algorithmic mechanism for generating and exchanging ephemeral cryptographic rolling proximity identifiers.
- Part C: Describe data storage, transmission encryption, and data expiration procedures that protect user anonymity.
- Part D: Evaluate the digital divide and voluntary adoption barriers that determine the system's public health efficacy.

Question 93 · Written Response Scenario

★★★★★ 9 POINTS

CROWDSOURCED GLOBAL BIODIVERSITY AND WILDLIFE CENSUS PLATFORM

A marine conservation consortium develops an open citizen science mobile app where divers, fishermen, and coastal residents submit geotagged photos of marine species. Computer vision models combined with peer community consensus identify species and monitor coral reef ecosystem health.

Design the computational architecture and safety protocols for this crowdsourcing platform:
- Part A: Explain why crowdsourcing is computationally and logistically superior to traditional scientific survey methods for this project.
- Part B: Construct an algorithmic consensus mechanism to validate species identification submitted by amateur contributors.
- Part C: Design a privacy and anti-poaching algorithm that protects endangered wildlife from illicit exploitation.
- Part D: Justify releasing the consolidated biodiversity dataset under an open Creative Commons CC BY license.

Question 94 · Written Response Scenario

★★★★★ 9 POINTS

CREATIVE COMMONS MULTI-MEDIA EDUCATIONAL REPOSITORY

A non-profit educational network creates an open platform where teachers upload, remix, and publish digital textbooks, interactive coding labs, and instructional media under various Creative Commons licenses (CC BY, CC BY-NC-SA, CC0).

Evaluate the legal, architectural, and educational dimensions of this digital repository:
- Part A: Contrast traditional automatic copyright with Creative Commons licensing and the Public Domain (CC0).
- Part B: Write an algorithmic licensing compatibility validation procedure for derivative remix works.
- Part C: Describe an automated metadata injection mechanism that formats legal attribution into exported materials.
- Part D: Analyze how open educational resources (OER) democratize access and bridge global learning divides.

Question 95 · Written Response Scenario

★★★★★ 9 POINTS

DIGITAL DIVIDE RURAL BROADBAND INFRASTRUCTURE ALLOCATOR

A state government allocates $500 million in public broadband infrastructure grants. Data scientists build an algorithmic decision-support tool to score and prioritize census tracts for fiber-optic cable deployment and municipal wireless installations.

Evaluate the socio-technical factors and computational logic of this infrastructure allocation tool:
- Part A: Analyze the geographic, economic, and educational dimensions of the digital divide in rural communities.
- Part B: Write an algorithmic multi-criteria scoring function to rank underserved communities.
- Part C: Describe the data abstraction (lists/records) representing regional demographic and infrastructure data.
- Part D: Critique potential algorithmic blindspots (e.g., flawed ISP broadband maps, neglect of tribal lands).

Question 96 · Written Response Scenario

★★★★★ 9 POINTS

HEURISTIC PHISHING DETECTION AND DOMAIN SPOOFING DEFENSE

A hospital network's cybersecurity team develops an automated email gateway inspection engine to detect spear-phishing attacks targeting healthcare workers with malicious credential-harvesting links.

Detail the cybersecurity engineering and algorithmic inspection heuristics of this defensive system:
- Part A: Analyze the social engineering psychological triggers and technical domain spoofing mechanisms (e.g., lookalike characters, typo-squatting).
- Part B: Write an algorithmic procedure that parses email headers, hyperlinks, and body text to generate a threat risk score.
- Part C: Evaluate the trade-offs between False Positives (blocking legitimate clinical emails) and False Negatives (allowing malware through).
- Part D: Propose employee training policies and technical multi-factor authentication controls to stop successful credential breaches.

Question 97 · Written Response Scenario

★★★★★ 9 POINTS

PII ANONYMIZATION AND DE-IDENTIFICATION PIPELINE FOR OPEN DATA

A metropolitan transit authority plans to release public transit smart-card transaction data to university researchers. To prevent passenger re-identification through spatial linkage attacks, engineers build an automated de-identification and k-anonymity pipeline.

Architect the data governance and computational anonymization pipeline for this public release:
- Part A: Define Personally Identifiable Information (PII) and explain how combining non-sensitive data points creates re-identification risks.
- Part B: Formulate an algorithmic data transformation process using generalization, bucketing, and perturbation.
- Part C: Write a verification procedure that calculates whether a transformed dataset satisfies k-anonymity.
- Part D: Formulate data governance and access control policies for ethical research partnerships.

Question 98 · Written Response Scenario

★★★★★ 9 POINTS

ALGORITHMIC FAIR LENDING AND CREDIT SCORING EVALUATION

A financial technology startup deploys an AI credit-scoring model that replaces traditional credit scores with alternative consumer data: smartphone operating system, frequency of battery recharging, social media graph connections, and shopping timestamp patterns.

Examine the equity, regulatory compliance, and computational mechanisms of this lending algorithm:
- Part A: Analyze how alternative non-financial variables function as discriminatory proxies for protected demographic classes.
- Part B: Formulate an algorithmic procedure to calculate disparate impact and statistical parity across demographic cohorts.
- Part C: Design an explainability procedure that generates legally mandated Adverse Action notices for rejected applicants.
- Part D: Evaluate the systemic macroeconomic consequences of automated algorithmic credit barriers.

Question 99 · Written Response Scenario

★★★★★ 9 POINTS

MULTI-FACTOR AUTHENTICATION (MFA) TOKEN VERIFICATION SYSTEM

A hospital develops a Time-based One-Time Password (TOTP) multi-factor authentication engine for doctors accessing electronic health records (EHR) containing sensitive patient medical histories, complying with HIPAA cybersecurity requirements.

Design the technical authentication protocols and accessibility safeguards for this clinical system:
- Part A: Analyze the three fundamental authentication factor categories in a clinical hospital environment.
- Part B: Write the algorithmic verification procedure for validating Time-based One-Time Passwords (TOTP) with clock drift tolerance.
- Part C: Detail brute-force rate-limiting and account lockout safeguards against credential stuffing attacks.
- Part D: Formulate accessible fallback mechanisms for staff with disabilities or lost hardware tokens without weakening security.

Question 100 · Written Response Scenario

★★★★★ 9 POINTS

ASSISTIVE SCREEN-READER AND ACCESSIBLE UI AUDIT ENGINE

A state university develops an automated web accessibility compliance checker based on WCAG 2.1 AA standards to audit all campus public portals, course management websites, and student registration interfaces.

Design the technical evaluation algorithms and evaluate the social impact of web accessibility:
• Part A: Analyze the ethical, legal, and educational imperative of web accessibility for individuals with sensory and motor disabilities.
• Part B: Write an algorithmic DOM analysis procedure that verifies non-text element alternative text (`alt`) and calculates relative color contrast luminance.
• Part C: Describe a data abstraction managing accessibility violation reports and automated severity triage.
• Part D: Evaluate the universal usability benefits (curb-cut effect) of accessibility standards for the entire student body.

SECTION II ◆ EXAMINATION KEY & SOLUTIONS

Answers, Worked Solutions & Scoring Rubrics

The following pages contain the complete 90-question Multiple Choice Answer Key Matrix, comprehensive step-by-step worked solutions with distractor failure analysis, and official 9-point College Board scoring rubrics for all 10 Create Performance Task Written Response questions.

Section I Answer Key Matrix

Quick-reference answer matrix for Multiple Choice Questions 1 through 90. All questions are weighted equally.

Q1 – Q23		Q24 – Q46		Q47 – Q68		Q69 – Q90	
Q01	A	Q24	A	Q47	A	Q69	A
Q02	C	Q25	B	Q48	B	Q70	D
Q03	D	Q26	C	Q49	B	Q71	D
Q04	D	Q27	C	Q50	C	Q72	B
Q05	A	Q28	D	Q51	C	Q73	B
Q06	C	Q29	A	Q52	D	Q74	A
Q07	C	Q30	B	Q53	D	Q75	C
Q08	B	Q31	C	Q54	B	Q76	D
Q09	D	Q32	C	Q55	C	Q77	D
Q10	D	Q33	A	Q56	C	Q78	A
Q11	C	Q34	D	Q57	A	Q79	A
Q12	C	Q35	B	Q58	D	Q80	C
Q13	A	Q36	C	Q59	D	Q81	B
Q14	A	Q37	A	Q60	C	Q82	B
Q15	D	Q38	A	Q61	C	Q83	D
Q16	D	Q39	B	Q62	B	Q84	A
Q17	C	Q40	D	Q63	B	Q85	C
Q18	A	Q41	C	Q64	D	Q86	B
Q19	A	Q42	A	Q65	D	Q87	B
Q20	B	Q43	A	Q66	C	Q88	A
Q21	C	Q44	B	Q67	B	Q89	D
Q22	D	Q45	B	Q68	B	Q90	C
Q23	A	Q46	D				

Question 01 · Beneficial vs. Harmful Effects of Computing Innovations

Priority ★★★★★

CORRECT ANSWER: (A) Computing innovations almost always produce both beneficial and harmful effects, some of which are intentional while others are completely unintended by the original developers.

- Analyze the fundamental concept of computing impacts in Big Idea 5 (IOC-1):
- • Every computing innovation changes how people live, work, and interact.
- • **Beneficial Effects:** Advances in communication, medical diagnostics, education, and economic productivity.
- • **Harmful Effects:** Privacy erosion, job displacement, automated bias, and cybersecurity vulnerabilities.
- • Crucially, many negative impacts are **unintended consequences** that arise only after an innovation is adopted at massive scale by diverse populations.

WHY OTHER CHOICES ARE INCORRECT:

✕ **(B):** Computing innovations profoundly affect entire societies, including non-users (e.g., automated traffic lights, municipal cameras, banking systems).
✕ **(C):** Harmful effects frequently arise from well-intentioned software through unforeseen systemic interactions, not just malicious intent.
✕ **(D):** Technology is created by humans and reflects human limitations; it is not inherently flawless or exclusively beneficial.

Question 02 · Unintended Consequences of Social Media Algorithms

Priority ★★★★★

CORRECT ANSWER: (C) The algorithm amplifies sensationalized, controversial, and false news stories because outrage and conflict generate higher user engagement and longer viewing times.

- Examine algorithmic optimization and unintended systemic consequences:
- • The developers' primary objective was neutral: maximize user retention and engagement.
- • However, behavioral psychology demonstrates that provocative, sensational, or divisive content elicits stronger emotional reactions than balanced facts.
- • As a result, the engagement-driven algorithm unintentionally promotes clickbait, conspiracy theories, and echo chambers, polarizing public discourse without human developers explicitly programming it to do so.

WHY OTHER CHOICES ARE INCORRECT:

✕ **(A):** Software recommendation algorithms have no physical connection to battery charging circuitry.
✕ **(B):** Algorithmic content feeds do not alter financial billing structures.
✕ **(D):** Social media engagement algorithms aim to retain users, not delete active accounts.

Question 03 · Socioeconomic Impacts of Ride-Sharing & Gig Economy Platforms

Priority ★★★★

CORRECT ANSWER: (D) Beneficial: Provides flexible on-demand income for drivers and affordable transit for passengers | Harmful: Contributes to urban traffic congestion and offers unstable earnings without traditional employee healthcare benefits.

- Evaluate the socioeconomic trade-offs of the platform/gig economy:
- • **Beneficial Effects:** Passengers gain convenient, cashless transportation; drivers gain flexible working hours without strict corporate schedules.
- • **Harmful Effects:** Traditional taxi industries experience severe economic disruption, drivers often face unpredictable sub-minimum wages after vehicle depreciation, and extra idle ride-share vehicles circulate in city centers, worsening traffic congestion and carbon emissions.

WHY OTHER CHOICES ARE INCORRECT:

✕ **(A):** Ride-sharing vehicles continue to rely heavily on physical roads and highways.
✕ **(B):** Mobile apps cannot alter the global petroleum commodity market or physical speed limits.
✕ **(C):** Wireless protocols cannot physically repair mechanical automobile tires.

Question 04 · Trade-offs in Smart Home Internet of Things (IoT) Technology

Priority ★★★★

CORRECT ANSWER: (D) Greater convenience and remote household automation come at the expense of creating network vulnerabilities that attackers could exploit to spy on private conversations or unlock physical doors.

- Analyze trade-offs in Internet of Things (IoT) adoption:
- • **Benefits:** Users can adjust thermostats remotely, monitor doorsteps, and turn off lights with voice commands.
- • **Risks / Trade-offs:** IoT hardware often ships with default passwords, weak encryption, and unpatched firmware.
- • If compromised, networked microphones and cameras can leak intimate household recordings, and cloud vendor data breaches expose personal daily schedules to commercial data brokers.

WHY OTHER CHOICES ARE INCORRECT:

✕ **(A):** Home automation has no capability to ensure students remember school assignments.
✕ **(B):** IoT hardware requires continuous electrical power (wired or battery).
✕ **(C):** Smart devices do not alter the physical optical visibility of a house.

Question 05 · Societal Impacts of Autonomous Vehicles

Priority ★★★★★

CORRECT ANSWER: (A) Beneficial: Drastically reduces automobile crashes caused by human driver fatigue and distraction | Economic Disruption: Threatens the livelihood of millions of commercial truck, taxi, and delivery drivers through job displacement.

- Analyze the societal impacts of autonomous transportation:
- • **Beneficial Effect:** Over 90% of motor vehicle accidents involve human error (speeding, alcohol, distraction). Autonomous perception systems have 360-degree vision and millisecond reaction times, saving thousands of lives annually.
- • **Economic Disruption:** Commercial driving is one of the most common occupations for non-college graduates. Rapid automation risks mass structural unemployment for truck, bus, and taxi drivers, requiring workforce retraining policies.

WHY OTHER CHOICES ARE INCORRECT:

✕ **(B):** Autonomous software does not alter acoustic physics or destroy computing infrastructure.
✕ **(C):** Autonomous vehicles do not overcome gravity or require gold paving.
✕ **(D):** Self-driving cars cannot control meteorological precipitation.

Question 06 · Target Population Design and Demographic Exclusion

Priority ★★★★

CORRECT ANSWER: (C) Low-income and elderly individuals who use older, inexpensive smartphones without biometric sensors or who live in areas lacking 5G cellular infrastructure.

- Examine target populations and accessibility in computing design:
- • When developers optimize software only for state-of-the-art hardware and high-speed broadband, they create an accessibility barrier for economically disadvantaged or elderly populations.
- • Those who cannot afford $1,000 smartphones or multi-gigabit data plans are effectively locked out of modern digital financial services.
- • Inclusive computing design requires backward compatibility with older devices, low-bandwidth modes, and accessible interfaces.

WHY OTHER CHOICES ARE INCORRECT:

✕ **(A):** Software engineers typically have access to modern hardware and high-speed internet.
✕ **(B):** Executives with the latest flagship devices are the exact demographic the app was designed for.
✕ **(D):** Computer science students generally possess current computing equipment.

Question 07 · Accessibility and Assistive Technologies in Software Design

Priority ★★★★★

CORRECT ANSWER: (C) To ensure that computing innovations can be effectively used by individuals with diverse physical, sensory, and cognitive abilities, including those who rely on screen readers.

- Understand accessibility and assistive technology principles in AP CSP:
- • **Accessibility:** Designing software and digital content so that people with disabilities (visual, auditory, motor, or cognitive impairments) can independently perceive, understand, navigate, and interact with the system.
- • Examples: Screen readers synthesize text into speech from `alt` attributes, closed captions provide video access for deaf users, and high-contrast color palettes assist individuals with color blindness.
- • Accessibility ensures equal participation in the digital economy and democratic life.

WHY OTHER CHOICES ARE INCORRECT:

✕ **(A):** Alt text and semantic HTML actually improve search engine indexing (SEO), not block it.
✕ **(B):** Accessibility features are lightweight text enhancements that do not bloat file sizes.
✕ **(D):** Accessibility guidelines are universal open standards intended for all public web users.

Question 08 · Impact of E-Commerce on Traditional Brick-and-Mortar Retail

Priority ★★★★

CORRECT ANSWER: (B) The decline and closure of local brick-and-mortar storefronts, leading to reduced municipal sales tax revenue and the loss of vibrant neighborhood commercial centers.

- Analyze the cultural and economic impacts of digital commerce:
- • **Benefits:** Convenience, price transparency, massive product selection, and rapid delivery.
- • **Adverse Impacts:** Local independent bookstores, clothing shops, and pharmacies cannot compete with the economies of scale and automation of giant e-commerce conglomerates.
- • When local shops close, downtown commercial districts empty out, community interaction decreases, and municipalities lose local tax revenue needed to fund public schools and emergency services.

WHY OTHER CHOICES ARE INCORRECT:

✕ **(A):** E-commerce heavily relies upon and encourages electronic credit card and digital wallet transactions.
✕ **(C):** Digital commerce has no specific causal relationship with paper dictionary prices.
✕ **(D):** Cardboard production has expanded globally to meet shipping demand; there is no permanent total shortage.

Question 09 · Healthcare Impacts: Telemedicine and Remote Patient Monitoring

Priority ★★★★

CORRECT ANSWER: (D) Benefit: Expands access to specialized doctors for rural patients who live far from major hospitals | Limitation: Cannot perform physical examinations or emergency surgical procedures, and excludes patients without reliable broadband internet.

- Evaluate the advantages and boundaries of telemedicine computing innovations:
- • **Benefits:** Patients in isolated rural communities or those with mobility impairments can consult top medical specialists without traveling hundreds of miles or sitting in contagion-prone waiting rooms.
- • **Limitations:** Video calls cannot replace physical palpation, stethoscope auscultation, or hands-on surgery.
- • Furthermore, patients affected by the **digital divide** (lacking high-speed internet or smartphones) cannot utilize the service, exacerbating health equity disparities.

WHY OTHER CHOICES ARE INCORRECT:

✕ **(A):** Telemedicine relies on human doctors and functions across all weather conditions via network connections.
✕ **(B):** Telemedicine is a communications platform; it cannot alter genetics, and doctors routinely e-prescribe medications.
✕ **(C):** Physicians still require rigorous medical school education; monitors do not emit ionizing medical radiation.

Question 10 · Comprehensive Innovation Impact Synthesis Table

Priority ★★★★★

CORRECT ANSWER: (D) Computing innovations designed with beneficial intentions can generate complex, widespread negative societal side-effects that require ethical evaluation and regulatory oversight.

- Synthesize Big Idea 5 (Impact of Computing):
- • All four innovations solve genuine human problems (navigation, authentication, entertainment, fitness).
- • Yet, every single one generated systemic unintended consequences affecting urban planning, racial equity, creator livelihoods, and healthcare privacy.
- • Responsible computer science requires anticipating these second-order impacts, conducting algorithmic audits, and implementing protective social and legal frameworks.
- • This core synthesis question reinforces the learning objectives across IOC-1 and IOC-2.

WHY OTHER CHOICES ARE INCORRECT:

✕ **(A):** Programming languages are tools; ethical and societal impacts arise from system design and deployment, not language choice.
✕ **(B):** All four technologies rely on computer processing and software code, meeting the formal definition of computing innovations.
✕ **(C):** Banning technology ignores the immense life-saving, educational, and economic benefits of computing innovations.

Question 11 · Definition of the Digital Divide

Priority ★★★★★

CORRECT ANSWER: (C) The gap between individuals, households, businesses, and geographic areas at different socioeconomic levels with regard both to their opportunities to access information and communication technologies and to their use of the Internet for a wide variety of activities.

- Define the Digital Divide in AP CSP Big Idea 5 (IOC-1.C):
- • **The Digital Divide:** Disparities in access to computing devices and high-speed Internet infrastructure.
- • It is driven by several interrelated factors:
- - **Socioeconomic Status:** Inability to afford computers, tablets, or monthly broadband fees.
- - **Geographic Location:** Lack of physical telecommunications lines in rural or mountainous areas.
- - **Demographics:** Age-related digital literacy gaps or lack of localized language software.
- • The divide creates structural inequities in education, employment, healthcare, and civic participation.

WHY OTHER CHOICES ARE INCORRECT:

✕ **(A):** Physical component spacing on a motherboard is hardware computer architecture, not the socioeconomic Digital Divide.
✕ **(B):** Address length math relates to network protocols (Unit 4), not the Digital Divide.
✕ **(D):** Frontend and backend separation is software engineering architecture, not the societal Digital Divide.

Question 12 · Geographic Inequities in Broadband Deployment

Priority ★★★★★

CORRECT ANSWER: (C) The high upfront financial cost of laying miles of physical fiber-optic cables across sparsely populated terrain yields too few paying customers to make the private infrastructure investment profitable.

- Analyze the economic drivers of the geographic digital divide:
- • In dense urban cities, laying one mile of fiber cable reaches thousands of potential residential and corporate subscribers (high return on investment / ROI).
- • In rural areas, laying one mile of cable may connect only two or three isolated farmhouses.
- • Commercial Internet Service Providers (ISPs) often refuse to deploy capital-intensive physical infrastructure without government subsidies or municipal broadband intervention, leaving rural communities stranded on legacy dial-up or high-latency satellite connections.

WHY OTHER CHOICES ARE INCORRECT:

✕ **(A):** Fiber-optic cables operate using total internal reflection of laser light inside glass cores, indifferent to rural geography.
✕ **(B):** Satellite signals operate on radio frequencies and have no physical effect on subterranean fiber cables.
✕ **(D):** There are no laws restricting rural citizens from learning computing or accessing networks.

Question 13 · The 'Homework Gap' and Educational Inequities

Priority ★★★★★

CORRECT ANSWER: (A) Students from low-income households who lack reliable home internet or dedicated personal laptops will fall behind academically compared to wealthier peers who have seamless home access.

- Examine the 'Homework Gap' in educational technology:
- • When curriculum shifts to digital-first assignments, home internet access becomes as essential as a textbook or pencil.
- • Students in households without broadband must often complete research on small smartphone screens, travel to fast-food parking lots for free Wi-Fi, or wait in long lines at public libraries.
- • This inequality in access directly harms academic achievement, GPAs, and college readiness for economically disadvantaged students.

WHY OTHER CHOICES ARE INCORRECT:

✕ **(B):** Access to high-speed home internet provides research advantages, not academic penalties.
✕ **(C):** Digital portals do not grant automatic perfect grades; effort and mastery are still required.
✕ **(D):** The core issue is lack of digital connectivity, not physical paper turn-in logistics.

Question 14 · Age, Demographics, and Digital Literacy Disparities

Priority ★★★★

CORRECT ANSWER: (A) Elderly senior citizens who may lack familiarity with web browsers, digital navigation skills, or personal computing devices.

- Analyze demographic factors in the digital divide:
- • The digital divide is not merely about physical wire infrastructure; it encompasses **digital literacy** and user interface accessibility.
- • Older generations who grew up before the Internet era often experience difficulty navigating complex multi-step web forms, two-factor authentication prompts, or mobile apps.
- • Eliminating physical paper offices and phone assistance cuts off vulnerable seniors from essential civic and social welfare services.

WHY OTHER CHOICES ARE INCORRECT:

✕ **(B):** Tech executives are highly adept at utilizing digital corporate and government platforms.
✕ **(C):** Gaming teenagers are digital natives comfortable with computer interfaces.
✕ **(D):** Software engineering students possess high technical literacy and easily navigate online portals.

Question 15 · The Global Digital Divide Between Nations

Priority ★★★★

CORRECT ANSWER: (D) Unreliable electrical power grids, high costs of cellular data relative to local average income, and a scarcity of educational content published in local indigenous languages.

- Examine the root causes of the global digital divide:
- • In developing countries, basic prerequisites for computing are often missing:
- - **Power:** Frequent blackouts prevent continuous server and router operation.
- - **Cost:** Mobile data bundles can consume a large fraction of a family's daily income.
- - **Language:** Over 50% of web content is in English, excluding billions of non-English speakers.
- • Bridging the global divide requires investments in clean energy, affordable spectrum, and localized digital educational resources.

WHY OTHER CHOICES ARE INCORRECT:

✕ **(A):** Modern systems use UTF-8/Unicode to flawlessly encode virtually all global languages and scripts.
✕ **(B):** International trade actively supports technological imports; there is no ban on silicon.
✕ **(C):** Electromagnetic radio waves propagate identically across all geographic latitudes.

Question 16 · Mobile-Only Access vs. Dedicated Broadband Computing

Priority ★★★★★

CORRECT ANSWER: (D) Smartphones have small screens, touch keyboards unsuited for long essays, restrictive mobile operating systems that hinder software development, and strict monthly cellular data caps that throttle video lectures.

- Distinguish mobile-only access from comprehensive digital inclusion:
- • Having a smartphone is far better than having zero connectivity, but it is not equivalent to a dedicated desktop or laptop with unlimited broadband.
- • Key limitations of mobile-only access:
- - Writing formal resumes, research papers, or computer programs is exceedingly difficult on a 6-inch touchscreen.
- - Cellular data caps frequently throttle speeds after a few gigabytes, cutting off video conferencing or online learning.
- - Complex productivity and programming tools often require desktop operating systems (Windows, macOS, Linux).

WHY OTHER CHOICES ARE INCORRECT:

✕ **(A):** Smartphones connect to the Internet and handle mobile email seamlessly.
✕ **(B):** Cellular networks transmit standard IP packets and display full web pages.
✕ **(C):** Mobile operating systems feature persistent non-volatile flash storage; they do not wipe files daily.

Question 17 · Community Wireless Mesh Networks as an Equity Solution

Priority ★★★★

CORRECT ANSWER: (C) A **Community Wireless Mesh Network.**

- Identify innovative grassroots interventions for digital equity:
- • A **Community Wireless Mesh Network** is a decentralized, locally owned networking model.
- • Instead of relying on a monopoly ISP to wire each house individually, neighborhood nodes forward data wirelessly to adjacent nodes.
- • Benefits: Drastically lowers monthly internet costs for low-income residents, builds community technical resilience, and ensures internet access even if single nodes go offline.

WHY OTHER CHOICES ARE INCORRECT:

✕ **(A):** A mainframe is a giant centralized corporate server, the opposite of a decentralized community mesh.
✕ **(B):** Telegraph lines are 19th-century Morse code communication channels.
✕ **(D):** Satellite television is a one-way entertainment broadcast, not a bidirectional internet network.

Question 18 · The Role of Public Libraries and Community Centers

Priority ★★★★

CORRECT ANSWER: (A) They serve as essential public access points and safety nets, ensuring that individuals without financial means can access government services, apply for jobs, and develop essential digital literacy skills.

- Recognize institutional anchors of digital inclusion:
- • Public libraries have evolved into critical community technology centers.
- • For individuals experiencing homelessness, low-income families, and recent immigrants, the library is often the *only* place with working computers, printers, and high-speed internet required to apply for jobs or file taxes.
- • Hotspot lending programs extend internet access directly into students' living rooms, narrowing the homework gap.

WHY OTHER CHOICES ARE INCORRECT:

✕ **(B):** Libraries complement formal public education; they do not replace schools or certified teachers.
✕ **(C):** Libraries purchase commercial computers; they do not stop companies from manufacturing hardware.
✕ **(D):** Libraries support both digital and print literacy; they do not confiscate personal technology.

Question 19 · Government Policies, Subsidies, and Universal Service Initiatives

Priority ★★★★

CORRECT ANSWER: (A) Because access to high-speed Internet is now essential for participating in modern economic life, obtaining an education, accessing healthcare, and exercising democratic rights, lack of access perpetuates systemic poverty.

- Understand the policy dimension of digital inclusion:
- • In the 20th century, universal service laws ensured that every home had electricity and landline telephone service regardless of remote location.
- • Today, high-speed Internet is recognized as an indispensable public utility.
- • Without broadband, citizens cannot apply for employment (most jobs accept online applications only), consult telemedicine doctors, file taxes, or attend remote classes.
- • Subsidizing access helps break intergenerational cycles of poverty.

WHY OTHER CHOICES ARE INCORRECT:

✕ **(B):** Digital bits are mathematical data representations; they are not finite physical consumables.
✕ **(C):** Universal service subsidies focus on expanding infrastructure access, not mass surveillance.
✕ **(D):** Digital infrastructure complements physical public infrastructure (water, roads); it does not replace it.

Question 20 · Analyzing Community Digital Divide Profiles and Interventions

Priority ★★★★★

CORRECT ANSWER: (B) Providing income-eligible families with direct monthly broadband financial subsidies or vouchers, because the physical fiber infrastructure is already present and only the financial cost is prohibitive.

- Analyze data tables to match digital divide barriers with targeted interventions:
- • Notice the critical distinction between **Infrastructure Gaps** and **Affordability Gaps**:
- - In Districts W and Z, physical wires do not exist; the solution requires physical construction (satellite, towers, or fiber laying).
- - In **District X**, the table states that fiber cables *already exist*, but families cannot pay the $80/month commercial fee due to low median income ($26,000).
- • Therefore, laying more cables would accomplish nothing. The precise solution is an **affordability subsidy or municipal voucher** to connect homes to existing lines.

WHY OTHER CHOICES ARE INCORRECT:

✕ **(A):** Reverting to paper encyclopedias abandons digital inclusion and widens the educational gap.
✕ **(C):** Banning ISPs removes the existing infrastructure entirely, harming the community.
✕ **(D):** District X already has physical fiber infrastructure; laying redundant cables does not solve the family affordability problem.

Question 21 · Definition of Computing Bias

Priority ★★★★★

CORRECT ANSWER: (C) Computing bias can be embedded directly by human programmer choices, but more frequently emerges when machine learning models are trained on historical datasets that reflect human prejudices or demographic underrepresentation.

- Analyze the nature of Computing Bias in AP CSP Big Idea 5 (IOC-1.D):
- • Computing bias does not require a developer to harbor personal prejudice or intentionally write discriminatory `IF/ELSE` statements.
- • Most modern bias arises through **machine learning on historical data**:
- - If society had historical hiring, lending, or policing disparities, feeding those historical records into a machine learning algorithm teaches the algorithm to replicate and codify those exact biases.
- - If certain demographic groups are underrepresented in the training sample, the model learns poor feature representations for them.
- • Software reflects the values, assumptions, and data of its creators and historical contexts.

WHY OTHER CHOICES ARE INCORRECT:

✕ **(A):** Algorithmic bias is prevalent across modern machine learning models and neural networks regardless of programming language.
✕ **(B):** The belief that algorithms are inherently objective is a dangerous misconception; algorithms learn and amplify human biases present in training data.
✕ **(D):** Microprocessor heat relates to physical thermodynamics, not algorithmic fairness.

Question 22 · Biased Training Data in Automated Resume Screening

Priority ★★★★★

CORRECT ANSWER: (D) The algorithm penalizes resumes containing words associated with women (such as 'women's chess club captain' or graduates of women's colleges) because it learned to associate past male candidates with hiring success.

- Understand training data bias in automated decision systems:
- • This is based on a famous real-world case study (Amazon's experimental recruiting AI).
- • Because historical hiring data reflected a male-dominated workforce, the machine learning model deduced a statistical correlation: 'successful candidates in the training set are male'.
- • Even when developers removed explicit gender fields, the model identified subtle linguistic proxies (e.g., women's sports teams, women's colleges, specific vocabulary) and downgraded female candidates.
- • This illustrates **historical bias**: automated systems learn to automate and amplify past societal inequities.

WHY OTHER CHOICES ARE INCORRECT:

✕ **(A):** The algorithm is selective and filters candidates; it does not grant universal employment.
✕ **(B):** Machine learning classifiers score applicant text; they do not sort by physical print layout.
✕ **(C):** Recruiting software operates internally and does not delete external corporate web domains.

Question 23 · Demographic Disparities in Facial Recognition Algorithms

Priority ★★★★★

CORRECT ANSWER: (A) The image datasets used to train the computer vision neural networks consisted predominantly of lighter-skinned male faces, leaving the algorithms with insufficient training samples to learn facial features of darker-skinned women.

- Examine demographic sampling bias in computer vision (Gender Shades study by Buolamwini & Gebru):
- • Deep learning algorithms learn patterns based on the statistical diversity of the images they are fed during training.
- • Landmark computer vision benchmarks historically scraped internet photo databases that were over 75% male and over 80% light-skinned.
- • Consequently, the neural networks optimized feature extraction for pale skin while failing to generalize to darker complexions, leading to dangerous real-world misidentifications and wrongful arrests.

WHY OTHER CHOICES ARE INCORRECT:

✕ **(B):** Human skin does not emit magnetic frequencies that disable optical digital camera sensors.
✕ **(C):** Facial recognition models process images continuously regardless of time of day.
✕ **(D):** Digital displays and camera sensors accurately render diverse skin tones when properly calibrated.

Question 24 · Algorithmic Risk Assessment and Proxy Variables in Criminal Justice

Priority ★★★★★

CORRECT ANSWER: (A) The algorithm uses other input variables (such as neighborhood zip code, socio-economic factors, and prior arrest history) that act as statistical **proxy variables closely correlated with systemic racial disparities.**

- Analyze proxy variables and algorithmic fairness in sensitive domains:
- • Simply deleting sensitive demographic attributes (like race or gender) is known as **fairness through unawareness**, and it almost never works.
- • In many societies, historical residential segregation and unequal policing mean that variables like **neighborhood zip code**, **family income**, and **number of prior police stops** are heavily correlated with race.
- • The algorithm unintentionally uses these correlated **proxy variables** to reconstruct the omitted protected attribute, leading to racially disparate risk predictions.

WHY OTHER CHOICES ARE INCORRECT:

✕ **(B):** Wi-Fi signals carry network packets; they cannot read human neurological thoughts.
✕ **(C):** Risk assessment tools run deterministic statistical regression models, not random numbers.
✕ **(D):** Risk assessment models execute complex weighted risk scoring equations, not coin flips.

Question 25 · Healthcare Inequities in Diagnostic Machine Learning

Priority ★★★★

CORRECT ANSWER: (B) Melanomas present with different visual contrast and pigmentation patterns on darker skin types, but the model never learned these patterns due to the homogeneous Scandinavian training data.

- Evaluate clinical dataset bias in medical computing:
- • Machine learning models exhibit **distribution shift** when deployed in environments that differ from their training environment.
- • Clinical presentations of dermatological conditions vary across Fitzpatrick skin phototypes.
- • Because the model was trained exclusively on fair-skinned Scandinavian patients, it fails to recognize malignant lesions on darker skin, leading to dangerous false negatives and delayed cancer diagnoses for non-white patients.

WHY OTHER CHOICES ARE INCORRECT:

✕ **(A):** Visual computer vision models process image pixels; patient spoken language does not alter image inference.
✕ **(C):** High-resolution digital photography can capture dermatological lesions on any skin tone globally.
✕ **(D):** Solar radiation has no ability to selectively delete application code from smartphone storage.

Question 26 · Software Engineering Team Diversity and Developer Blindspots

Priority ★★★★

CORRECT ANSWER: (C) A lack of diversity on software development teams can lead to **developer blindspots, where features essential to underrepresented groups are unintentionally overlooked during the design phase.**

- Understand the impact of team diversity on software innovation:
- • When engineering teams lack demographic diversity (gender, race, age, physical ability), developers naturally design for their own lived experiences and needs.
- • Homogeneous teams frequently overlook crucial use cases, accessibility features, and potential safety risks for populations not present in the design room.
- • Inclusive computing teams yield more robust, equitable, and globally useful software products.

WHY OTHER CHOICES ARE INCORRECT:

✕ **(A):** Step tracking is a lightweight background pedometer sensor process that consumes minimal battery.
✕ **(B):** Mobile operating systems do not restrict app downloads based on user gender.
✕ **(D):** Calendar tracking requires minimal bytes of memory, which modern smartphones have in billions (gigabytes).

Question 27 · Algorithmic Discrimination in Online Advertising Delivery

Priority ★★★★

CORRECT ANSWER: (C) The ad platform's automated bidding and optimization algorithm determines that female consumers are more commercially valuable to retail advertisers, thereby outbidding job ads and steering women toward consumer goods rather than career opportunities.

- Analyze market-driven algorithmic bias in ad auctions:
- • In automated ad networks, complex multi-market auction dynamics interact in unexpected ways.
- • Because women make a high percentage of retail purchasing decisions, consumer brands bid high prices to display retail and beauty ads to female demographics.
- • As a result, commercial ad algorithms prioritize expensive retail ads for female users, inadvertently pricing out high-paying executive job ads and creating an unintended systemic barrier to economic opportunity.

WHY OTHER CHOICES ARE INCORRECT:

✕ **(A):** Employment discrimination laws strictly prohibit hiding job opportunities based on sex.
✕ **(B):** Women have full legal and technical access to search engines and online job boards.
✕ **(D):** Low-level network protocols (TCP/IP) are gender-blind and route packets purely by numerical IP addresses.

Question 28 · Techniques for Auditing and Mitigating Algorithmic Bias

Priority ★★★★★

CORRECT ANSWER: (D) Disaggregate performance metrics (such as approval rates, false positive rates, and false negative rates) across distinct demographic subgroups (race, gender, age) to verify equitable accuracy and outcomes.

- Understand algorithmic auditing and fairness metrics in AP CSP:
- • An aggregate accuracy score (e.g. '95% overall accuracy') can completely mask extreme failure rates in smaller demographic subgroups.
- • In an **Algorithmic Audit**, engineers slice and evaluate the model's error matrix across disaggregated groups:
- - Are false rejection rates (denying loans to creditworthy applicants) significantly higher for minority groups?
- - Does the model rely on unfair proxy features?
- • Auditing requires continuous post-deployment monitoring, representative benchmarking, and ethical review.

WHY OTHER CHOICES ARE INCORRECT:

✕ **(A):** Paper printing has zero relevance to digital algorithmic auditing or fairness metrics.
✕ **(B):** Testing only on developers introduces severe sampling bias and fails to reflect the diverse general public.
✕ **(C):** Syntax compilation only proves code is syntactically valid; it says nothing about algorithmic fairness or ethics.

Question 29 · Deconstructing the Myth of Algorithmic Neutrality

Priority ★★★★

CORRECT ANSWER: (A) While computers do not have feelings, algorithms are designed by humans, use datasets selected by humans, and learn from historical human records that contain deeply ingrained societal biases.

- Deconstruct the myth of algorithmic objectivity:
- • Math itself is neutral, but the *application* of mathematical models to human social systems is never neutral.
- • If police historically targeted low-income minority neighborhoods for drug arrests at higher rates despite equal drug usage across populations, the historical arrest dataset reflects that policing practice.
- • Feeding those numbers into a predictive algorithm causes the model to predict 'more crimes in minority neighborhoods', sending more officers there, generating more arrests, and creating a vicious, self-reinforcing feedback loop masquerading as 'objective math'.

WHY OTHER CHOICES ARE INCORRECT:

✕ **(B):** Mathematical numbers and operations have precise, deterministic definitions in computing.
✕ **(C):** Computers are electronic silicon devices without human sentience, emotional feelings, or consciousness.
✕ **(D):** Government officials routinely review algorithmic dashboard outputs.

Question 30 · Comparative Algorithmic Bias Scenarios and Mitigations

Priority ★★★★★

CORRECT ANSWER: (B) Retraining and validating the models using comprehensively diverse, representative, and balanced datasets that accurately reflect all demographic and linguistic subgroups.

- Synthesize core mitigations for computing bias across domains:
- • In every single system in the table, the bias stems from an **unbalanced, non-representative, or historically prejudiced dataset**:
- - Non-diverse audio (System 2) misses accents.
- - Gender-skewed text corpora (System 3) reinforce patriarchal occupational stereotypes.
- - Non-representative academic data (System 1) penalizes innovative schools.
- • The primary remedy is rigorous data curation: collecting balanced training datasets, evaluating models against diverse benchmarks, and continually auditing algorithmic outputs for disparate impact.

WHY OTHER CHOICES ARE INCORRECT:

✕ **(A):** Human cultural and linguistic diversity is natural; technology must adapt to humans, not force human homogenization.
✕ **(C):** Shutting down electricity is a catastrophic overreaction that would paralyze modern society.
✕ **(D):** Reverting to paper fails to leverage the immense beneficial power of computation while keeping all human biases intact.

Question 31 · Definition and Mechanics of Crowdsourcing

Priority ★★★★★

CORRECT ANSWER: (C) The practice of obtaining needed services, data, ideas, or content by soliciting contributions from a large, distributed group of online people rather than relying solely on traditional employees or contractors.

- Define Crowdsourcing in AP CSP Big Idea 5 (IOC-1.E):
- • **Crowdsourcing:** Distributing tasks, research, or funding across an open, decentralized network of Internet participants.
- • Instead of relying on a small internal team, organizations tap into the collective intelligence, diverse perspectives, and distributed processing power of thousands or millions of contributors.
- • It enables rapid data collection, creative problem-solving, and massive-scale categorization that would be financially or physically impossible for a single organization.

WHY OTHER CHOICES ARE INCORRECT:

✕ **(A):** Crowdsourcing operates digitally across networks, not via physical human crowding in hardware rooms.
✕ **(B):** Memory allocation is computer hardware management, not crowdsourcing.
✕ **(D):** A botnet traffic flood is a Distributed Denial of Service (DDoS) cyberattack, not crowdsourcing.

Question 32 · Citizen Science and Collaborative Research

Priority ★★★★★

CORRECT ANSWER: (C) It allows researchers to analyze massive, petabyte-scale datasets that are far too large for a small team of scientists to manually examine, accelerating scientific discovery at minimal financial cost.

- Analyze the role of Citizen Science in modern computational research:
- • Modern telescopes and environmental sensors produce more raw observational data than professional scientists could review in multiple lifetimes.
- • **Citizen Science** harnesses human pattern recognition (which often outperforms automated computer vision on noisy or unusual astronomical images).
- • By distributing small image slices to thousands of passionate volunteers, projects classify millions of celestial objects or migratory patterns in months rather than decades.

WHY OTHER CHOICES ARE INCORRECT:

✕ **(A):** Professional scientists are essential to design the research protocols, calibrate models, and publish formal peer-reviewed findings.
✕ **(B):** Computers transmit digital informational bits; physical human interstellar space travel is not possible through cables.
✕ **(D):** Volunteer data inevitably contains noise, typos, and mistakes, requiring statistical filtering and cross-verification.

Question 33 · Real-Time Crowdsourced Traffic and Sensor Telemetry

Priority ★★★★★

CORRECT ANSWER: (A) Passively collecting real-time GPS speed telemetry from all active user smartphones to detect traffic slow-downs, while actively allowing drivers to report road hazards, construction, and police presence.

- Distinguish between passive and active crowdsourcing mechanisms:
- • **Passive Crowdsourcing:** The app runs in the background, automatically relaying real-time velocity and coordinate telemetry. When hundreds of users slow down to 5 mph on a highway, the server instantly infers a traffic jam.
- • **Active Crowdsourcing:** Users intentionally tap on-screen buttons to submit discrete hazard reports (e.g. stalled vehicle, pothole, road closure).
- • The combination creates an ultra-responsive, dynamic routing map far more current than static municipal street databases.

WHY OTHER CHOICES ARE INCORRECT:

✕ **(B):** Navigation apps maintain and update maps rather than erasing them, and have no connection to battery circuitry.
✕ **(C):** Consumer navigation apps do not possess municipal authorization or wireless protocols to hack traffic signals.
✕ **(D):** Crowdsourced navigation provides directions; it does not tamper with vehicle lighting systems or bill hidden fees.

Question 34 · Decentralized Peer Production vs. Traditional Publishing

Priority ★★★★

CORRECT ANSWER: (D) Strength: Rapidly incorporates new global events and covers niche topics in hundreds of languages | Challenge: Vulnerable to intentional vandalism, misinformation, and edit wars that require active moderation.

- Evaluate open collaborative peer production models:
- • **Strengths:** Extreme agility (breaking news articles appear within minutes), massive breadth across millions of obscure topics, and democratization of global knowledge across diverse languages.
- • **Challenges:** Anyone can edit, opening the door to malicious misinformation, bias, propaganda, and vandalism.
- • Wikipedia combats this through transparent revision histories, bot verification algorithms, and dedicated community volunteer moderators who lock high-conflict articles.

WHY OTHER CHOICES ARE INCORRECT:

✕ **(A):** Wikipedia is hosted on massive web server clusters and accessed digitally via HTTP/HTTPS.
✕ **(B):** Wikipedia is completely free to read and edit and relies on community editors for orthographic proofreading.
✕ **(C):** Wikipedia articles are continuously updated and openly readable by the entire global public.

Question 35 · Economic Democratization Through Crowdfunding

Priority ★★★★

CORRECT ANSWER: (B) It allows creators to raise funds through small financial contributions from thousands of everyday backers worldwide, validating market demand without giving up company ownership to traditional gatekeepers.

- Analyze the socioeconomic impact of crowdfunding innovations:
- • Traditional finance (banks, venture capital) has historic barriers that disproportionately exclude women, minority founders, and niche community projects.
- • **Crowdfunding** bypasses traditional gatekeepers by connecting creators directly with end consumers.
- • A creator can raise $100,000 by convincing 2,000 people to pre-order a $50 device.
- • This democratizes entrepreneurship, charitable disaster relief, and grassroots creative arts.

WHY OTHER CHOICES ARE INCORRECT:

✕ **(A):** Many crowdfunded hardware projects experience supply chain and manufacturing delays; funding does not guarantee flawless execution.
✕ **(C):** Crowdfunding gathers private voluntary donations and purchases; it does not alter national monetary printing.
✕ **(D):** Crowdfunding funds product development; backers usually pay for the reward or pre-order.

Question 36 · Data Verification and Quality Control in Crowdsourced Systems

Priority ★★★★★

CORRECT ANSWER: (C) Implementing multi-contributor consensus algorithms, community reputation scores, and automated statistical anomaly detection to cross-verify submissions before publishing.

- Understand verification and data hygiene in crowdsourcing:
- • To maintain high data quality, platforms employ automated and algorithmic checks:
- - **Redundant Consensus:** Requiring 3 or more independent volunteers to classify the same astronomical image before marking it confirmed.
- - **Reputation Systems:** Weighting contributions more heavily from users with a verified track record of accuracy.
- - **Anomaly Detection:** Flagging sudden spikes in contradictory reports (e.g., someone reporting a blizzard in Miami in July).
- • These algorithmic defenses protect crowdsourced databases from trolls, spam, and human error.

WHY OTHER CHOICES ARE INCORRECT:

✕ **(A):** Deleting weekend data discards valid submissions and does not verify weekday data quality.
✕ **(B):** Filtering by user name is arbitrary and has zero correlation with data accuracy.
✕ **(D):** Randomly discarding data defeats the core purpose of collecting crowdsourced information.

Question 37 · Risks of Crowdsource Manipulation: Astroturfing and Fake Reviews

Priority ★★★★

CORRECT ANSWER: (A) **Astroturfing (Sockpuppeting), which creates a deceptive, manufactured illusion of grassroots public support or consensus through coordinated fake accounts.**

- Analyze the vulnerabilities of open crowdsourced feedback mechanisms:
- • **Astroturfing:** Masking the sponsors of an organization or message to make it appear as though it originates from genuine, grassroots participant consensus.
- • E-commerce review platforms, restaurant ratings, and social media trending topics are highly vulnerable to astroturfing by paid click farms and botnets.
- • This corrupts consumer decision-making and undermines public trust in crowdsourced ratings, necessitating machine-learning fraud detection filters.

WHY OTHER CHOICES ARE INCORRECT:

✕ **(B):** Protein folding simulations are legitimate biomedical research (Folding@home), not review fraud.
✕ **(C):** Optical character recognition (OCR) converts scanned paper text into digital characters.
✕ **(D):** Open source attribution is intellectual property compliance, not deceptive feedback manipulation.

Question 38 · Incentive Structures in Crowdsourcing Platforms

Priority ★★★★

CORRECT ANSWER: (A) Gamification relies on intrinsic hobbyist enthusiasm and produces high-quality voluntary engagement for specialized causes, whereas micro-payments enable rapid completion of massive repetitive tasks but risk attracting workers who rush through tasks to maximize hourly earnings.

- Evaluate human incentives in computational crowdsourcing systems:
- • **Gamified / Citizen Science platforms (Foldit, eBird):** Participants are motivated by curiosity, altruism, and community status. They spend hours carefully reviewing complex tasks.
- • **Paid Micro-task platforms (MTurk):** Great for labeling 500,000 photos for machine learning in 48 hours. However, financial incentives can incentivize speed over precision, requiring aggressive automated validation checks to filter low-effort responses.

WHY OTHER CHOICES ARE INCORRECT:

✕ **(B):** Displaying web badges and scoreboards consumes negligible computer memory.
✕ **(C):** Micro-payments are commercial transactions fully compliant with software distribution laws.
✕ **(D):** Both models operate entirely through digital representations (virtual profile badges or electronic bank transfers).

Question 39 · Data Privacy Concerns in Crowdsourced Telemetry

Priority ★★★★

CORRECT ANSWER: (B) Aggregating and publishing crowdsourced location telemetry, even when anonymized, can inadvertently expose sensitive personal habits, military installations, and confidential security perimeters.

- Analyze privacy risks in crowdsourced location aggregation (Strava military base heatmap case study):
- • Even when individual names are stripped from crowdsourced datasets (**anonymization**), dense patterns of real-world activity leave undeniable spatial signatures.
- • Soldiers jogging around secret military outposts created bright glowing loops on an otherwise pitch-black satellite map, revealing hidden supply roads and base layouts.
- • Crowdsourcing systems must incorporate differential privacy, geographic masking, and strict opt-in location policies to prevent unintended intelligence leaks.

WHY OTHER CHOICES ARE INCORRECT:

✕ **(A):** Outdoor physical exercise is legal and independent of telecommunications regulation.
✕ **(C):** GPS constellations provide 24/7 continuous radio positioning regardless of lunar cycles.
✕ **(D):** Fitness apps are designed for mobile wearable devices; restricting them to desktop PCs destroys their primary function.

Question 40 · Comprehensive Synthesis of Crowdsourcing Models

Priority ★★★★★

CORRECT ANSWER: (D) All four models demonstrate that distributed human collaboration over the Internet can achieve tasks that exceed the capacity of single institutions, provided appropriate verification, moderation, and privacy safeguards are enforced.

- Synthesize crowdsourcing principles in AP CSP Big Idea 5:
- • Crowdsourcing harnesses the power of the Internet's open, connected architecture to aggregate human labor, intelligence, and financial resources.
- • Across scientific research, knowledge curation, live transit, and capital generation, the collective power of the crowd achieves monumental scale.
- • However, success depends upon robust platform governance: peer review, consensus mechanisms, fraud detection, and privacy controls.
- • This synthesis question solidifies mastery of CED Topic 5.4.

WHY OTHER CHOICES ARE INCORRECT:

✕ **(A):** Crowdsourcing models are global international phenomena embraced across all continents.
✕ **(B):** The primary power of crowdsourcing is that it empowers everyday citizens of all educational backgrounds to contribute.
✕ **(C):** Crowdsourcing relies primarily on human participation and web connectivity, not facial recognition AI.

Question 41 · Forms of Intellectual Property in Computer Science

Priority ★★★★★

CORRECT ANSWER: (C) **Copyright, which grants the creator exclusive legal rights to reproduce, distribute, display, and create derivative works of an original creative expression.**

- Distinguish forms of Intellectual Property (IP) in computing:
- • **Copyright:** Protects original works of authorship fixed in a tangible medium, including written computer programs, documentation, music, images, and text.
- • **Patent:** Protects novel, non-obvious functional inventions or mechanical processes (e.g., a specific hardware chip design).
- • **Trademark:** Protects brand names, logos, and slogans ('QuickStream') to prevent consumer confusion in commerce.
- • Software source code is legally treated as a literary/creative work and is shielded by copyright law.

WHY OTHER CHOICES ARE INCORRECT:

✕ **(A):** Trademarks protect commercial brand identifiers, not United Nations cash collection.
✕ **(B):** Patents protect functional inventions, not passive cables, and do not automatically protect written code lines.
✕ **(D):** The Public Domain means works are free of copyright restrictions, not illegal to execute.

Question 42 · Automatic Inception of Digital Copyright

Priority ★★★★★

CORRECT ANSWER: (A) The code is **automatically copyrighted by the student the moment it is created and saved; other programmers cannot legally copy or republish the code without the student's permission.**

- Understand the automatic inception of copyright protection:
- • Under international treaties (such as the Berne Convention), copyright is secured **automatically** the instant a work is created and fixed in a tangible medium (such as being saved to a hard drive or published on a website).
- • Formal government registration and the '©' symbol are helpful for legal lawsuits, but they are *not required* to establish copyright ownership.
- • Just because an image or code snippet is publicly viewable online does NOT mean it is free to reuse without permission.

WHY OTHER CHOICES ARE INCORRECT:

✕ **(B):** Publicly visible works are not automatically Public Domain; copyright remains intact unless explicitly waived.
✕ **(C):** ISPs are passive transit carriers; they do not acquire ownership of user-authored code.
✕ **(D):** Any individual author, student, or creator holds full legal copyright over their original creations.

Question 43 · Creative Commons Licensing: CC BY (Attribution)

Priority ★★★★★

CORRECT ANSWER: (A) The student may freely copy, distribute, modify, and even use the icons commercially, provided that the original designer is credited and given proper attribution.

- Examine Creative Commons (CC) licensing structures:
- • Creative Commons provides standardized copyright licenses that grant the public permission to use creative work under specified conditions.
- • **CC BY (Attribution):** The most accommodating CC license.
- - Anyone can share, adapt, remix, and build upon the work for any purpose (even commercially).
- - The only mandatory legal condition is giving appropriate **Attribution** (credit) to the original author.

WHY OTHER CHOICES ARE INCORRECT:

✕ **(B):** CC BY licenses are free public licenses that require zero royalty fees.
✕ **(C):** CC licenses do not contain calendar-day restrictions.
✕ **(D):** CC BY explicitly permits modifications, adaptations, and remixes (unlike CC BY-ND).

Question 44 · Creative Commons Modifiers: ShareAlike (SA) and NonCommercial (NC)

Priority ★★★★★

CORRECT ANSWER: (B) Yes; the company violated the **NC (NonCommercial) clause by selling the book for commercial profit, and violated the **SA (ShareAlike)** clause if their derivative translation was not licensed under identical terms.**

- Analyze Creative Commons license modifiers:
- • **NC (NonCommercial):** Restricts use strictly to non-commercial purposes; commercial sale or monetization is explicitly prohibited.
- • **SA (ShareAlike):** Dictates that any adaptations, translations, or derivative works must be distributed under the exact same CC license terms ('copyleft' mechanism).
- • The tutoring company committed two direct violations: selling the work for financial profit (breaching NC) and failing to preserve open ShareAlike terms.

WHY OTHER CHOICES ARE INCORRECT:

✕ **(A):** Translations are legally considered 'derivative works' and remain subject to the original license.
✕ **(C):** Copyright terms govern all physical and digital distribution media.
✕ **(D):** Creative Commons licenses do not mandate commercial sales; they offer flexible open sharing.

Question 45 · Creative Commons: NoDerivatives (ND) vs. Public Domain (CC0)

Priority ★★★★★

CORRECT ANSWER: (B) Cropping the photo, applying color filters, and incorporating it into a remixed digital collage without permission.

- Contrast CC BY-ND with CC0 Public Domain:
- • **CC BY-ND (NoDerivatives):** Allows others to copy and distribute the work in its *original, unedited form only*. Remixing, altering, filtering, cropping, or creating derivative collages is strictly prohibited.
- • **CC0 (Public Domain Dedication):** The creator completely waives all copyright rights worldwide. Anyone can alter, remix, commercialize, or build upon the image without attribution or restriction.
- • Therefore, remixing or modifying is legal for Photo 2, but a copyright violation for Photo 1.

WHY OTHER CHOICES ARE INCORRECT:

✕ **(A):** Sharing links or sharing unaltered copies is permitted under both licenses.
✕ **(C):** Downloading for personal view is permitted under both licenses.
✕ **(D):** Viewing both photos is fully permitted under both licenses.

Question 46 · Open Source Software vs. Commercial Proprietary Software

Priority ★★★★★

CORRECT ANSWER: (D) The underlying source code is publicly accessible, allowing programmers worldwide to inspect for security vulnerabilities, fix bugs collaboratively, and customize features freely.

- Examine Open Source Software (OSS) vs Proprietary Software:
- • **Open Source Software:** The human-readable source code is made freely available to the public under licenses like MIT, Apache, or GPL.
- - *Advantages:* Transparency, peer review (Linus's Law: 'given enough eyeballs, all bugs are shallow'), community innovation, and vendor independence.
- • **Proprietary Software:** The source code is a guarded trade secret compiled into binary executables; users purchase a license to run it, but cannot view or modify the underlying code.

WHY OTHER CHOICES ARE INCORRECT:

✕ **(A):** Open source software runs on consumer laptops, smartphones, servers, and embedded microcontrollers worldwide.
✕ **(B):** Open source software can still have software bugs; open access facilitates finding and fixing them.
✕ **(C):** Most of the Internet's core infrastructure (Linux, Apache, BIND) is powered by open source software.

Question 47 · Plagiarism vs. Ethical Attribution in Programming

Priority ★★★★

CORRECT ANSWER: (A) No; the student engaged in ethical code reuse because they adhered to the open-source license terms and provided explicit, honest **attribution identifying the original author.**

- Distinguish Plagiarism from Legitimate Open Source Code Reuse:
- • **Plagiarism:** Presenting someone else's intellectual work, code, or ideas as one's own original creation without acknowledgment.
- • **Ethical Code Reuse:** Software engineering thrives on modular libraries. Utilizing open-source packages is standard industry practice *provided* the developer complies with the license and explicitly cites the source.
- • Because the student included the MIT license notice and gave attribution, they acted with complete academic and professional integrity.

WHY OTHER CHOICES ARE INCORRECT:

✕ **(B):** Software engineering relies on modular abstractions; nobody writes entire operating systems from scratch for every app.
✕ **(C):** Open-source licensing is completely legal and forms the foundation of modern technology.
✕ **(D):** Integrity is determined by transparent citation, not by whether judges noticed.

Question 48 · The Digital Millennium Copyright Act (DMCA) and Safe Harbor

Priority ★★★★

CORRECT ANSWER: (B) The platform operates as a passive online service provider that does not prescreen every upload, and maintains immunity as long as it promptly removes or disables access to the infringing material upon receiving a formal takedown notice from the copyright owner.

- Analyze the DMCA Safe Harbor framework in digital platforms:
- • The **DMCA (1998)** established **Safe Harbor** protections for online intermediaries (like YouTube, Reddit, or cloud hosts).
- • If platforms were personally liable for every copyright violation committed by users, modern social media and cloud storage would be legally impossible.
- • To maintain Safe Harbor immunity, platforms must:
- - Expeditiously remove infringing content upon receiving a valid **DMCA Takedown Notice**.
- - Maintain a policy to terminate repeat copyright infringers.

WHY OTHER CHOICES ARE INCORRECT:

✕ **(A):** Copyright protection endures for decades (author's life + 70 years); uploading does not forfeit rights.
✕ **(C):** The DMCA protects all digital creative media, including music, movies, software, and text.
✕ **(D):** Platforms cannot knowingly host or profit from piracy without losing Safe Harbor protections.

Question 49 · The Fair Use Doctrine in Digital Media

Priority ★★★★

CORRECT ANSWER: (B) The **Fair Use Doctrine, which permits limited use of copyrighted material without permission for purposes such as criticism, commentary, news reporting, teaching, and scholarship.**

- Understand the Fair Use Doctrine in copyright law:
- • **Fair Use** provides a critical exception to copyright monopoly, ensuring freedom of speech and education.
- • Courts evaluate four statutory factors:
- 1. **Purpose and character:** Transformative educational/critical commentary favors fair use.
- 2. **Nature of work:** Factual works have wider latitude than pure fictional entertainment.
- 3. **Amount used:** Using small, necessary excerpts (10 seconds) favors fair use.
- 4. **Market effect:** The short review does not substitute for buying or watching the full movie.
- • The student's critique is a textbook example of transformative fair use.

WHY OTHER CHOICES ARE INCORRECT:

✕ **(A):** Patents protect functional mechanical inventions, not creative artistic film excerpts.
✕ **(C):** There is no legal statute granting a 'right to piracy'.
✕ **(D):** Broadcasting mandates do not provide copyright exemptions for educational videos.

Question 50 · Comprehensive Software and Media Licensing Synthesis Table

Priority ★★★★★

CORRECT ANSWER: (C) Modern digital copyright spans a broad continuum from total closed restriction (Proprietary) to conditional open collaboration (Creative Commons / Open Source) and complete open freedom (Public Domain).

- Synthesize Intellectual Property and digital licensing models in AP CSP:
- • Licensing is not an all-or-nothing binary between strict copyright and outright piracy.
- • Instead, modern computing offers a flexible spectrum:
- - **Proprietary (All Rights Reserved):** Maximum commercial control.
- - **Open Source / Copyleft (GPL / CC BY-SA):** Open collaboration with viral sharing terms.
- - **Permissive Open Source (MIT / CC BY):** Open use with attribution.
- - **Public Domain (CC0 / NASA):** Unrestricted universal heritage.
- • Understanding these legal abstractions is essential for modern software architects.

WHY OTHER CHOICES ARE INCORRECT:

✕ **(A):** Creators are completely free to choose open source, public domain, or proprietary licensing.
✕ **(B):** Open source licenses actively promote computer programming education through open source access.
✕ **(D):** Public domain NASA space imagery is universally legal and encouraged for educational projects.

Question 51 · Definition and Scope of Personally Identifiable Information (PII)

Priority ★★★★★

CORRECT ANSWER: (C) Information about an individual that can be used directly or indirectly to distinguish, trace, or identify a specific person's identity, such as Social Security Numbers, biometric records, full legal names, and home addresses.

- Define Personally Identifiable Information (PII) in AP CSP Big Idea 5 (IOC-2.A):
- • **PII:** Any piece of data (or combination of data points) that identifies a specific unique human being.
- • **Direct Identifiers:** Social Security Number (SSN), driver's license number, biometric prints, passport number, full legal name.
- • **Indirect / Combinatorial Identifiers:** Date of birth, mother's maiden name, precise GPS coordinates, IP address, medical records.
- • PII requires strong technical safeguards (encryption, salted hashing, access controls) because compromised PII enables severe identity theft and fraud.

WHY OTHER CHOICES ARE INCORRECT:

✕ **(A):** Keyboard casing color is a generic industrial product property.
✕ **(B):** Municipal park counts are public civic geographic metrics unrelated to individual personal identity.
✕ **(D):** CPU clock frequency is dynamic hardware performance telemetry, not personal human identity.

Question 52 · Data Aggregation and Re-Identification Risks

Priority ★★★★★

CORRECT ANSWER: (D) Through **Data Aggregation, combining just Date of Birth, Gender, and 5-digit Zip Code is mathematically sufficient to uniquely re-identify approximately 87% of the entire United States population when matched against public voter registration databases.**

- Analyze data aggregation and re-identification (landmark Dr. Latanya Sweeney study):
- • Many organizations mistakenly believe that simply stripping names and SSNs creates complete 'anonymity'.
- • **Data Aggregation:** The process of gathering data from multiple sources to combine into summary or enriched profiles.
- • Because few people in a single zip code share the exact same birthdate and gender, cross-referencing this 'anonymized' health data with public records (like voter rolls containing names and birthdates) allows researchers to re-attach specific medical diagnoses to specific citizens.

WHY OTHER CHOICES ARE INCORRECT:

✕ **(A):** Relational database tables retain records regardless of which specific columns are populated.
✕ **(B):** Zip code data processing does not cause thermal combustion in storage hardware.
✕ **(C):** Numerical postal zip codes are geographic numbers; they cannot generate biological or digital viruses.

Question 53 · Active vs. Passive Digital Footprints

Priority ★★★★

CORRECT ANSWER: (D) An active footprint is created when a user deliberately shares data (such as posting on social media or filling out an online form); a passive footprint is collected automatically without the user's explicit action (such as server IP logging, browsing history, and device telemetry).

- Contrast Active and Passive Digital Footprints in AP CSP:
- • **Active Footprint:** Data the user consciously and intentionally emits into the digital ecosystem (e.g., uploading a photo to Instagram, publishing a blog, sending an email, submitting a survey).
- • **Passive Footprint:** Data collected silently in the background by servers and ad trackers without the user's active awareness (e.g., web server access logs tracking IP addresses, device screen dimensions, clickstream paths, session durations).
- • Together, these trails build comprehensive, permanent dossiers on individual web citizens.

WHY OTHER CHOICES ARE INCORRECT:

✕ **(A):** Digital footprints relate to electronic data transactions, not physical athletic exercise.
✕ **(B):** Both active and passive footprints can be transmitted over encrypted protocols (HTTPS/TLS).
✕ **(C):** Both types of footprints are generated on all computing platforms (phones, tablets, PCs, smart TVs).

Question 54 · Continuous Geolocation Tracking and Intimate Privacy Exposure

Priority ★★★★★

CORRECT ANSWER: (B) Continuous location data reveals intimate personal details of a user's life—such as home address, workplace, places of religious worship, medical clinics visited, and political protests attended—allowing third parties to infer sensitive personal beliefs and health conditions.

- Evaluate the privacy risks of mobile geolocation tracking:
- • A flashlight app has zero legitimate technical need for precise GPS satellite coordinates (the flash hardware is controlled purely by local device APIs).
- • Continuous spatial tracking over weeks creates a comprehensive map of a human being's private life:
- - Sleeping location = home address.
- - Daytime location = employer.
- - Recurring visits to oncology centers, rehabilitation clinics, mosques, or political rallies reveal intimate medical, religious, and political affiliations without the user ever typing a word.
- • This data is frequently sold on open data exchanges without user consent.

WHY OTHER CHOICES ARE INCORRECT:

✕ **(A):** LED flashlight diodes cannot transform into coherent high-energy lasers through software.
✕ **(C):** Radio GPS receivers do not create physical macroscopic magnetic fields.
✕ **(D):** Background location services operate on separate threads and do not block incoming cellular phone calls.

Question 55 · Third-Party Tracking Cookies and Cross-Site Web Beacons

Priority ★★★★

CORRECT ANSWER: (C) **Third-Party Tracking Cookies and Web Beacons (Pixel Tags) embedded in third-party advertising scripts across multiple websites, allowing the ad network to build a persistent browsing profile tied to a unique browser identifier.**

- Analyze cross-site tracking mechanisms:
- • When multiple independent websites embed tracking scripts or invisible 1x1 tracking pixels from the same commercial advertising network (like Google or Meta), the ad server reads and writes **third-party cookies** on the user's browser.
- • Each time the user loads a new site containing that ad network's code, the browser sends the tracking cookie back to the ad server.
- • The ad network stitches together a comprehensive chronological log of the user's web browsing activity across thousands of separate domains to serve hyper-targeted behavioral ads.

WHY OTHER CHOICES ARE INCORRECT:

✕ **(A):** Silicon RAM memory stores arbitrary binary bits and has no cognitive bias toward hiking boots.
✕ **(B):** Monitors are passive display hardware; display glass does not store or process digital data.
✕ **(D):** Web tracking is executed digitally in milliseconds via HTTP request headers, not physical postcards.

Question 56 · Device and Browser Fingerprinting Without Cookies

Priority ★★★★★

CORRECT ANSWER: (C) **Browser Fingerprinting, which queries browser APIs for unique configurations (such as screen resolution, installed system fonts, audio hardware context, canvas graphics rendering quirks, and browser plugins) to generate a statistically unique device fingerprint.**

- Examine Browser Fingerprinting techniques in modern web tracking:
- • Unlike cookies (which can be deleted), **Browser Fingerprinting** exploits the vast diversity of personal computing environments.
- • A client-side JavaScript script measures dozens of system attributes:
- - Exact operating system version and build
- - Installed system fonts and browser extensions
- - Canvas/WebGL rendering micro-differences caused by GPU hardware drivers
- - Audio buffer latency quirks
- • When combined, this configuration entropy is so high that only 1 in millions of computers shares the identical combination, creating a persistent, undeletable identifier.

WHY OTHER CHOICES ARE INCORRECT:

✕ **(A):** Fingerprinting happens entirely within browser JavaScript execution, not via orbital satellite imagery.
✕ **(B):** Browser fingerprinting is a software metadata calculation; it does not require physical biometric sensors.
✕ **(D):** Undersea cable maintenance involves subsea vessels, but periscopes cannot inspect encrypted packet contents inside glass fibers.

Question 57 · The Role of Commercial Data Brokers and Consumer Profiling

Priority ★★★★

CORRECT ANSWER: (A) They harvest, purchase, and aggregate data from public records, smartphone app trackers, loyalty store cards, and credit reporting bureaus to construct detailed consumer dossiers that they sell to advertisers, insurers, and political campaigns.

- Understand the commercial data broker industry:
- • Most consumers have never heard of major data brokers (e.g. Acxiom, Experian, LexisNexis), yet these firms hold thousands of data points on almost every adult.
- • Sources: Real estate property records, motor vehicle registries, mobile app telemetry, store discount cards, and web browsing logs.
- • Uses: Targeted commercial marketing, insurance risk underwriting, employment background checks, and predictive political voter mobilization.
- • The lack of consumer visibility and regulatory oversight creates major privacy and civil liberty concerns.

WHY OTHER CHOICES ARE INCORRECT:

✕ **(B):** Data brokers are for-profit data monetization enterprises, not rural broadband charities.
✕ **(C):** Data brokers rely on automated digital tracking and scraping, not manual in-person interviews.
✕ **(D):** Data brokers deal in digital information aggregation, not hardware computer manufacturing.

Question 58 · Consequences of Enterprise PII Data Breaches

Priority ★★★★★

CORRECT ANSWER: (D) Victims face long-term risks of **Identity Theft, where criminals use the permanent stolen PII to fraudulently open bank accounts, obtain credit cards, take out loans, and file fraudulent tax returns in the victims' names.**

- Analyze the long-term harms of PII data breaches (Equifax breach case study):
- • Unlike passwords (which can be changed in 30 seconds), core PII—like Social Security Numbers, dates of birth, and legal names—is permanent and virtually impossible to alter.
- • Once stolen and sold on dark web marketplaces, this data gives criminals everything necessary to impersonate the victim, bypass identity verification questions, destroy credit scores, and divert tax refunds.
- • Victims often spend years and thousands of dollars repairing the financial damage.

WHY OTHER CHOICES ARE INCORRECT:

✕ **(A):** Stolen server databases have no physical effect on victim smartphone battery chemistry.
✕ **(B):** Digital data theft does not transform physical paper currency into digital cryptocurrency.
✕ **(C):** Data breaches do not alter constitutional freedom of movement or residential relocation rights.

Question 59 · Modern Privacy Regulations: GDPR and CCPA Consumer Rights

Priority ★★★★

CORRECT ANSWER: (D) The right to know what personal data companies collect about them, the right to demand that companies delete their personal data (the 'Right to be Forgotten'), and the right to opt out of having their data sold.

- Understand modern data privacy legal frameworks in AP CSP:
- • **GDPR (EU) & CCPA (California):** Pioneer comprehensive consumer data privacy rights.
- • Core provisions:
- - **Right to Access / Transparency:** Companies must disclose what PII they collect and with whom they share it.
- - **Right to Erasure ('Right to be Forgotten'):** Consumers can demand permanent deletion of their personal records.
- - **Consent Requirements:** Requires explicit opt-in for data processing and gives users the right to forbid data sales.
- - Heavy financial penalties for non-compliant corporations.

WHY OTHER CHOICES ARE INCORRECT:

✕ **(A):** Deploying malware remains a serious criminal offense worldwide; privacy laws provide legal rights, not cyberattack authorization.
✕ **(B):** Privacy regulations do not subsidize physical utility power bills.
✕ **(C):** Privacy laws regulate data governance; they do not provide free consumer electronics.

Question 60 · Evaluating Legitimate Need vs. Privacy Overreach in App Permissions

Priority ★★★★★

CORRECT ANSWER: (C) Basic arithmetic calculations require only local CPU mathematical operations; demanding access to the user's GPS location, private address book, and camera indicates unauthorized data harvesting unrelated to the app's functionality.

- Apply the Principle of Least Privilege to mobile application permissions:
- • **Principle of Least Privilege:** An application should only be granted access to the specific resources and data strictly necessary to perform its legitimate function.
- • Evaluating App 3:
- - Performing $5 + 5$ or calculating percentages executes locally in volatile memory.
- - It has zero technical need to know where the user is standing (GPS), who the user's friends are (Contacts), or what the user looks like (Camera).
- • Requesting these invasive permissions reveals that the calculator is spyware designed to harvest user PII for ad broker monetization.

WHY OTHER CHOICES ARE INCORRECT:

✕ **(A):** Modern processors perform billions of numerical operations per second; number volume is not a permission issue.
✕ **(B):** Software calculators fully support division, multiplication, addition, and subtraction.
✕ **(D):** Software calculator apps are standard across all mobile operating systems.

Question 61 · The Fundamental Tension: Digital Privacy vs. Public Safety

Priority ★★★★★

CORRECT ANSWER: (C) Robust end-to-end encryption shields innocent citizens and dissidents from authoritarian surveillance and identity theft, but simultaneously prevents law enforcement from intercepting communications between terrorists and violent criminals.

- Examine the privacy vs. security policy trade-off in AP CSP:
- • **Pro-Privacy Perspective:** Unbreakable end-to-end encryption protects human rights, journalists, confidential healthcare, banking integrity, and personal privacy from rogue hackers and corrupt governments.
- • **Pro-Security Perspective:** Law enforcement agencies argue that 'going dark' makes it impossible to monitor illicit drug cartels, human trafficking rings, or terror plots even with a court-approved judicial warrant.
- • This irreconcilable tension is at the heart of modern technology legislation worldwide.

WHY OTHER CHOICES ARE INCORRECT:

✕ **(A):** Intelligence and defense agencies operate some of the world's largest supercomputing data centers.
✕ **(B):** The debate is over software cryptography and legal surveillance policies, not physical paint colors.
✕ **(D):** 'Nothing to hide, nothing to fear' is a recognized fallacy; law-abiding citizens require privacy to protect finances, medical records, and freedom of thought.

Question 62 · The Cryptographic Impossibility of 'Safe' Encryption Backdoors

Priority ★★★★★

CORRECT ANSWER: (B) Mathematically, there is no way to create an encryption backdoor that only 'good people' can use; any intentional architectural vulnerability or master key will inevitably be discovered, leaked, or exploited by hostile cybercriminals and foreign adversaries.

- Understand the computer science consensus on encryption backdoors:
- • Mathematics does not recognize legal jurisdictions, intent, or badges.
- • If an algorithm is weakened with an intentional vulnerability (a backdoor) or master escrow key, that vulnerability exists in the mathematics.
- • History proves that master keys and leaked agency exploits (e.g., EternalBlue stolen from the NSA and weaponized into the WannaCry ransomware) will inevitably be leaked or brute-forced by adversaries, leaving the entire global banking and civilian infrastructure dangerously compromised.

WHY OTHER CHOICES ARE INCORRECT:

✕ **(A):** Cryptographic software routines have no physical mechanical impact on display glass integrity.
✕ **(C):** Backdoor debates center on software code compliance, not shutting down all consumer manufacturing.
✕ **(D):** Cryptographic algorithms perform mathematical modular arithmetic without increasing hardware voltage.

Question 63 · Mass Bulk Data Collection and Telephony Metadata

Priority ★★★★★

CORRECT ANSWER: (B) Metadata patterns over time reveal deeply intimate personal facts—such as calls to an addiction helpline at 2:00 AM, followed by a divorce attorney at 9:00 AM, and an oncologist at 11:00 AM—painting an unmistakable picture of an individual's private life without hearing a single word spoken.

- Evaluate the privacy invasiveness of communications metadata:
- • Metadata is data *about* data (phone numbers, timestamps, duration, cell tower geolocation).
- • Former NSA General Counsel Stewart Baker famously remarked: '*Metadata absolutely tells you everything about somebody's life. If you have enough metadata, you don't really need content.*'
- • Analyzing connection graphs and call frequency provides automated, machine-readable behavioral profiles that are far easier to query and datamine at scale than unstructured audio recordings.

WHY OTHER CHOICES ARE INCORRECT:

✕ **(A):** Digital telecommunications metadata remains electronic database entries.
✕ **(C):** Metadata is processed using automated supercomputer graph analytics and machine learning algorithms.
✕ **(D):** Metadata collection occurs silently at telecom routing switches without causing speaker static.

Question 64 · Constitutional Law and Digital Search Warrants: Riley v. California

Priority ★★★★

CORRECT ANSWER: (D) A modern smartphone is not a mere physical container, but a digital repository holding millions of pages of personal text messages, financial records, medical histories, photos, and location tracking spanning years, holding vastly more intimate information than an entire physical house.

- Analyze the constitutional dimension of digital privacy in AP CSP:
- • Chief Justice John Roberts noted: *'Modern cell phones are now such a pervasive and insistent part of daily life that the proverbial visitor from Mars might conclude they were an important feature of human anatomy.'*
- • Traditional search incident to arrest allowed police to look inside a physical wallet or cigarette pack for weapons or evidence.
- • But a smartphone contains gigabytes of private thoughts, search queries, intimate photos, and location histories—far more data than could ever be found in a physical home search.
- • Therefore, the Fourth Amendment mandates a judicial warrant based on probable cause.

WHY OTHER CHOICES ARE INCORRECT:

✕ **(A):** Forensic software tools extract phone data automatically without manual assembly programming.
✕ **(B):** Smartphones are standard consumer electronics without self-destruct mechanisms.
✕ **(C):** Maritime law has no jurisdiction over terrestrial smartphone digital searches.

Question 65 · Chilling Effect of Public Facial Recognition Surveillance

Priority ★★★★★

CORRECT ANSWER: (D) Citizens may become fearful of participating in peaceful political protests, attending controversial religious services, or visiting legal medical clinics because their presence is permanently logged in a government tracking database.

- Examine the 'Chilling Effect' on democratic participation:
- • Historically, walking in public afforded practical **anonymity in a crowd**.
- • Continuous, automated facial recognition strips away this public anonymity.
- • When citizens know that attending a labor union rally, civil rights protest, or controversial political gathering will permanently link their biometric identity to a government police dossier, they alter their behavior, self-censor, or stay home.
- • This 'chilling effect' undermines fundamental democratic freedoms of speech and assembly.

WHY OTHER CHOICES ARE INCORRECT:

✕ **(A):** Municipal surveillance cameras have no relationship to footwear regulations.
✕ **(B):** Passive optical camera sensors capture light; they do not physically alter human facial anatomy.
✕ **(C):** Outdoor security cameras are enclosed in weatherproof heated housings that withstand sub-zero temperatures.

Question 66 · Automated License Plate Readers (ALPR) and Pervasive Vehicular Tracking

Priority ★★★★

CORRECT ANSWER: (C) It allows government agencies and private investigators to reconstruct the complete historical travel routes, habits, and associations of innocent citizens who are suspected of no crime.

- Analyze the surveillance implications of Automated License Plate Readers (ALPR):
- • When ALPRs only scanned for stolen cars on a local 'hotlist', privacy impacts were limited.
- • However, modern ALPR networks store billions of scans of *all* vehicles indefinitely.
- • By querying a license plate in an ALPR database, an analyst can see everywhere a citizen drove over the past three years: their doctor visits, romantic partners, political meetings, and worship centers.
- • This amounts to warrantless, retroactive location tracking of the entire population.

WHY OTHER CHOICES ARE INCORRECT:

✕ **(A):** License plates remain standard stamped metal plates visible to the human eye.
✕ **(B):** Optical cameras photograph passing vehicles; they have zero interaction with internal engine oil.
✕ **(D):** ALPRs capture surveillance imagery; they do not levy municipal turn fees.

Question 67 · Workplace Digital Surveillance and 'Bossware'

Priority ★★★★

CORRECT ANSWER: (B) Increased employee stress, burnout, and anxiety, as workers feel constantly distrusted and perform performative busywork (like jiggling the mouse) rather than focusing on creative, high-value problem solving.

- Evaluate the human and psychological impacts of workplace digital surveillance ('Bossware'):
- • Excessive digital surveillance destroys workplace trust and psychological safety.
- • Measuring employee value by raw keystroke counts or mouse movements penalizes thoughtful reflection, reading, and creative planning.
- • It leads to profound workplace stress, high turnover rates, and invasive privacy violations as home webcam captures record family members and private domestic spaces.

WHY OTHER CHOICES ARE INCORRECT:

✕ **(A):** Surveillance software has no biological effect on human circadian sleep needs.
✕ **(C):** Keyboard usage does not cause chemical bonding between human skin and plastic keys.
✕ **(D):** Background monitoring software consumes bandwidth uploading logs; it does not accelerate network throughput.

Question 68 · Predictive Policing Software and Self-Fulfilling Feedback Loops

Priority ★★★★

CORRECT ANSWER: (B) Because police are dispatched more heavily to those specific blocks, they naturally observe and make more arrests there, which feeds new arrest records back into the model, causing the algorithm to predict even more crime in those same neighborhoods.

- Analyze systemic algorithmic feedback loops in computational sociology:
- • Predictive policing systems (like PredPol) confuse *crime rate* with *policing rate*.
- • If police spend 80% of their time patrolling Neighborhood A, they will naturally detect more low-level infractions (e.g. jaywalking, drug possession) in Neighborhood A than in unpatrolled Neighborhood B.
- • The database records more arrests in Neighborhood A, reinforcing the algorithm's prediction, which sends even more officers there.
- • The model does not predict crime; it predicts where police will go to make arrests, codifying systemic bias as mathematical truth.

WHY OTHER CHOICES ARE INCORRECT:

✕ **(A):** Predictive software does not affect radio hardware communications.
✕ **(C):** Software models output patrol route coordinates; they do not physically move architectural structures.
✕ **(D):** Predictive routing directs police cars; it has no role in criminal justice vocational training.

Question 69 · Psychographic Profiling, Micro-Targeting, and Democratic Integrity

Priority ★★★★★

CORRECT ANSWER: (A) Voters were segmented and served hyper-targeted, fear-based political advertisements and disinformation tailored precisely to their specific psychological vulnerabilities and anxieties to depress voter turnout or manipulate vote choices.

- Analyze the impact of psychographic profiling on democratic systems:
- • By combining harvested Facebook data (likes, friendships, quizzes) with demographic records, political firms built detailed psychological models of individual voters.
- • **Micro-targeting:** Rather than running broad public campaign ads subject to journalistic scrutiny, campaigns delivered dark, personalized digital ads:
- - Fearful voters received alarmist messaging about crime.
- - Specific voter groups received misleading messages designed to suppress their motivation to vote.
- • This weaponization of personal data poses grave threats to informed democratic deliberation.

WHY OTHER CHOICES ARE INCORRECT:

✕ **(B):** Constitutional law requires human candidates; servers cannot be elected to public office.
✕ **(C):** Psychographic targeting influences voter sentiment; it cannot erase human memory.
✕ **(D):** Physical paper ballots inside sealed boxes cannot be altered by wireless radio signals.

Question 70 · Comprehensive Synthesis: Surveillance Technologies and Civil Liberties

Priority ★★★★★

CORRECT ANSWER: (D) Every surveillance technology entails profound trade-offs between public security and fundamental civil liberties; responsible computer scientists must analyze both technical capabilities and societal consequences when designing and deploying systems.

- Synthesize Big Idea 5 ethical and societal principles:
- • Technology is never deployed in a moral vacuum.
- • Every innovation in the table presents a genuine public safety rationale, balanced against a severe constitutional or human rights peril.
- • The core mission of the AP CSP curriculum is to develop technologists who understand that engineering choices have profound human consequences, requiring ethical accountability, democratic oversight, and rigorous algorithmic audits.

WHY OTHER CHOICES ARE INCORRECT:

✕ **(A):** The IETF develops open technical internet protocols; it has no authority over constitutional law.
✕ **(B):** All four technologies are software-driven computational systems relying on advanced algorithms.
✕ **(C):** Professional software engineering codes of ethics (such as the ACM Code of Ethics) mandate that computing professionals prioritize human wellbeing and ethical impacts.

Question 71 · Social Engineering and the Human Factor in Cybersecurity

Priority ★★★★★

CORRECT ANSWER: (D) The psychological manipulation of human individuals into divulging confidential information, clicking malicious links, or performing actions that compromise security systems.

- Define Social Engineering in AP CSP Big Idea 5 (IOC-2.B):
- • **Social Engineering:** Exploiting human psychology rather than software technical vulnerabilities.
- • Attackers leverage emotions such as fear, urgency, curiosity, authority, or helpfulness (e.g., impersonating an IT technician demanding a password to 'fix an urgent email crash').
- • Even the most mathematically secure military encryption cannot protect a system if a human administrator willingly gives away their master credentials to an impostor.
- • Robust security requires ongoing human training alongside technical firewalls.

WHY OTHER CHOICES ARE INCORRECT:

✕ **(A):** Robotics manufacturing relates to industrial hardware assembly, not cybersecurity manipulation.
✕ **(B):** Civil engineering builds bridges; social engineering manipulates human trust.
✕ **(C):** Text-to-speech synthesis is accessibility software, not social engineering.

Question 72 · Mechanics and Indicators of Phishing Attacks

Priority ★★★★★

CORRECT ANSWER: (B) A **Phishing attack, designed to trick the student into entering their private banking username and password on an attacker-controlled clone website.**

- Analyze the mechanics of Phishing attacks in AP CSP:
- • **Phishing:** The fraudulent attempt to obtain sensitive information (credentials, credit card numbers) by disguising as a trustworthy entity in an electronic communication.
- • Classic hallmarks of phishing:
- - Manufactured artificial urgency ('Account suspended in 24 hours!')
- - Generic greetings or emotional pressure
- - Deceptive spoofed domain URLs (`secure-bank-login-update.biz` instead of `bank.com`)
- • Once the victim enters their credentials, the attacker logs the password and accesses the real account.

WHY OTHER CHOICES ARE INCORRECT:

✕ **(A):** Citizen science crowdsources research data; it does not steal private banking credentials.
✕ **(C):** Legitimate OS updates are delivered through operating system system panels, not suspicious urgent emails.
✕ **(D):** Banks never demand that users click external links to verify account passwords under threat of instant deletion.

Question 73 · Spear-Phishing: Targeted Social Engineering

Priority ★★★★★

CORRECT ANSWER: (B) **Spear-Phishing, which is a highly targeted, personalized phishing attack directed at a specific individual or organization using researched personal information.**

- Distinguish Spear-Phishing from Generic Phishing:
- • **Generic Phishing:** Casts a wide net with identical generic lures to millions of random users (e.g., 'Dear customer, click here').
- • **Spear-Phishing:** Highly customized and dangerous. The attacker spends days researching the victim on LinkedIn, social media, and corporate press releases.
- • By mentioning real colleagues, conferences, and believable project names, the email easily bypasses suspicion, tricking high-value executives into clicking malicious malware payloads or approving fraudulent wire transfers.

WHY OTHER CHOICES ARE INCORRECT:

✕ **(A):** Software license audits inspect code legality, not cyberattack penetration.
✕ **(C):** DDoS attacks flood network ports with junk traffic; they do not craft personalized executive emails.
✕ **(D):** Physical cable severance physically cuts wires; spear-phishing manipulates humans digitally.

Question 74 · Rogue Wi-Fi Access Points and Evil Twin Exploits

Priority ★★★★★

CORRECT ANSWER: (A) The attacker can eavesdrop on and capture all unencrypted data packets transmitted by the laptop (packet sniffing), intercept cleartext passwords, and redirect web queries to malicious phishing sites.

- Analyze Rogue Access Point (Evil Twin) wireless attacks:
- • A **Rogue Access Point** broadcasts a familiar, trusted network name (SSID) in a public venue to lure victims into connecting.
- • Because the attacker controls the router, all passenger traffic flows through the attacker's network interface.
- • The attacker can use packet inspection tools (like Wireshark) to harvest unencrypted credentials, record visited domains via DNS snooping, and inject malicious scripts.
- • Defenses: Avoid public open Wi-Fi, enforce HTTPS everywhere, and tunnel all traffic through an encrypted VPN.

WHY OTHER CHOICES ARE INCORRECT:

✕ **(B):** Connecting to a network does not permanently change the color physics of display panels.
✕ **(C):** Network signals cannot vaporize physical silicon or aluminum laptop hardware.
✕ **(D):** Wi-Fi hotspots interact with connecting user devices, not heavy industrial railway locomotive controls.

Question 75 · Keyloggers: Keystroke Logging Malware and Hardware Devices

Priority ★★★★★

CORRECT ANSWER: (C) A **Hardware Keylogger; it captures passwords, usernames, and private messages at the physical input level before the operating system or software encryption can protect the data.**

- Examine the operation and threats of Keyloggers in AP CSP:
- • **Keyloggers (Hardware or Software):** Log every discrete keystroke typed on a keyboard.
- • Why keyloggers are exceptionally dangerous:
- - Encryption (like HTTPS) protects data *after* it leaves the application.
- - But a keylogger captures credentials *at the moment of input* as the human types them.
- • Hardware keyloggers cannot be detected by antivirus software because they operate transparently at the physical electrical interface.
- • Defenses: Visual inspection of public PC ports, virtual on-screen scrambled keyboards, and Multi-Factor Authentication (MFA).

WHY OTHER CHOICES ARE INCORRECT:

✕ **(A):** The device records private keystrokes maliciously; it has no sanitary cleaning properties.
✕ **(B):** USB keyboard interceptors are not satellite telecommunications routers.
✕ **(D):** The dongle intercepts keyboard scan codes; it does not accelerate mouse velocity.

Question 76 · Ransomware Attacks and Critical Infrastructure Extortion

Priority ★★★★★

CORRECT ANSWER: (D) **Ransomware, which locks and encrypts a victim's critical data, holding it hostage until an extortion ransom is paid.**

- Analyze the mechanics and societal harms of Ransomware:
- • **Ransomware:** A category of extortion malware that uses strong symmetric/asymmetric encryption to deny access to user files or entire operating systems.
- • Attacks on healthcare systems, city governments, and energy pipelines (e.g. Colonial Pipeline) endanger human lives by canceling emergency surgeries and halting critical municipal infrastructure.
- • Primary defenses:
- - Maintaining air-gapped, offline, immutable data backups.
- - Strict network segmentation and employee phishing training.
- - Rapid incident response and zero-trust access architecture.

WHY OTHER CHOICES ARE INCORRECT:

✕ **(A):** Spreadsheets organize tabular data; they do not extort hospitals using cryptographic locks.
✕ **(B):** Broadband subsidies are government funding initiatives for digital equity.
✕ **(C):** Galaxy classification is citizen science (Galaxy Zoo), not extortion malware.

Question 77 · Distinguishing Malware Typologies: Trojan, Virus, and Worm

Priority ★★★★

CORRECT ANSWER: (D) A **Trojan Horse, which disguises itself as desirable, legitimate software while concealing malicious hidden functionality.**

- Classify malware typologies according to AP CSP standards:
- • **Trojan Horse (Trojan):** Named after the ancient Greek myth; software that misleads users of its true intent by masquerading as a legitimate, useful program (game, utility, cracked software) while concealing a hostile payload (backdoor, spyware).
- • **Virus:** Malicious code that attaches itself to an existing host file and replicates only when executed by a user.
- • **Worm:** Standalone malware that self-replicates and spreads autonomously across networks by exploiting software vulnerabilities without human intervention.
- • The video game installer in the scenario is a classic Trojan Horse.

WHY OTHER CHOICES ARE INCORRECT:

✕ **(A):** Digital certificates verify authentic cryptographic identities; they do not install webcam backdoors.
✕ **(B):** Electrical fuses are passive circuit safety hardware, not deceptive software programs.
✕ **(C):** Creative Commons licenses govern legal copyright permissions, not malicious executables.

Question 78 · Cryptojacking: Unauthorized Computational Resource Hijacking

Priority ★★★★

CORRECT ANSWER: (A) **Cryptojacking, the unauthorized hijacking of a victim's computing hardware (CPU/GPU) and electricity to mine cryptocurrency for an attacker.**

- Understand Cryptojacking as an emerging computational threat vector:
- • **Cryptojacking:** Attackers inject malicious JavaScript into compromised websites or install background malware that turns victim devices into cryptocurrency miners.
- • While it does not typically steal or delete personal files, it robs the victim of valuable resources:
- - Maxes out CPU and GPU utilization.
- - Drastically shortens hardware lifespan through sustained high temperatures.
- - Inflates electrical utility bills and paralyzes user multitasking performance.
- • Defenses: Browser ad-blockers, script blockers, and monitoring process CPU usage.

WHY OTHER CHOICES ARE INCORRECT:

✕ **(B):** Cryptojacking severely slows down user computer processing; it does not accelerate hardware.
✕ **(C):** Compression algorithms optimize storage space without continuously red-lining CPU cores for financial theft.
✕ **(D):** Software code cannot physically wipe dust particles from mechanical plastic fan blades.

Question 79 · Watering Hole Cyberattacks

Priority ★★★★

CORRECT ANSWER: (A) A **Watering Hole Attack, which targets a specific group of users by compromising a trusted third-party website that the targets are known to visit regularly.**

- Analyze Watering Hole cyberattack strategies:
- • The metaphor comes from a predator waiting by a watering hole where prey gathers to drink.
- • When high-value targets (defense contractors, banks, government agencies) have impenetrable direct perimeter defenses, attackers observe which secondary websites their employees frequent (local restaurants, trade news blogs).
- • The attackers compromise the weaker third-party website and wait for the targets to visit, exploiting browser vulnerabilities to establish an initial foothold inside the enterprise network.

WHY OTHER CHOICES ARE INCORRECT:

✕ **(B):** Scientific publishing disseminates research openly; it does not deploy stealth malware.
✕ **(C):** The attack is executed entirely through digital web code injection, not physical bank burglary.
✕ **(D):** Spell checkers correct typography; they do not compromise web servers with zero-day exploits.

Question 80 · Comprehensive Synthesis of Cybersecurity Threat Vectors

Priority ★★★★★

CORRECT ANSWER: (C) Cybersecurity requires **Defense-in-Depth—a multi-layered defense combining technical cryptography, network security architecture, automated monitoring, and continuous human user education.**

- Synthesize Defense-in-Depth across cybersecurity threat vectors:
- • No single software tool or policy can defend against all cyber threats.
- • **Defense-in-Depth:** A holistic security architecture where multiple layered safeguards protect the system:
- - Human layer (phishing awareness, credential hygiene).
- - Physical layer (securing ports against hardware keyloggers).
- - Cryptographic layer (VPNs against rogue Wi-Fi, MFA against credential theft).
- - Operational layer (immutable offline backups against ransomware).
- • If one defensive layer fails, adjacent layers prevent catastrophic compromise.

WHY OTHER CHOICES ARE INCORRECT:

✕ **(A):** Leaving written passwords on physical sticky notes creates massive physical security vulnerability.
✕ **(B):** All four threat vectors are documented, real-world attack vectors responsible for billions in global economic losses.
✕ **(D):** Antivirus software is only one component of defense and cannot protect against social engineering, hardware keyloggers, or rogue Wi-Fi.

Question 81 · Multi-Factor Authentication (MFA) and Authentication Factors

Priority ★★★★★

CORRECT ANSWER: (B) It combines two fundamentally different categories of authentication factors: 'something you know' (the password) and 'something you have' (the physical smartphone running the authenticator application).

- Analyze Multi-Factor Authentication (MFA) under AP CSP Big Idea 5 (IOC-2.B):
- • **Authentication Factor Categories:**
- 1. *Something you know:* Knowledge factor (e.g., password, PIN, mother's maiden name).
- 2. *Something you have:* Possession factor (e.g., physical smartphone, hardware security key like a YubiKey, smartcard).
- 3. *Something you are:* Inherence factor (e.g., biometric fingerprint, facial geometry, iris scan).
- • **Why Two of the Same Factor is NOT MFA:** Requiring a password *plus* a security question is only 'Two-Step' using *one* factor (both are 'something you know'). If an attacker steals your notes or phishes you, both are compromised.
- • **True MFA:** Combines at least two distinct categories. Even if an attacker steals your password across the internet, they cannot log in without physically possessing your hardware token or phone.

WHY OTHER CHOICES ARE INCORRECT:

✕ **(A):** MFA operates at the identity and access management layer; it works alongside encryption rather than eliminating it.
✕ **(C):** MFA enhances protection and never stores or transmits passwords in unencrypted plain text.
✕ **(D):** Using two items from the same category (e.g., two passwords) is two-step verification of a single factor, not true multi-factor authentication.

Question 82 · Password Security: Combinatorial Entropy and Brute-Force Resistance

Priority ★★★★★

CORRECT ANSWER: (B) Password search space (combinatorial entropy) increases exponentially with string length, making longer passphrases exponentially more resistant to automated brute-force cracking tools while remaining easier for humans to remember.

- Examine Password Security and Mathematical Entropy in AP CSP:
- • **Brute-Force Search Space Formula:** The total possible combinations equal C^L, where C is the size of the character set and L is password length.
- • **Exponential Growth with Length:** Increasing length (L) increases entropy far faster than slightly expanding the character set (C):
- - An 8-character password from an expanded 94-character set has $94^8 \approx 6.1 \times 10^{15}$ combinations (crackable in seconds to minutes on modern multi-GPU rigs).
- - A 16-character passphrase from a 7,776-word dictionary has $7776^4 \approx 3.6 \times 10^{15}$ just from 4 words, and with 16 characters from lowercase letters alone has $26^{16} \approx 4.36 \times 10^{22}$ combinations.
- • **Human Usability:** Complex short passwords lead users to write them on sticky notes; memorable long passphrases offer superior entropy and practical usability.

WHY OTHER CHOICES ARE INCORRECT:

✕ **(A):** Attackers specifically utilize automated dictionary and hybrid rule-based attacks; they are not constrained by any laws.
✕ **(C):** Passphrases authenticate users; they do not alter or disable standard HTTP/network communications.
✕ **(D):** 8-character passwords with special symbols are easily cracked by modern automated GPU-accelerated hash-cracking software within hours.

Question 83 · Credential Stuffing and the Danger of Password Reuse Across Services

Priority ★★★★★

CORRECT ANSWER: (D) A **Credential Stuffing attack, enabled by users reusing the exact same username and password combination across multiple unrelated websites and services.**

- Analyze Credential Stuffing in AP CSP Big Idea 5 (IOC-2.B):
- • **Credential Stuffing:** An automated cyberattack where threat actors take massive databases of compromised username/password pairs stolen from one breached site and systematically test them against hundreds of other popular web services (banking, social media, shopping).
- • **The Vulnerability (Password Reuse):** Because many users find memorizing dozens of passwords difficult, they use the identical password across many websites.
- • **The Consequence:** A security flaw in a low-importance website (e.g., a small hobby forum) instantly compromises the user's high-stakes accounts (email, banking).
- • **Mitigation:** Using a password manager to generate distinct, unique credentials for every single account, combined with multi-factor authentication (MFA).

WHY OTHER CHOICES ARE INCORRECT:

✕ **(A):** The attack occurs purely through automated network authentication requests, not physical cable alteration.
✕ **(B):** Citizen science engages volunteer computation or observation for scientific discovery, not illegal account hijacking.
✕ **(C):** Electromagnetic pulses are physical phenomena, not automated software-based credential abuse.

Question 84 · Biometric Authentication: Benefits, Limitations, and Irrevocability

Priority ★★★★

CORRECT ANSWER: (A) Unlike passwords, biometric identifiers (such as fingerprints or facial geometry) cannot be reset or revoked once the underlying digital template data is compromised or breached.

- Evaluate Biometric Authentication Trade-offs in AP CSP:
- • **Advantages:** High user convenience, cannot be forgotten, left at home, or easily guessed via dictionary attacks.
- • **The Irrevocability Hazard:** If a password is leaked in a database breach, you simply change your password to a new string. However, you cannot change your biological fingerprint, iris pattern, or facial structure. Once biometric hash templates are stolen, that credential is permanently compromised.
- • **False Match / False Rejection Rates:** Biometric systems rely on probability thresholds rather than exact character matching, introducing potential false acceptances (impostors accepted) or false rejections (legitimate user locked out).
- • **Legal Compulsion:** In many jurisdictions, law enforcement can compel physical biometric unlocking more easily than compelling a user to reveal a memorized passcode.

WHY OTHER CHOICES ARE INCORRECT:

✕ **(B):** Sensors require active power and the device operating system to capture and verify biometric scans.
✕ **(C):** Biometrics are processed locally on device hardware security enclaves without requiring government fiber connections.
✕ **(D):** Biometric authentication is an access control mechanism; it does not grant image copyrights to hardware vendors.

Question 85 · Cryptographic Password Storage: One-Way Hashing and Salting

Priority ★★★★★

CORRECT ANSWER: (C) It ensures that two users with identical passwords will produce completely different stored hash values, preventing attackers from using precomputed Rainbow Tables to instantly crack large batches of passwords.

- Understand Password Hashing and Salting in AP CSP:
- • **Cryptographic One-Way Hash:** Converts arbitrary input into a fixed-length string (digest). It is computationally infeasible to invert (cannot calculate original password from hash).
- • **The Rainbow Table Threat:** Attackers precompute millions of hashes for common passwords (e.g., hash of 'password123'). If a stolen database contains unsalted hashes, looking up matches takes milliseconds.
- • **Role of the Cryptographic Salt:**
- - A salt is a unique, random string appended to each password *before* hashing (`hash(password + salt)`).
- - Even if 1,000 users have the password 'P@ssword1', each receives a unique salt, generating 1,000 distinct hash outputs.
- - Precomputed tables (Rainbow Tables) become useless; the attacker must calculate hashes individually for each unique salt, drastically increasing computational cost.

WHY OTHER CHOICES ARE INCORRECT:

✕ **(A):** Cryptographic hash functions are strictly one-way mathematical operations; salting does not allow reversing hashes to plain text.
✕ **(B):** Salting secures passwords inside private databases; it never publishes them to public search engines.
✕ **(D):** Hashing with salts adds necessary cryptographic security checks; it does not disable verification.

Question 86 · Password Managers and Modern Credential Hygiene

Priority ★★★★★

CORRECT ANSWER: (B) It generates, encrypts, and stores unique, high-entropy passwords for every distinct website, requiring the user to memorize only a single strong master passphrase.

- Examine Password Managers in AP CSP Cybersecurity Principles:
- • **Core Problem:** Humans cannot memorize dozens of 20-character random passwords, leading to weak passwords and dangerous password reuse.
- • **Password Manager Architecture:**
- - Uses strong local cryptographic encryption (e.g., AES-256) to secure a credential database ('vault').
- - The user only needs to memorize one strong, high-entropy master passphrase.
- - Generates cryptographically pseudorandom passwords of arbitrary length for every site.
- • **Anti-Phishing Benefit:** Password managers match the stored domain URL against the browser's active address bar. If a user lands on a phishing site (`g00gle.com`), the manager detects the domain mismatch and refuses to autofill credentials.

WHY OTHER CHOICES ARE INCORRECT:

✕ **(A):** Password managers use zero-knowledge encryption; transmitting credentials in unencrypted plain text email would create a catastrophic vulnerability.
✕ **(C):** Password managers manage client authentication; they cannot prevent remote server hardware failures.
✕ **(D):** Password managers are client-side credential tools; they do not alter remote website server firewalls.

Question 87 · Network Security on Untrusted Networks: HTTPS and Virtual Private Networks (VPNs)

Priority ★★★★★

CORRECT ANSWER: (B) It creates an encrypted tunnel that encapsulates all data packets transmitted between the employee's computer and the VPN server, preventing local eavesdroppers from viewing packet payloads and browsing activity.

- Analyze Untrusted Networks and VPN Security in AP CSP Big Idea 5 (IOC-2.B):
- • **Public Wi-Fi Vulnerabilities:** Open Wi-Fi networks transmit wireless radio packets in the clear. Anyone within radio range running packet sniffing software (e.g., Wireshark) can capture unencrypted packets.
- • **Role of HTTPS:** Encrypts application-layer data between browser and server, but local eavesdroppers can still observe destination IP addresses and domain name lookups (DNS).
- • **Role of a VPN:**
- - Encapsulates *all* network traffic from the device inside an encrypted tunnel.
- - The local coffee shop router only sees encrypted gibberish traveling between the device and the VPN gateway server.
- - Shields DNS queries, destination websites, and all non-web application traffic from local snooping.

WHY OTHER CHOICES ARE INCORRECT:

✕ **(A):** VPNs use cryptographic digital packet tunneling over IP networks, not public analog FM radio.
✕ **(C):** A VPN operates purely at the software networking level; it does not physically cut local router power lines.
✕ **(D):** VPN is a software networking protocol; it does not block battery electrical power.

Question 88 · Software Updates, Patch Management, and Vulnerability Lifecycle

Priority ★★★★

CORRECT ANSWER: (A) When a software vendor releases a security patch, malicious actors analyze the patch code to discover the underlying vulnerability and immediately launch automated attacks against unpatched systems.

- Analyze Patch Management and Vulnerabilities in AP CSP:
- • **Vulnerability Disclosure Window:** When a vendor issues a patch, they publicly describe or fix a specific flaw (e.g., a buffer overflow or remote code execution bug).
- • **Reverse-Engineering Patches:** Cybercriminals inspect the patch difference ('diff') to see exactly what code was modified, allowing them to reverse-engineer an exploit within hours or days.
- • **Automated Scanners:** Attackers deploy automated botnets scanning the public internet for servers and workstations running older, unpatched software versions.
- • **Zero-Day Vulnerability:** A flaw exploited by attackers *before* the vendor develops a patch. Once a patch exists, failure to install it converts a known vulnerability into an easy target.

WHY OTHER CHOICES ARE INCORRECT:

✕ **(B):** Unpatched devices face exploitation by hackers, not physical confiscation by regulatory agencies.
✕ **(C):** Security patches repair vulnerabilities in operating system components; they do not erase personal user files.
✕ **(D):** Software patching relates to binary logic and executable code; it does not degrade physical hardware display crystals.

Question 89 · Principle of Least Privilege in System Administration

Priority ★★★★

CORRECT ANSWER: (D) The **Principle of Least Privilege, which minimizes the potential damage ('blast radius') if an employee accidentally executes malware or falls victim to phishing.**

- Examine the Principle of Least Privilege in AP CSP System Security:
- • **Principle of Least Privilege (PoLP):** Every user, program, or system process should be granted only the minimum permissions strictly necessary to perform its legitimate function.
- • **Defense in Action:**
- - If a user running as 'Administrator' opens a malicious email attachment, the malware inherits administrative rights and can disable antivirus, install kernel rootkits, or encrypt system-wide backups.
- - If the same user runs under a restricted 'Standard User' account, the malware cannot write to system directories, modify OS files, or compromise other network users.
- • **Limiting Blast Radius:** Isolates compromised accounts and dramatically reduces lateral movement during cyber incidents.

WHY OTHER CHOICES ARE INCORRECT:

✕ **(A):** Hardware redundancy involves duplicate disks (RAID) for fault tolerance, not account permission limits.
✕ **(B):** Restricting administrator access protects system files; it does not broadcast keystrokes to public social media.
✕ **(C):** Access control accounts govern security rights; they do not overclock CPU clock frequencies.

Question 90 · Comprehensive Synthesis: Multi-Layer Cybersecurity Defenses

Priority ★★★★★

CORRECT ANSWER: (C) No single defensive control can mitigate all threat categories; an attacker can bypass MFA by exploiting an unpatched software vulnerability (Layer 3) or eavesdropping on unencrypted local network traffic (Layer 2) without needing user credentials.

- Synthesize Defense-in-Depth in AP CSP Big Idea 5 (IOC-2.B):
- • **Defense-in-Depth Strategy:** Cybersecurity is an asymmetric discipline. Attackers look for the single weakest link, whereas defenders must protect every vector.
- • **Limitations of Single Controls:**
- - MFA (Layer 1) protects user authentication against credential theft, but does not stop malware exploiting a remote code execution bug in an unpatched operating system (Layer 3).
- - Encryption (Layer 2) shields data in transit, but cannot prevent an employee from willingly transferring funds to a scammer during an executive impersonation spear-phishing attack (Layer 4).
- • **Layered Synergy:** Layering administrative policies, technical access controls, network encryption, and human training creates overlapping protections so that if one fails, others prevent catastrophe.

WHY OTHER CHOICES ARE INCORRECT:

✕ **(A):** Layered security strengthens protection; it does not justify disabling firewalls or eliminating passwords.
✕ **(B):** MFA is universally recommended by NIST and global cybersecurity agencies, not illegal.
✕ **(D):** Software patching updates code; it does not replace network transmission encryption (TLS/VPN).

Question 91 · Model Solution & Scoring Rubric

Algorithmic Bias Audit in AI Hiring Software ❖ Total Available Points: 9 Points

COMPREHENSIVE EXEMPLARY STUDENT RESPONSE:

Part A: Program Purpose and Computing Bias / Impact

Root Cause: The algorithm was trained on historical company hiring decisions spanning 10 years, during which human recruiters predominantly hired male and non-minority candidates. The algorithm optimized to reproduce historical patterns, treating past disparities as ground-truth criteria for success.

Demographic Proxies: Simply omitting explicit demographic labels (e.g., 'Gender' or 'Race') does not eliminate bias. The algorithm identified correlated proxy variables, such as attendance at women's colleges, participation in specific affinity organizations, or subtle linguistic markers in resume phrasing, effectively reconstructing protected categories.

Part B: Algorithmic Procedure & Data Mitigation

Data Preprocessing Mitigation: The engineering team must audit and scrub training datasets using feature selection to remove proxy variables with high demographic correlation. Furthermore, they should apply statistical re-weighting or synthetic oversampling to balance candidate representations across all experience bands before model training.

Adversarial Debiasing: Implement adversarial debiasing during neural network training, where an auxiliary predictor attempts to infer the candidate's demographic group from internal representations; the main model penalizes representations that allow demographic prediction.

Part C: Data Abstraction & Complexity Management

Data Abstraction: Define a structured record `Applicant` with fields `applicantID`, `qualificationsList`, `assignedScore`, and `cohortGroup`. Store all candidates in a dynamic list `applicantPool`.

Algorithmic Audit Procedure: Develop a procedure `AuditDisparateImpact(applicantPool, passThreshold)` that iterates through `applicantPool`, tallies total candidates and qualified candidates per `cohortGroup`, and calculates the selection rate for each group. If the selection rate for any group is less than 80% of the highest group's selection rate (Four-Fifths rule), the procedure flags the model and halts automated filtering.

Part D: Testing, Edge Cases & Ethical Governance

Ethical Responsibility & Accountability: Software engineers have a professional obligation under the ACM Code of Ethics to avoid harm and promote fairness. Companies must establish ongoing algorithmic oversight committees and mandate annual third-party algorithmic audits.

Human-in-the-Loop: Automated scoring should serve only as an advisory decision-support tool, not an autonomous gatekeeper. Candidates must retain the right to contest algorithmic rejections and request human review.

OFFICIAL COLLEGE BOARD 9-POINT SCORING RUBRIC:

Criteria / Evaluated Performance Skill	Points Allocated
Row 1: Bias Identification: Correctly identifies that the machine learning algorithm learned and reinforced historical human hiring disparities embedded in past training data.	**1 pt**
Row 2: Training Data Flaw: Explains how demographic proxy variables (e.g., college attended, extracurriculars, linguistic patterns) reflect historical bias even if gender/race fields are removed.	**1 pt**
Row 3: Technical Mitigation: Specifies a valid preprocessing intervention (e.g., re-weighting datasets, synthetic data balancing, or stripping proxy features).	**1 pt**
Row 4: Algorithmic Procedure: Describes a programmatic audit algorithm that calculates disaggregated acceptance rates across distinct applicant demographic cohorts.	**1 pt**
Row 5: Data Abstraction: Outlines a data abstraction (e.g., list of applicant candidate records with structured attributes) used to manage scoring evaluations.	**1 pt**
Row 6: Disparity Metric: Identifies a mathematical threshold or fairness metric (e.g., Four-Fifths rule, disparate impact ratio, or equal opportunity difference).	**1 pt**
Row 7: Human-in-the-Loop: Describes the role of human recruiters reviewing algorithmic edge cases rather than relying on full automation.	**1 pt**
Row 8: Societal Consequence: Explains the broader economic harm to underrepresented groups and loss of innovation when diverse talent is systematically excluded.	**1 pt**
Row 9: Corporate Accountability: Proposes an ongoing governance mechanism (e.g., annual third-party bias audits, public transparency reports, or regulatory compliance logs).	**1 pt**
TOTAL SCORING MAXIMUM	**9 / 9 Points**

Question 92 · Model Solution & Scoring Rubric

Privacy-Preserving Contact Tracing and Geolocation Tracking ❖ Total Available Points: 9 Points

COMPREHENSIVE EXEMPLARY STUDENT RESPONSE:

Part A: Program Purpose and Computing Bias / Impact

Centralized GPS vs Decentralized Bluetooth: Centralized GPS logging uploads continuous timestamped geographic coordinates to a central database. This reveals sensitive personal locations (home, medical clinics, protests) and creates a honeypot vulnerable to rogue government surveillance or data breaches.

Decentralized Proximity: Bluetooth Low Energy (BLE) beacons do not record physical location. Instead, phones broadcast anonymous random numbers to nearby devices within 2 meters, recording only proximity duration without ever knowing *where* the encounter occurred.

Part B: Algorithmic Procedure & Data Mitigation

Ephemeral Rolling Identifiers: The app uses a daily cryptographic seed key to derive pseudorandom Rolling Proximity Identifiers (RPIs) that change every 15 minutes. This prevents eavesdroppers from tracking a user as they move between locations.

Contact Exchange Algorithm: When Device A and Device B remain within BLE range for at least 15 minutes, they log each other's current RPI into an encrypted local list `localEncounterLog` stored solely on the device hardware.

Part C: Data Abstraction & Complexity Management

Local Exposure Matching: If a user tests positive, they voluntarily upload their daily diagnosis keys (not location logs) to a secure health authority server over TLS 1.3. Other devices regularly download the list of positive keys and run a local matching algorithm to determine if any stored RPI matches. No central authority knows who matched.

Data Expiration: A scheduled background routine deletes all records in `localEncounterLog` older than 14 days, reflecting the biological pathogen incubation window and eliminating unnecessary data retention.

Part D: Testing, Edge Cases & Ethical Governance

Digital Divide Barriers: Lower-income individuals, elderly citizens, and rural populations frequently use older smartphones that lack Bluetooth Low Energy hardware or cannot support modern operating system updates, leaving vulnerable communities unprotected.

Civic Trust and Network Effects: Epidemiological efficacy requires roughly 60% community adoption. Without ironclad transparency, open-source code verification, and strict legal guarantees against repurposing data for law enforcement, public skepticism suppresses adoption below useful thresholds.

OFFICIAL COLLEGE BOARD 9-POINT SCORING RUBRIC:

Criteria / Evaluated Performance Skill	Points Allocated
Row 1: Centralized GPS Risk: Accurately articulates the severe privacy risks of centralized GPS logging (continuous location tracking, state surveillance, database breach hazards).	**1 pt**
Row 2: Decentralized Bluetooth Concept: Explains how decentralized Bluetooth exchanges anonymous proximity beacons without recording spatial coordinates.	**1 pt**
Row 3: Ephemeral Key Generation: Describes the algorithmic generation of rotating pseudo-random cryptographic identifiers (rolling proximity identifiers) refreshed periodically.	**1 pt**
Row 4: Local Storage Verification: Outlines how exposure matching occurs locally on the user's smartphone device rather than on a central government server.	**1 pt**
Row 5: Data Transmission Security: Specifies secure cryptographic protocols (TLS/HTTPS) for downloading diagnosis keys from the public health server.	**1 pt**
Row 6: Data Retention & Expiration: Details an automated data purging policy (e.g., deleting rolling identifiers after 14 to 21 days corresponding to incubation periods).	**1 pt**
Row 7: Re-Identification Defense: Explains how the protocol defends against timing, linkage, or triangulation re-identification attacks.	**1 pt**
Row 8: Digital Divide Barriers: Identifies digital divide barriers (older phone models lacking BLE hardware, battery drain, lack of mobile connectivity).	**1 pt**
Row 9: Voluntary Adoption Threshold: Discusses the network effect requirement (e.g., 60%+ public adoption needed for epidemiological efficacy) and civic trust.	**1 pt**
TOTAL SCORING MAXIMUM	**9 / 9 Points**

Question 93 · Model Solution & Scoring Rubric

Crowdsourced Global Biodiversity and Wildlife Census Platform ❖ Total Available Points: 9 Points

COMPREHENSIVE EXEMPLARY STUDENT RESPONSE:

Part A: Program Purpose and Computing Bias / Impact

Scale and Distributed Coverage: Traditional marine surveys are constrained by limited research grants, small academic teams, and seasonal expeditions.

Crowdsourcing leverages tens of thousands of active divers, fishermen, and eco-tourists globally, generating year-round, real-time observational data across millions of square kilometers of ocean.

Cost-Effective Data Pipeline: Crowdsourcing transforms a logistically prohibitive undertaking into an asynchronous, distributed computational dataset, capturing sudden environmental events (e.g., coral bleaching spikes) immediately.

Part B: Algorithmic Procedure & Data Mitigation

Quality Control and Consensus Logic: To eliminate unreliable amateur observations, the platform implements a weighted consensus algorithm. When an observation is uploaded, it is assigned `status = 'Pending'`.

Reputation Algorithm: Contributor accounts have a `trustScore` between 0.0 and 1.0. When community members submit species identifications, votes are weighted by `trustScore`. If weighted agreement exceeds 85% across at least three independent reviewers, `status` updates to `'Research Grade'` and the contributor's `trustScore` increments.

Part C: Data Abstraction & Complexity Management

Anti-Poaching Threat: Publicly displaying exact GPS coordinates for rare or high-value marine life (e.g., seahorses, endangered sea turtles) creates a digital roadmap for illegal wildlife traffickers and poachers.

Spatial Obfuscation Algorithm: The platform maintains an `EndangeredTaxaList`. When an uploaded record matches an endangered species, the algorithm automatically strips high-precision EXIF GPS metadata and snaps the published coordinate to a randomized 10x10 kilometer grid cell, hiding the precise location while preserving regional ecological validity.

Part D: Testing, Edge Cases & Ethical Governance

Open Science Impact: Releasing the anonymized, aggregated dataset accelerates global climate and ecological research, allowing scientists worldwide to train climate resilience models without redundant data collection.

Creative Commons CC BY Licensing: CC BY ensures maximum utility—researchers, NGOs, and governments can freely copy, analyze, and remix the dataset, while legally mandating that they attribute the citizen science consortium and its volunteer community.

OFFICIAL COLLEGE BOARD 9-POINT SCORING RUBRIC:

Criteria / Evaluated Performance Skill	Points Allocated
Row 1: Crowdsourcing Scale: Explains how crowdsourcing achieves geographic scale and continuous longitudinal monitoring impossible for small scientific research teams.	**1 pt**
Row 2: Cost & Computation Efficiency: Details how distributed volunteer data collection drastically reduces financial costs and accelerates data ingestion.	**1 pt**
Row 3: Quality Control / Noise: Identifies the risk of misidentified submissions, blurred photos, or malicious vandalism inherent to crowdsourced inputs.	**1 pt**
Row 4: Consensus Algorithm: Specifies an algorithmic consensus rule (e.g., 3 independent verified expert reviews or 80% community agreement threshold).	**1 pt**
Row 5: Trust/Reputation Scoring: Incorporates a contributor reputation weighting score based on historical verification accuracy.	**1 pt**
Row 6: Poaching Vulnerability: Explains how publishing exact high-resolution GPS coordinates of endangered species enables commercial poaching or habitat destruction.	**1 pt**
Row 7: Spatial Obfuscation Algorithm: Describes an algorithmic spatial generalization procedure (e.g., fuzzing coordinates to a 10 km bounding box or removing EXIF GPS metadata).	**1 pt**
Row 8: Open Science Benefit: Articulates how open data enables global researchers to cross-reference climate change models and ecosystem dynamics.	**1 pt**
Row 9: CC BY Attribution Terms: Accurately specifies that CC BY permits free reuse, remixing, and commercial application provided the conservation platform is credited.	**1 pt**
TOTAL SCORING MAXIMUM	**9 / 9 Points**

Question 94 · Model Solution & Scoring Rubric

Creative Commons Multi-Media Educational Repository ❖ Total Available Points: 9 Points

COMPREHENSIVE EXEMPLARY STUDENT RESPONSE:

Part A: Program Purpose and Computing Bias / Impact

Copyright vs Creative Commons vs CC0: Traditional copyright applies automatically the instant a creative work is fixed in a tangible medium, granting exclusive 'All Rights Reserved' rights that prohibit copying or remixing without explicit written permission.

Creative Commons provides a standardized, legally binding framework allowing creators to grant advance permissions ('Some Rights Reserved'). CC0 (Public Domain Dedication) fully waives all copyright, allowing unrestricted use, commercialization, and modification without even requiring attribution.

Part B: Algorithmic Procedure & Data Mitigation

License Compatibility Matrix: A media remix that incorporates an asset marked **ND (NoDerivatives)** violates the license if modified or integrated into a new textbook chapter. Furthermore, combining a **CC BY-NC** asset with a **CC BY-SA** asset creates a licensing conflict because ShareAlike mandates releasing the entire derivative under CC BY-SA, which permits commercial use, directly conflicting with the NC restriction.

Algorithmic Compatibility Procedure: Define `CheckRemixValidity(assetA, assetB)`. If `assetA.hasND` or `assetB.hasND`, return `False (Cannot remix NoDerivatives works)`. If `assetA.hasNC` and `assetB.hasSA`, return `False (Incompatible license combination)`. If valid, calculate the most restrictive compatible license (e.g., combining CC BY and CC BY-NC yields CC BY-NC) and assign it to the composite project.

Part C: Data Abstraction & Complexity Management

Automated Attribution Generator: When a teacher compiles an educational module, the export pipeline parses all embedded media records and generates a standardized TASL (Title, Author, Source, License) HTML metadata footer.

Example Output: 'Photo: Solar Flare by Dr. Elena Ramos, licensed under CC BY 4.0, available at https://edu-repo.org/solar-flare'. This ensures legal compliance automatically without requiring educators to memorize copyright formatting.

Part D: Testing, Edge Cases & Ethical Governance

Democratizing Education: Traditional college and high school textbooks cost hundreds of dollars per student, severely disadvantaging low-income schools and families. Open Educational Resources (OER) dismantle this economic barrier, granting equal access to high-quality learning tools.

Global Localization and Remixing: Teachers in non-English speaking or developing nations can translate, adapt, and update open textbooks to reflect local cultural contexts, scientific discoveries, and curricula without paying exorbitant licensing fees.

OFFICIAL COLLEGE BOARD 9-POINT SCORING RUBRIC:

Criteria / Evaluated Performance Skill	Points Allocated
Row 1: Copyright Comparison: Distinguishes traditional 'All Rights Reserved' copyright (automatic upon fixation) from permissive Creative Commons frameworks.	**1 pt**
Row 2: CC0 Public Domain: Explains that CC0 waives all copyright interests, allowing anyone to use the work without attribution or restrictions.	**1 pt**
Row 3: License Modifier Rules: Accurately details the operational constraints of NC (NonCommercial), ND (NoDerivatives), and SA (ShareAlike).	**1 pt**
Row 4: Derivative Incompatibility: Identifies that a work with ND (NoDerivatives) cannot be remixed, and SA (ShareAlike) requires identical downstream licensing.	**1 pt**
Row 5: Compatibility Procedure: Constructs a programmatic logic function that checks whether combining two source media licenses produces a valid derivative license.	**1 pt**
Row 6: Data Representation: Uses a structured list or record format to represent media asset licensing metadata.	**1 pt**
Row 7: Attribution Generation: Specifies an automated attribution generator that formats Author, Title, Source URL, and License string.	**1 pt**
Row 8: Educational Equity Impact: Discusses how zero-cost open educational resources eliminate textbook financial barriers for underprivileged students.	**1 pt**
Row 9: Global Localization: Explains how the freedom to remix and translate materials under CC BY enables culturally responsive local education.	**1 pt**
TOTAL SCORING MAXIMUM	**9 / 9 Points**

Question 95 · Model Solution & Scoring Rubric

Digital Divide Rural Broadband Infrastructure Allocator ❖ Total Available Points: 9 Points

COMPREHENSIVE EXEMPLARY STUDENT RESPONSE:

Part A: Program Purpose and Computing Bias / Impact

Dimensions of the Digital Divide: In rural areas, the vast geographic distances between households and difficult terrain (mountains, dense forests) make laying fiber-optic cables financially unviable for private, profit-driven Internet Service Providers (ISPs).

Socio-Economic Penalties: Disconnected rural families suffer from the 'homework gap' (students sitting in fast-food parking lots to complete homework), inability to access remote work opportunities, and severe barriers to modern telemedicine services.

Part B: Algorithmic Procedure & Data Mitigation

Data Abstraction: Create a record `CensusTract` with fields `tractID`, `currentSpeedMbps`, `medianIncome`, `studentCount`, `terrainDifficultyFactor`, and `calculatedPriorityScore`. Store tracts in a list `stateTracts`.

Scoring Function: Define `CalculateNeedScore(tract)`: `priorityScore = (100 - tract.currentSpeedMbps) 0.45 + (100000 - tract.medianIncome) / 1000 0.35 + (tract.studentCount / 100) * 0.20`. A sorting algorithm arranges `stateTracts` in descending order of `priorityScore` to allocate funding to highest-need communities first.

Part C: Data Abstraction & Complexity Management

Filtering and Allocation Routine: The program implements `AllocateGrants(sortedTracts, totalBudget)`: Initialize `allocatedFunds = 0`. Iterate through `sortedTracts`. If `tract.currentSpeedMbps < 25`, estimate installation cost based on distance and terrain, subtract from `totalBudget`, and flag tract for grant award until funds are fully committed.

Part D: Testing, Edge Cases & Ethical Governance

Algorithmic Blindspots & Flawed Maps: A major flaw in government broadband planning has been reliance on self-reported ISP data (such as historical FCC Form 477 maps), where an ISP could report an entire census block as 'connected' if they served merely one house at the edge of the border.

Equitable Remediation: To eliminate this blindspot and protect marginalized indigenous reservations and remote communities, the algorithm must ingest ground-truth crowdsourced mobile speed tests and mandate on-the-ground physical verification before disqualifying any region from funding.

OFFICIAL COLLEGE BOARD 9-POINT SCORING RUBRIC:

Criteria / Evaluated Performance Skill	Points Allocated
Row 1: Geographic Factors: Explains how low population density and rugged terrain make rural fiber deployment economically unprofitable for commercial ISPs.	**1 pt**
Row 2: Socio-Economic Impact: Details the severe educational (homework gap), economic, and healthcare (telemedicine) penalties suffered by disconnected areas.	**1 pt**
Row 3: Multi-Criteria Factors: Incorporates multiple distinct metrics (current average download/upload speed, median household income, school-age child density).	**1 pt**
Row 4: Algorithmic Scoring Logic: Constructs a weighted mathematical ranking formula that prioritizes communities with low speeds and low median incomes.	**1 pt**
Row 5: Data Abstraction Design: Defines a structured record `CensusTract` containing relevant connectivity and economic attributes stored in a collection list.	**1 pt**
Row 6: Filtering Procedure: Specifies a filtering procedure that identifies tracts failing minimum broadband speed standards (e.g., <25 Mbps download).	**1 pt**
Row 7: ISP Mapping Blindspots: Identifies the fatal flaw of relying on self-reported commercial ISP coverage maps that mark entire census blocks as served if a single home has access.	**1 pt**
Row 8: Marginalized Communities: Explains how indigenous tribal lands, migrant worker communities, and mobile home parks are frequently misclassified or overlooked.	**1 pt**
Row 9: Equitable Policy Remedy: Proposes integrating ground-truth crowdsourced speed test data and community public comment periods before final funding approval.	**1 pt**
TOTAL SCORING MAXIMUM	**9 / 9 Points**

Question 96 · Model Solution & Scoring Rubric

Heuristic Phishing Detection and Domain Spoofing Defense ❖ Total Available Points: 9 Points

COMPREHENSIVE EXEMPLARY STUDENT RESPONSE:

Part A: Program Purpose and Computing Bias / Impact

Psychological Exploitation: Phishing emails exploit high-stress clinical environments by weaponizing artificial urgency and authority (e.g., 'URGENT: Payroll suspended—verify credentials immediately' or 'Emergency ICU Protocol Update from the Medical Director').

Technical Domain Spoofing: Attackers employ typo-squatting (`hosp1tal-portal.com` instead of `hospital-portal.com`) or internationalized domain homoglyph attacks (substituting Latin 'a' with identical-looking Cyrillic 'a') so the destination URL appears authentic to the naked eye.

Part B: Algorithmic Procedure & Data Mitigation

Algorithmic Threat Heuristic Procedure: Define `AnalyzeEmail(emailRecord)`: Initialize `riskScore = 0`. 1. Run SPF/DKIM verification: If sender IP fails DMARC authentication, `riskScore += 40`. 2. Iterate through all hyperlinks in `emailRecord.body`: If visible text displays `hospital.org` but actual `href` points to external domain `update-portal.biz`, `riskScore += 35`. 3. Scan body for urgency keywords ('immediately', 'suspend', 'verify password'): If present, `riskScore += 15`. 4. Check domain creation date via WHOIS API: If domain was registered within past 14 days, `riskScore += 30`. If `riskScore >= 70`, move to `Quarantine` and alert security; if between 35 and 69, inject prominent red external warning banner.

Part C: Data Abstraction & Complexity Management

False Positives vs False Negatives: In a hospital setting, setting heuristic thresholds involves critical life-safety trade-offs:

- **False Positives:** If the filter is overly aggressive, critical communications (e.g., external organ donor match notifications, lab reports) are quarantined, potentially endangering patients.

- **False Negatives:** If the filter is too lenient, a single nurse or doctor clicking a phishing link allows ransomware to encrypt patient databases, taking surgical monitoring systems offline.

Part D: Testing, Edge Cases & Ethical Governance

Defensive Policies and Controls: Relying on human vigilance alone is insufficient. The hospital must deploy hardware-based, phishing-resistant Multi-Factor Authentication (e.g., FIDO2/WebAuthn physical security keys), where cryptographic token binding prevents credentials from being intercepted even if entered on a fake site.

Continuous Simulated Training: Conduct regular, unannounced phishing simulations with instant feedback and safe learning remediation for staff.

OFFICIAL COLLEGE BOARD 9-POINT SCORING RUBRIC:

Criteria / Evaluated Performance Skill	Points Allocated
Row 1: Social Engineering Triggers: Identifies psychological triggers used in phishing emails (urgency, fear, administrative authority, curiosity).	**1 pt**
Row 2: Technical Spoofing Mechanics: Explains technical evasion tactics: homoglyph attacks (lookalike Cyrillic characters), typo-squatting, and hidden link redirection.	**1 pt**
Row 3: Header Parsing Inspection: Details checking sender domain alignment against authentication standards (SPF, DKIM, and DMARC verification).	**1 pt**
Row 4: Hyperlink Text Mismatch: Specifies an algorithmic check verifying whether visible anchor text matches the actual underlying destination URL (`href`).	**1 pt**
Row 5: Threat Scoring Algorithm: Constructs a procedural heuristic function that aggregates suspicion points from multiple indicators into an overall risk score.	**1 pt**
Row 6: Quarantine Logic: Defines automated action thresholds (e.g., score > 75 routes email to quarantine; score between 40-75 displays prominent warning banner).	**1 pt**
Row 7: False Positive Medical Harm: Explains how high false positives can delay urgent patient medical transfers or laboratory results.	**1 pt**
Row 8: False Negative Breach Risk: Explains how a false negative can lead to hospital-wide ransomware extortion and shutdown of life-saving medical devices.	**1 pt**
Row 9: Defense-in-Depth Controls: Recommends phishing-resistant hardware MFA (FIDO2/WebAuthn) and periodic simulated phishing training exercises.	**1 pt**
TOTAL SCORING MAXIMUM	**9 / 9 Points**

Question 97 · Model Solution & Scoring Rubric

PII Anonymization and De-Identification Pipeline for Open Data ❖ Total Available Points: 9 Points

COMPREHENSIVE EXEMPLARY STUDENT RESPONSE:

Part A: Program Purpose and Computing Bias / Impact

PII and the Mosaic Effect: Personally Identifiable Information (PII) includes any data that directly identifies an individual (name, social security number, card ID) or can be combined with other data to identify them. Even without names, individual location trajectories are highly unique.

Re-Identification via Linkage: Latanya Sweeney proved that 87% of the US population can be uniquely identified by just {5-digit ZIP, gender, date of birth}. In transit, if an attacker knows an individual's home subway stop and arrival time at work, matching these two quasi-identifiers against an 'anonymized' log re-identifies their entire travel history.

Part B: Algorithmic Procedure & Data Mitigation

Data Transformation Pipeline: The raw dataset passes through three transformation stages:

1. **Direct Identifier Suppression:** Remove card serial numbers, billing names, and phone numbers.
2. **Timestamp Bucketing:** Generalize precise timestamps (e.g., `2026-10-14 08:03:41`) into 30-minute intervals (`2026-10-14 08:00-08:30`).
3. **Spatial Aggregation:** Group specific entrance turnstile turnstiles into generalized municipal borough transit zones (e.g., `Zone 4 - Midtown East`).

Part C: Data Abstraction & Complexity Management

k-Anonymity Definition & Verification: A dataset satisfies k-anonymity if every distinct combination of quasi-identifier attributes appears in at least k distinct individual records, ensuring no person can be singled out from a crowd of fewer than k peers.

Verification Procedure: Define `VerifyKAnonymity(transitDataset, quasiIdentifierColumns, k)`: Group `transitDataset` by `quasiIdentifierColumns` and compute the record count for each group. Iterate through group counts: if any group has `count < k`, return `False` and flag the non-compliant groups for further generalization or record suppression. Return `True` only when all cohorts meet or exceed k.

Part D: Testing, Edge Cases & Ethical Governance

Ethical Governance & Legal Safeguards: Technical de-identification must be accompanied by administrative legal agreements. University researchers must execute binding Data Use Agreements (DUAs) that explicitly prohibit any attempts to de-anonymize individuals or link datasets with commercial databases.

Tiered Access: Raw aggregate queries should be mediated through a secure API that applies Differential Privacy noise, rather than publishing raw individual micro-data files on the public internet.

OFFICIAL COLLEGE BOARD 9-POINT SCORING RUBRIC:

Criteria / Evaluated Performance Skill	Points Allocated
Row 1: PII Definition: Accurately defines Personally Identifiable Information (PII) as data that directly or indirectly identifies, contacts, or locates an individual.	**1 pt**
Row 2: Mosaic Re-Identification: Explains how combining non-PII quasi-identifiers (timestamp, station, fare type) with external public records re-identifies specific individuals.	**1 pt**
Row 3: Direct Identifier Removal: Specifies complete stripping of direct identifiers (card serial numbers, passenger names, credit card tokens).	**1 pt**
Row 4: Generalization / Bucketing: Details an algorithmic transformation that buckets exact timestamps (e.g., 8:04:22 AM) into generalized time intervals (e.g., 8:00 AM - 8:30 AM).	**1 pt**
Row 5: Spatial Aggregation: Details spatial aggregation (grouping individual turnstiles into broader transit zone sectors).	**1 pt**
Row 6: k-Anonymity Concept: Explains that a dataset satisfies k-anonymity if each unique combination of quasi-identifiers is shared by at least k distinct individuals.	**1 pt**
Row 7: Verification Algorithm: Outlines an algorithmic grouping and counting procedure that flags any quasi-identifier cohort with count < k.	**1 pt**
Row 8: Differential Privacy / Noise: Discusses the addition of mathematical Laplace/Gaussian noise to prevent reconstruction attacks.	**1 pt**
Row 9: Governance & Legal Agreements: Proposes strict Data Use Agreements (DUAs) prohibiting researchers from attempting to de-anonymize or re-identify subjects.	**1 pt**
TOTAL SCORING MAXIMUM	**9 / 9 Points**

Question 98 · Model Solution & Scoring Rubric

Algorithmic Fair Lending and Credit Scoring Evaluation ❖ Total Available Points: 9 Points

COMPREHENSIVE EXEMPLARY STUDENT RESPONSE:

Part A: Program Purpose and Computing Bias / Impact

Alternative Data as Discriminatory Proxies: While the fintech claims its model is objective because it excludes race and gender, alternative features act as direct proxies for socioeconomic status and race. For example, using expensive iPhone models vs budget Android devices reflects household wealth, and social media graph connections reflect historical residential segregation.

Facially Neutral Disparate Impact: Even with no intentional malicious discrimination, an algorithm that penalizes late-night phone charging or specific cell towers disproportionately denies credit to shift-working hourly laborers and minority neighborhoods, creating illegal disparate impact under the Equal Credit Opportunity Act (ECOA).

Part B: Algorithmic Procedure & Data Mitigation

Statistical Parity and Disparate Impact Audit Procedure: Define `AuditFairLending(loanApplicationData, demographicAttribute)`: 1. For each group G in `demographicAttribute`: `acceptanceRate[G] = totalApproved[G] / totalApplicants[G]`. 2. Find `maxRate = max(acceptanceRate.values())`. 3. For each group G: `impactRatio = acceptanceRate[G] / maxRate`. If `impactRatio < 0.80`, record regulatory violation flag ('Disparate impact detected: Group ' + G + ' falls below 80% benchmark'). 4. Return audit report for compliance review.

Part C: Data Abstraction & Complexity Management

Adverse Action Explainability: Financial regulations mandate that any denied loan applicant receive a clear Adverse Action notice stating the principal reasons for denial. Deep learning 'black box' models cannot legally be deployed if they cannot explain why a credit decision was made.

Explainability Procedure: Implement feature attribution analysis (such as SHAP - Shapley Additive exPlanations). When a borrower is denied, the system calculates the mathematical negative weight contributed by each input feature and outputs the top 3 factors in plain English (e.g., '1. High debt-to-income ratio; 2. History of late utility payments; 3. Insufficient stable employment history') rather than uninterpretable mathematical weights.

Part D: Testing, Edge Cases & Ethical Governance

Macroeconomic and Intergenerational Consequences: Credit scoring determines access to mortgages, small business loans, vehicle financing, and rental housing. When automated algorithms entrench biased credit denials, they systematically prevent marginalized families from accumulating intergenerational home equity and building businesses, perpetuating racial and economic inequality through a facade of mathematical objectivity.

OFFICIAL COLLEGE BOARD 9-POINT SCORING RUBRIC:

Criteria / Evaluated Performance Skill	Points Allocated
Row 1: Proxy Variable Identification: Explains how alternative data (e.g., smartphone model, battery charging habits, zip codes) strongly correlates with income, race, and socioeconomic class.	1 pt
Row 2: Disparate Impact Concept: Distinguishes facially neutral algorithmic rules from legally impermissible disparate impact on protected demographic groups.	1 pt
Row 3: Statistical Parity Metric: Defines demographic statistical parity (acceptance rate for cohort A equals acceptance rate for cohort B).	1 pt
Row 4: Parity Audit Algorithm: Constructs an algorithm that evaluates loan approval rates per demographic group and calculates the ratio against benchmark rates.	1 pt
Row 5: Four-Fifths Rule Threshold: Applies regulatory standards (e.g., EEOC Four-Fifths rule / 80% threshold) to trigger algorithmic compliance flags.	1 pt
Row 6: Black-Box Explainability Need: Articulates why complex neural networks / deep learning models create unexplainable 'black-box' credit decisions.	1 pt
Row 7: Adverse Action Procedure: Designs a procedure that extracts the top mathematical contributing negative factors from the scoring model.	1 pt
Row 8: Actionable Recourse: Ensures the generated explanation provides the applicant with transparent, actionable steps to improve their score.	1 pt
Row 9: Systemic Inequality Feedback: Explains how predatory credit denial locks disadvantaged populations out of homeownership, cementing generational wealth gaps.	1 pt
TOTAL SCORING MAXIMUM	**9 / 9 Points**

Question 99 · Model Solution & Scoring Rubric

Multi-Factor Authentication (MFA) Token Verification System ❖ Total Available Points: 9 Points

COMPREHENSIVE EXEMPLARY STUDENT RESPONSE:

Part A: Program Purpose and Computing Bias / Impact

The Three Authentication Factor Categories:

1. **Something You Know (Knowledge):** Alphanumeric passwords, passphrases, or PINs.
2. **Something You Have (Possession):** Hardware security keys (YubiKey), smartcards, or smartphone authenticator apps generating time-based tokens.
3. **Something You Are (Inherence):** Biometric fingerprints, iris scans, or facial recognition.

Clinical Requirement: Hospital EHR systems handle sensitive HIPAA-protected health records. Relying solely on passwords leaves systems vulnerable to credential stuffing and phishing; true MFA requires at least two separate factor categories.

Part B: Algorithmic Procedure & Data Mitigation

TOTP Verification Procedure with Clock Drift: TOTP calculates codes using HMAC-SHA1 on a shared secret `K` and a counter `T = floor(currentTime / 30)`. Because server and smartphone clocks can be slightly out of sync (clock drift), the server checks adjacent time windows.

Algorithmic Verification Logic: `function VerifyToken(userSecret, submittedCode, currentTime)`: 1. `currentWindow = floor(currentTime / 30)`. 2. For `offset` from -1 to +1 (covering $t-1$, t, and $t+1$): `calculatedCode = GenerateTOTP(userSecret, currentWindow + offset)` If `submittedCode == calculatedCode`: If `IsCodeRecentlyUsed(submittedCode)` return `False` (Replay defense); `MarkCodeAsUsed(submittedCode)`; Return `True (Access Granted)`; 3. Return `False (Invalid Code)`.

Part C: Data Abstraction & Complexity Management

Brute-Force and Rate-Limiting Defenses: Because a 6-digit code has only 1,000,000 possibilities, an attacker with a stolen password could attempt rapid brute-force guesses within the 30-second window.

Rate-Limiting Rule: Maintain an attempt counter per account and IP address. If 5 incorrect MFA attempts occur within 5 minutes, enforce an exponential delay (e.g., 30-second lockout doubling on each failure) and alert the security operations team.

Part D: Testing, Edge Cases & Ethical Governance

Accessibility & Emergency Break-Glass: Medical staff with visual impairments or motor tremors may struggle with typing 6-digit codes every 30 seconds. The hospital should deploy tap-to-authenticate FIDO2 USB/NFC badges that authenticate with a single physical touch.

Break-Glass Procedure: In life-threatening trauma situations where a doctor's MFA device is broken or unavailable, a supervised 'break-glass' protocol allows emergency bypass with dual biometric authorization from a supervising nurse, generating an automatic audit log for immediate retrospective review.

OFFICIAL COLLEGE BOARD 9-POINT SCORING RUBRIC:

Criteria / Evaluated Performance Skill	Points Allocated
Row 1: Three Factors Identification: Identifies and contrasts the three factor categories: something you know, something you have, and something you are.	**1 pt**
Row 2: True MFA Compliance: Explains why combining two distinct factors (e.g., password + physical authenticator) is mandatory for HIPAA compliance.	**1 pt**
Row 3: TOTP Mathematical Concept: Explains how TOTP uses a shared secret key and current UNIX epoch time divided into 30-second time steps.	**1 pt**
Row 4: Clock Drift Algorithm: Designs an algorithm verifying the submitted code against the current time window, previous window ($t-1$), and next window ($t+1$).	**1 pt**
Row 5: One-Time Token Invalidation: Ensures an accepted 6-digit code is immediately invalidated in memory to prevent replay attacks.	**1 pt**
Row 6: Brute-Force Rate Limiting: Specifies an exponential backoff or rate-limiting algorithm that locks attempts after 5 consecutive failures.	**1 pt**
Row 7: Emergency Clinical Override: Details secure procedures for emergency 'break-glass' patient trauma access during acute hospital crises.	**1 pt**
Row 8: Accessibility Considerations: Provides accessible hardware options (e.g., FIDO2 NFC keys with large physical buttons, biometric fingerprint readers for motor disabilities).	**1 pt**
Row 9: Account Recovery Protocol: Establishes secure, out-of-band identity verification for resetting lost MFA tokens without relying on easily phished SMS.	**1 pt**
TOTAL SCORING MAXIMUM	**9 / 9 Points**

Question 100 · Model Solution & Scoring Rubric

Assistive Screen-Reader and Accessible UI Audit Engine ❖ Total Available Points: 9 Points

COMPREHENSIVE EXEMPLARY STUDENT RESPONSE:

Part A: Program Purpose and Computing Bias / Impact

Ethical, Legal, and Educational Imperative: Under the Americans with Disabilities Act (ADA) and Section 508 of the Rehabilitation Act, public educational institutions must provide equal access to electronic information and technology. Denying accessible course registration, textbooks, or assignment submissions effectively excludes students with visual, auditory, cognitive, or motor impairments from higher education.

Assistive Technology Mechanics: Screen readers (e.g., NVDA, JAWS) parse the web page's Document Object Model (DOM) and convert text and structural HTML elements into synthetic speech or refreshable braille displays. If developers omit semantic tags, headings, or alternative text, the interface becomes an impenetrable wall of silence.

Part B: Algorithmic Procedure & Data Mitigation

Algorithmic DOM Accessibility Inspection: Define `AuditPageAccessibility(domDocument)`: 1. **Alt Text Check:** Query all `<img>` elements. If `!img.hasAttribute('alt')` or `img.alt.trim() == ''` (and image is not purely decorative `aria-hidden='true'`), or if `img.alt` contains generic placeholders like 'image.jpg' or 'photo', flag `Violation: Missing meaningful alt text`. 2. **Contrast Ratio Check:** For each text node, extract foreground color C_1 and background color C_2. Compute relative luminance L_1 and L_2 using standard sRGB gamma formulas. Calculate contrast ratio $R = (L_1 + 0.05) / (L_2 + 0.05)$ where $L_1 > L_2$. If normal text has $R < 4.5$ (or large text has $R < 3.0$), flag `Violation: Insufficient color contrast ($R:1$ fails WCAG AA standard)`.

Part C: Data Abstraction & Complexity Management

Data Abstraction and Triage: Define a record `AccessibilityViolation` with fields `elementID`, `wcagRule`, `severity` (Critical / Major / Minor), `description`, and `remediationGuide`. Store all flagged issues in a list `auditLog`. Triage Algorithm: Prioritize violations: Interactive elements that cannot be focused via keyboard navigation or forms without submit labels are categorized as 'Critical' because they completely block user workflows. Low contrast text is ranked 'Major'. Sort and export `auditLog` to development sprint backlogs.

Part D: Testing, Edge Cases & Ethical Governance

The Curb-Cut Effect and Universal Usability: In architecture, sidewalk curb cuts built for wheelchair users ultimately benefited parents with strollers, delivery workers with hand trucks, and travelers with rolling luggage. Similarly, digital accessibility improvements enhance usability for everyone ('the digital curb-cut effect'):

- High-contrast themes allow students to read course portals on mobile devices outdoors under direct sunlight.
- Video closed captions assist non-native English speakers and students studying in quiet libraries.
- Semantic heading hierarchies help search engine spiders index content efficiently and allow all users to skim long articles quickly.

OFFICIAL COLLEGE BOARD 9-POINT SCORING RUBRIC:

Criteria / Evaluated Performance Skill	Points Allocated
Row 1: Accessibility Legal & Moral Imperative: Articulates legal standards (ADA, Section 508, WCAG 2.1) and the moral right to equal access in public education.	**1 pt**
Row 2: Assistive Technology Interaction: Explains how blind and low-vision students navigate web content using screen readers (JAWS, NVDA, VoiceOver) via structured HTML semantics.	**1 pt**
Row 3: Image Alt Text Inspection: Details an algorithmic check ensuring `<img>` elements possess descriptive, non-empty, non-redundant `alt` attributes.	**1 pt**
Row 4: Luminance Calculation Formula: Specifies the mathematical calculation of relative luminance and contrast ratio between foreground text and background color.	**1 pt**
Row 5: WCAG 4.5:1 Ratio Check: Applies the WCAG AA standard requiring at least a 4.5:1 contrast ratio for normal text (3:1 for large text).	**1 pt**
Row 6: Keyboard Navigation Audit: Details programmatic checks for keyboard accessibility (`tabindex`, visible focus indicators, avoiding keyboard traps).	**1 pt**
Row 7: Data Abstraction for Violations: Designs a structured record `AccessibilityViolation` containing element selector, violation type, severity, and remediation guidance.	**1 pt**
Row 8: Priority Triage Logic: Implements a triage sorting procedure prioritizing blockers (e.g., inaccessible submit buttons) over minor cosmetic warnings.	**1 pt**
Row 9: Universal Curb-Cut Effect: Explains how accessibility features (captions, high contrast, clean semantic hierarchy) benefit non-disabled users, mobile users in sunlight, and English learners.	**1 pt**
TOTAL SCORING MAXIMUM	**9 / 9 Points**

★ ❖ ★

Thank You, Future Technologists, Ethicists & Innovators!

YOU HAVE COMPLETED THE 100-QUESTION UNIT 5 MASTER EXAMINATION

Congratulations on rigorously conquering all 100 questions of **Unit 5: Impact of Computing**. From diagnosing algorithmic bias and dismantling demographic proxy variables, to exploring the socioeconomic roots of the Digital Divide, protecting intellectual property with Creative Commons, and engineering robust cybersecurity defenses with Multi-Factor Authentication, you have mastered the foundational intersections of technology, ethics, and human society.

Computing is not merely about writing algorithms—it is about shaping the human experience. As future computer scientists, software architects, and ethical leaders, you hold the extraordinary power to design systems that empower communities, protect human dignity, and expand equitable opportunity across the globe.

"Technology is neither good nor bad; nor is it neutral. The future of computing depends on the integrity, empathy, and ethical vigilance of its creators."